WSQ COURSE · MICROSOFT AZURE

Microsoft Azure AI Fundamentals (AI-900)

Course Code: TGS-2023021100
ICT-TEM-3034-1.1 · 2 days · 16 hours

Version 10.0 · 2 September 2026

Tertiary Infotech Academy Pte Ltd
UEN 201200696W



LEGACY SLIDE 002 · ADMIN

Digital Attendance Is Mandatory

1 INPUT / SIGNAL

Scan the SSG-issued QR code shown by the trainer.

2 MECHANISM

Submit AM, PM and assessment attendance separately.

3 OUTPUT / EVIDENCE

Use the enrolled learner identity and correct course run.

4 CONTROL / VERIFY

Show the confirmation state before closing the form.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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About the Trainer



Trainer Profile Template

Complete before delivery

NAME / DESIGNATION _____

QUALIFICATIONS _____

EXPERTISE _____

EXPERIENCE _____

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About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE
Principal trainer and course developer

DOMAIN
Cloud, AI, automation and software engineering

DELIVERY
Applied WSQ technical training

CONTACT
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Ground Rules

1 INPUT / SIGNAL

Participate actively and ask technical questions.

2 MECHANISM

Keep devices silent except during guided lab work.

3 OUTPUT / EVIDENCE

Use only trainer-approved Azure subscriptions and data.

4 CONTROL / VERIFY

Complete attendance, evidence and assessment independently.

Control: define the threshold or failure that routes to human review.

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Lesson Plan

1	2	3	4
INPUT / SIGNAL	MECHANISM	OUTPUT / EVIDENCE	CONTROL / VERIFY
Day 1 AM: administration, outcomes and AI workload foundations.	Day 1 AM: Responsible AI, Foundry and machine learning concepts.	Day 1 PM: regression, classification, clustering and metrics.	Day 1 close: preserve evidence and stop billable compute.

Control: define the threshold or failure that routes to human review.

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Lesson Plan

1 INPUT / SIGNAL	2 MECHANISM	3 OUTPUT / EVIDENCE	4 CONTROL / VERIFY
Day 2 AM: computer vision, OCR and Document Intelligence.	Day 2 AM: language, speech, translation and conversational AI.	Day 2 PM: anomaly evidence, safety and Azure AI Search.	Day 2 close: feedback, digital attendance and final assessment.

Control: define the threshold or failure that routes to human review.

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Skills Framework and TSC

1 INPUT / SIGNAL

TSC: Artificial Intelligence Application in Product Development.

2 MECHANISM

TSC code: ICT-TEM-3034-1.1.

3 OUTPUT / EVIDENCE

Context: implement and review AI applications with service providers.

4 CONTROL / VERIFY

Evidence spans deployment, issue reporting and data quality.

Control: define the threshold or failure that routes to human review.

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TSC Abilities and Knowledge

1 INPUT / SIGNAL

A1-A2: implement applications and deploy planned workflows.

2 MECHANISM

A3: identify and report application or collected-data issues.

3 OUTPUT / EVIDENCE

A4-A5: maintain interoperability and perform data cleaning.

4 CONTROL / VERIFY

K1-K11 cover AI, ML, data, implementation and clean datasets.

Control: define the threshold or failure that routes to human review.

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Learning Outcomes

1 INPUT / SIGNAL

LO1: implement AI applications with Microsoft Azure services.

2 MECHANISM

LO2: deploy computer vision and language workflows.

3 OUTPUT / EVIDENCE

LO3: identify and report anomaly, safety and data issues.

4 CONTROL / VERIFY

LO4: maintain interoperability and clean data for knowledge mining.

Control: define the threshold or failure that routes to human review.

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Course Outline

1 INPUT / SIGNAL

Topic 1: AI workloads, Responsible AI and Microsoft Foundry.

2 MECHANISM

Topic 2: ML fundamentals, metrics, designer and AutoML.

3 OUTPUT / EVIDENCE

Labs 1-3 produce workload, metric and lifecycle evidence.

4 CONTROL / VERIFY

Knowledge: K1-K4, K6-K7 · Abilities: A1-A2.

Control: define the threshold or failure that routes to human review.

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Course Outline

1 INPUT / SIGNAL

Topic 3: computer vision, image analysis and OCR.

2 MECHANISM

Topic 4: language, speech, translation and conversational AI.

3 OUTPUT / EVIDENCE

Topics 5-6: anomaly evidence, governance and knowledge mining.

4 CONTROL / VERIFY

Knowledge: K5, K8-K11 · Abilities: A3-A5.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 013 · ADMIN

Course Outline

1 INPUT / SIGNAL

Final blueprint: AI workloads, ML, vision, language and generative AI.

2 MECHANISM

AI-900 certification exam retired on 30 June 2026.

3 OUTPUT / EVIDENCE

The registered WSQ title and learning outcomes remain unchanged.

4 CONTROL / VERIFY

Current labs use Microsoft Foundry and supported Azure AI capabilities.

Control: define the threshold or failure that routes to human review.

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Prerequisite

1 INPUT / SIGNAL

Use a Microsoft account approved for the class.

2 MECHANISM

Use only the trainer-approved Azure subscription and region.

3 OUTPUT / EVIDENCE

Bring a modern browser; Python 3 supports local validation.

4 CONTROL / VERIFY

Never place keys, personal data or confidential content in evidence.

Control: define the threshold or failure that routes to human review.

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Final Assessment

1 INPUT / SIGNAL

WA-SAQ: 11 open-ended questions in 60 minutes.

2 MECHANISM

PP: four practical tasks in 90 minutes.

3 OUTPUT / EVIDENCE

Open-book materials are limited to trainer-approved courseware.

4 CONTROL / VERIFY

Appeals follow the documented assessment and reassessment process.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 016 · ADMIN

Briefing for Assessment

1 INPUT / SIGNAL

Place unapproved materials and phones away before assessment.

2 MECHANISM

Do not photograph, record or discuss assessment scripts.

3 OUTPUT / EVIDENCE

Use the required pen or approved digital submission method.

4 CONTROL / VERIFY

Submit every script and evidence artifact when time is called.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 017 · ADMIN

Criteria for Funding

1 INPUT / SIGNAL

Meet the attendance requirement for the funded course run.

2 MECHANISM

Complete the course feedback and TRAQOM survey.

3 OUTPUT / EVIDENCE

Attempt every registered assessment component.

4 CONTROL / VERIFY

Keep confirmation evidence until the course administrator verifies it.

Control: define the threshold or failure that routes to human review.

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LMS / TMS · VISUAL ACCESS PATH

Download Course Material

1 Sign in

lms-tms.tertiaryinfotech.com

2 Open course

Microsoft Azure AI Fundamentals

3 Download

Slides · Learner Guide · Labs

Verify course code TGS-2023021100 before downloading. Detailed access instructions are in the Learner Guide.

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LEGACY COVERAGE · SLIDE 019

Topic 1 Get started with Artificial Intelligence

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Final AI-900 Skills Domains

1 INPUT / SIGNAL

AI-900 Azure AI Fundamentals.

2 MECHANISM

Identify the input contract for Final AI-900 Skills Domains.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

Capture a named output, metric or observable artifact.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 021 · FORMULA

What is AI?

1 INPUT / SIGNAL

Simply put, AI is the creation of software that imitates human behaviors and capabilities.

2 MECHANISM

Anomaly detection - The capability to automatically detect errors or unusual activity in a system.

3 OUTPUT / EVIDENCE

Computer vision - The capability of software to interpret the world visually through cameras, video, and images.

4 CONTROL / VERIFY

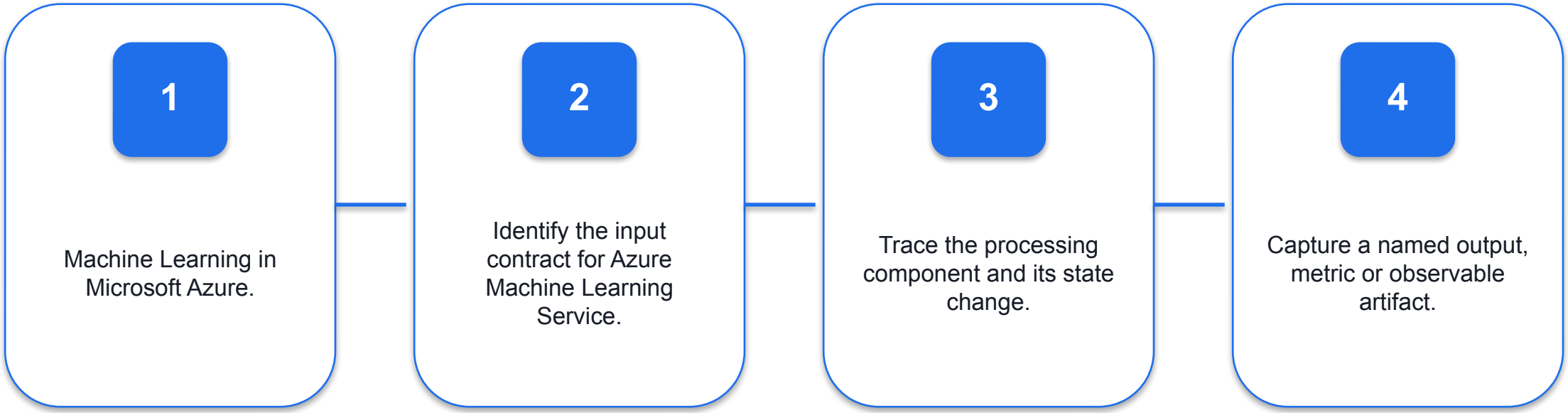
Natural language processing - The capability for a computer to interpret written or spoken language, and respond in kind.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Azure Machine Learning Service



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 023 · FORMULA

Time-Series Anomaly Detection

1 INPUT / SIGNAL

Anomaly Detection.

2 MECHANISM

Expected value and margins define an evidence boundary.

3 OUTPUT / EVIDENCE

Sensitivity changes how tightly the boundary follows history.

4 CONTROL / VERIFY

A flagged point is a review signal, not a root-cause diagnosis.

Control: define the threshold or failure that routes to human review.

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Computer Vision Workloads

1 INPUT / SIGNAL

Computer Vision.

2 MECHANISM

Identify the input contract for Computer Vision Workloads.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

Capture a named output, metric or observable artifact.

Control: define the threshold or failure that routes to human review.

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Image Classification

1 INPUT / SIGNAL

Features map to one of two or more categorical labels.

2 MECHANISM

The model emits class scores or probabilities.

3 OUTPUT / EVIDENCE

A threshold converts scores into predicted labels.

4 CONTROL / VERIFY

Validate with a confusion matrix and class-specific metrics.

Control: define the threshold or failure that routes to human review.

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Object Detection

1 INPUT / SIGNAL

The model returns a label, confidence and bounding box per object.

2 MECHANISM

Multiple objects can appear in one image.

3 OUTPUT / EVIDENCE

Validate both classification and localization quality.

4 CONTROL / VERIFY

Route low-confidence or sensitive detections for review.

Control: define the threshold or failure that routes to human review.

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Semantic Segmentation

1 INPUT / SIGNAL

Every pixel receives a class label.

2 MECHANISM

The output mask preserves object shape and boundaries.

3 OUTPUT / EVIDENCE

Validate class overlap, missed regions and edge quality.

4 CONTROL / VERIFY

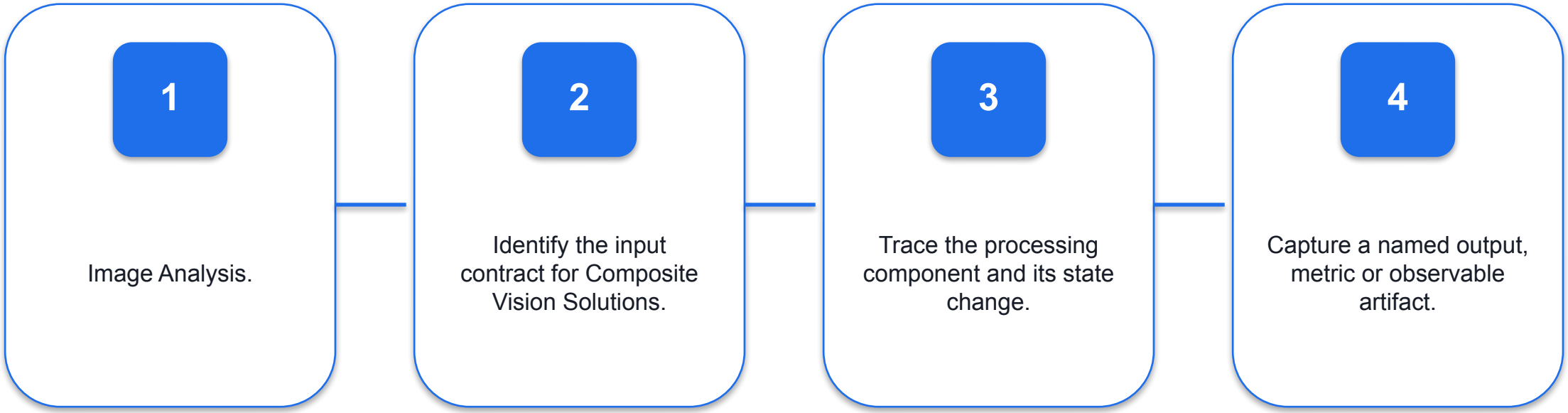
Use when pixel-level area matters more than a box.

Control: define the threshold or failure that routes to human review.

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Composite Vision Solutions



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Face Detection

1 INPUT / SIGNAL

Detection returns a face region; identification asserts identity.

2 MECHANISM

Consent, purpose and legal authority must be explicit.

3 OUTPUT / EVIDENCE

Do not infer sensitive traits from facial appearance.

4 CONTROL / VERIFY

Use human review for high-impact decisions.

Control: define the threshold or failure that routes to human review.

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Optical Character Recognition

1 INPUT / SIGNAL

OCR.

2 MECHANISM

Identify the input contract for Optical Character Recognition.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

Capture a named output, metric or observable artifact.

Control: define the threshold or failure that routes to human review.

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Natural Language Processing

1

Interpret spoken language, and synthesize speech responses.

2

Automatically translate spoken or written phrases between languages.

3

Interpret commands and determine appropriate actions.

4

ROC plots true-positive rate against false-positive rate.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Natural Language Processing in Microsoft Azure

1

ROC plots true-positive rate against false-positive rate.

2

Each point represents a different classification threshold.

3

A curve near the upper-left indicates stronger separation.

4

Choose an operating point; AUC alone does not set the threshold.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 033 · EVIDENCE

Microsoft AI Principles

1 **INPUT / SIGNAL**

Fairness checks outcomes across relevant groups.

2 **MECHANISM**

Reliability requires edge-case tests and safe failure.

3 **OUTPUT / EVIDENCE**

Privacy limits data use; transparency explains boundaries.

4 **CONTROL / VERIFY**

Accountable owners approve, monitor and remediate.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 034 · MODEL

Microsoft AI Principles

1 INPUT / SIGNAL

Fairness: AI systems shouldn't make decisions that discriminate against or support bias of a group or individual.

2 MECHANISM

Reliability and Safety: AI systems should respond safely to new situations and potential manipulation.

3 OUTPUT / EVIDENCE

Privacy and Security: AI systems should be secure and respect data privacy.

4 CONTROL / VERIFY

Inclusiveness: AI systems should empower everyone and engage people.

Control: define the threshold or failure that routes to human review.

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Fairness

1 INPUT / SIGNAL

Evaluate comparable groups using the same quality criteria.

2 MECHANISM

Check labels and sampling for representation gaps.

3 OUTPUT / EVIDENCE

Compare error rates across relevant learner-approved segments.

4 CONTROL / VERIFY

Document mitigation, residual risk and an accountable owner.

Control: define the threshold or failure that routes to human review.

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Reliability and Safety

1 INPUT / SIGNAL

Test expected, edge and deliberately adverse inputs.

2 MECHANISM

Set confidence and failure thresholds before deployment.

3 OUTPUT / EVIDENCE

Use monitoring, rollback and human escalation for unsafe states.

4 CONTROL / VERIFY

Record incident evidence without exposing protected data.

Control: define the threshold or failure that routes to human review.

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Privacy and Security

1 INPUT / SIGNAL

Collect only data required for the declared purpose.

2 MECHANISM

Protect keys and endpoints with identity-based access and RBAC.

3 OUTPUT / EVIDENCE

Redact personal data from prompts, logs and screenshots.

4 CONTROL / VERIFY

Define retention, deletion and access-review controls.

Control: define the threshold or failure that routes to human review.

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Inclusiveness

1 INPUT / SIGNAL

Test accessibility, language and interaction assumptions.

2 MECHANISM

Provide alternatives when speech, vision or text interfaces fail.

3 OUTPUT / EVIDENCE

Include affected users in evaluation and feedback.

4 CONTROL / VERIFY

Record who may be excluded and how the design responds.

Control: define the threshold or failure that routes to human review.

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Transparency and Accountability

1 INPUT / SIGNAL

State model limits, provenance and confidence clearly.

2 MECHANISM

Keep decision logs, owners and approval evidence.

3 OUTPUT / EVIDENCE

Explain when automation defers to a human reviewer.

4 CONTROL / VERIFY

Reassess controls after model, prompt or data changes.

Control: define the threshold or failure that routes to human review.

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Conversational AI

1

Conversational AI is the term used to describe solutions where AI agents participate in conversations with humans.

2

Most commonly, conversational AI solutions use bots to manage dialogs with users.

3

These dialogs can take place through web site interfaces, email, social media platforms, messaging systems, phone calls, and other channels.

4

Bots can be the basis of AI solutions for:
Customer support for products or services.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Azure Conversational AI Services

1

Conversational AI in Microsoft Azure.

2

An utterance is mapped to an intent and extracted entities.

3

Confidence and competing intents determine the route.

4

Fallback handles unsupported or ambiguous requests.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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TOPIC 1 · HANDS-ON BRIDGE

Lab 01 · AI Workloads, Responsible AI, and Microsoft Foundry

55 MIN

Contoso Retail needs a defensible service map before its implementation partner provisions any resource.

SCENARIO

Classify AI workloads, map them to Azure services, inspect a Foundry project, and record responsible-AI controls.

MECHANISM

Microsoft Foundry portal; Azure portal; worksheet

EVIDENCE

A workload-to-service decision map, responsible-AI control register, and Foundry project evidence pack.

VERIFY

Six workload rows classified · Six responsible-AI principles addressed

Full step-by-step: Learner Guide + labs/lab-01-ai-workloads-responsible-ai-foundry/README.md

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LEGACY SLIDE 042 · CASE

Conversational AI

1 INPUT / SIGNAL

Activity: Conversational AI.

2 MECHANISM

An utterance is mapped to an intent and extracted entities.

3 OUTPUT / EVIDENCE

Confidence and competing intents determine the route.

4 CONTROL / VERIFY

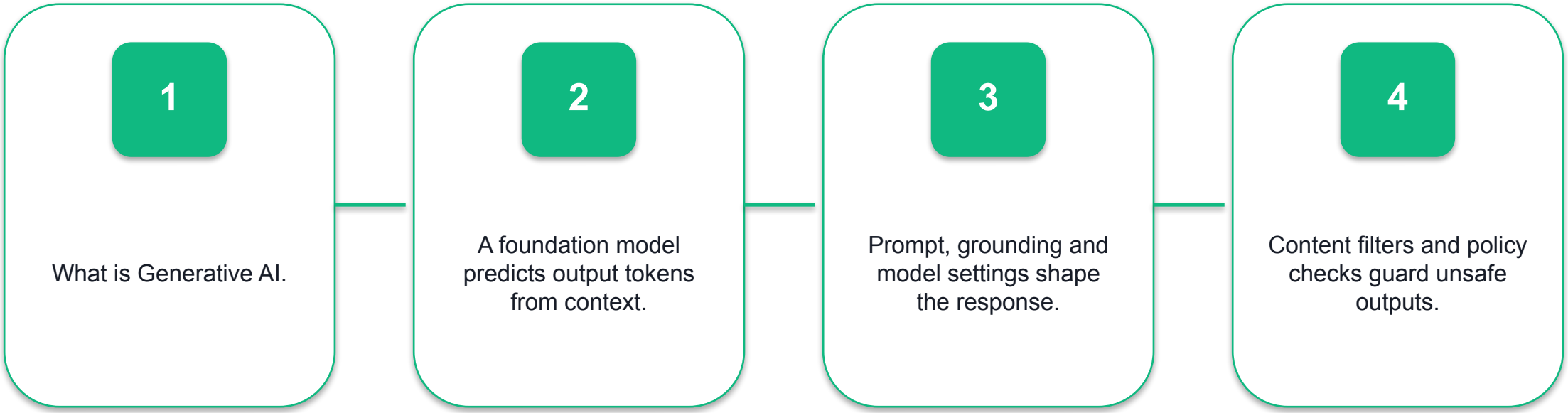
Fallback handles unsupported or ambiguous requests.

Control: define the threshold or failure that routes to human review.

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Foundation Models for Applications



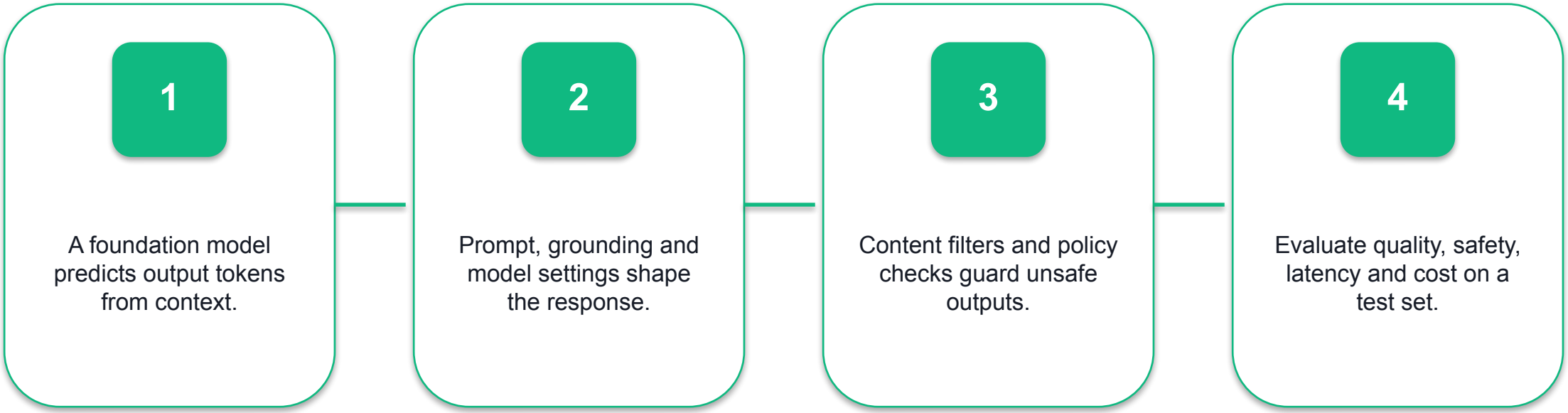
Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 044 · MODEL

Azure OpenAI



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 045 · EVIDENCE

Grounding and Retrieval

1 INPUT / SIGNAL

Ground answers in approved enterprise sources.

2 MECHANISM

Retrieve relevant passages before generating a response.

3 OUTPUT / EVIDENCE

Return citations and refuse when evidence is absent.

4 CONTROL / VERIFY

Measure groundedness, relevance, latency and safety.

Control: define the threshold or failure that routes to human review.

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Prompt, Model and Safety Controls

1 **INPUT / SIGNAL**

Prompt defines the task, context and output contract.

2 **MECHANISM**

Model and deployment determine capability, latency and cost.

3 **OUTPUT / EVIDENCE**

Safety filters and policy checks inspect inputs and outputs.

4 **CONTROL / VERIFY**

Evaluation gates promotion and records regression evidence.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 047 · EVIDENCE

Microsoft Foundry model playground

1

In the Microsoft Foundry model playground, you can build AI models and deploy them for public consumption in software applications.

2

Azure OpenAI's capabilities are made possible by specific generative AI models.

3

These Azure OpenAI models fall into a few main families: GPT-4 current chat model family Codex Embeddings image generation model.

4

Identify the input contract for Microsoft Foundry model playground.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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GitHub Copilot

1 INPUT / SIGNAL

OpenAI partnered with GitHub to create GitHub Copilot, which they call an AI pair programmer.

2 MECHANISM

GitHub Copilot integrates the power of OpenAI Codex into a plugin for developer environments like Visual Studio Code.

3 OUTPUT / EVIDENCE

Repository and editor context condition code suggestions.

4 CONTROL / VERIFY

Comments and function signatures constrain the completion.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 049 · EVIDENCE

Code Completion from Context

1 INPUT / SIGNAL

GitHub Copilot.

2 MECHANISM

Identify the input contract for Code Completion from Context.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

Capture a named output, metric or observable artifact.

Control: define the threshold or failure that routes to human review.

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GitHub Copilot Suggestion Controls

1 INPUT / SIGNAL

GitHub Copilot.

2 MECHANISM

Repository and editor context condition code suggestions.

3 OUTPUT / EVIDENCE

Comments and function signatures constrain the completion.

4 CONTROL / VERIFY

Review correctness, security, licensing and tests.

Control: define the threshold or failure that routes to human review.

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OpenAI's NLP Capabilities

1 INPUT / SIGNAL

A foundation model predicts output tokens from context.

2 MECHANISM

Prompt, grounding and model settings shape the response.

3 OUTPUT / EVIDENCE

Content filters and policy checks guard unsafe outputs.

4 CONTROL / VERIFY

Evaluate quality, safety, latency and cost on a test set.

Control: define the threshold or failure that routes to human review.

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Image Generation with Generative AI

1 INPUT / SIGNAL

image generation model.

2 MECHANISM

A foundation model predicts output tokens from context.

3 OUTPUT / EVIDENCE

Prompt, grounding and model settings shape the response.

4 CONTROL / VERIFY

Content filters and policy checks guard unsafe outputs.

Control: define the threshold or failure that routes to human review.

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TOPIC 1 · HANDS-ON BRIDGE

Lab 08 · Generative AI and Agents in Microsoft Foundry

65 MIN

A policy assistant must answer only from approved material and escalate when evidence is absent.

SCENARIO

Deploy a model, configure instructions and grounding, evaluate responses, and record safety controls.

MECHANISM

Microsoft Foundry; model catalog; chat playground; agent playground

EVIDENCE

A model deployment and agent configuration with grounded tests, refusal evidence and model-card controls.

VERIFY

Model version and deployment recorded · Instructions constrain scope

Full step-by-step: Learner Guide + labs/lab-08-generative-ai-foundry-agent/README.md

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LEGACY SLIDE 053 · EVIDENCE

Knowledge Quiz - Fundamental AI Concepts

1 INPUT / SIGNAL

Match classification, regression and clustering to outputs.

2 MECHANISM

Distinguish vision, language, speech and generative workloads.

3 OUTPUT / EVIDENCE

Choose a Responsible AI principle for a stated risk.

4 CONTROL / VERIFY

Name the input and evidence needed to verify the choice.

Control: define the threshold or failure that routes to human review.

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Knowledge Quiz - Fundamentals of machine learning

1 INPUT / SIGNAL

Identify features, labels, training and validation data.

2 MECHANISM

Choose regression, classification or clustering.

3 OUTPUT / EVIDENCE

Interpret MAE, RMSE, accuracy or a confusion matrix.

4 CONTROL / VERIFY

Explain why held-out validation evidence matters.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Knowledge Quiz - Fundamentals of Azure AI services

1 INPUT / SIGNAL

Map a workload to the current Azure AI capability.

2 MECHANISM

Choose single-service or multi-service resource boundaries.

3 OUTPUT / EVIDENCE

Name the endpoint, authentication and output contract.

4 CONTROL / VERIFY

Apply RBAC, monitoring and cost-cleanup controls.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

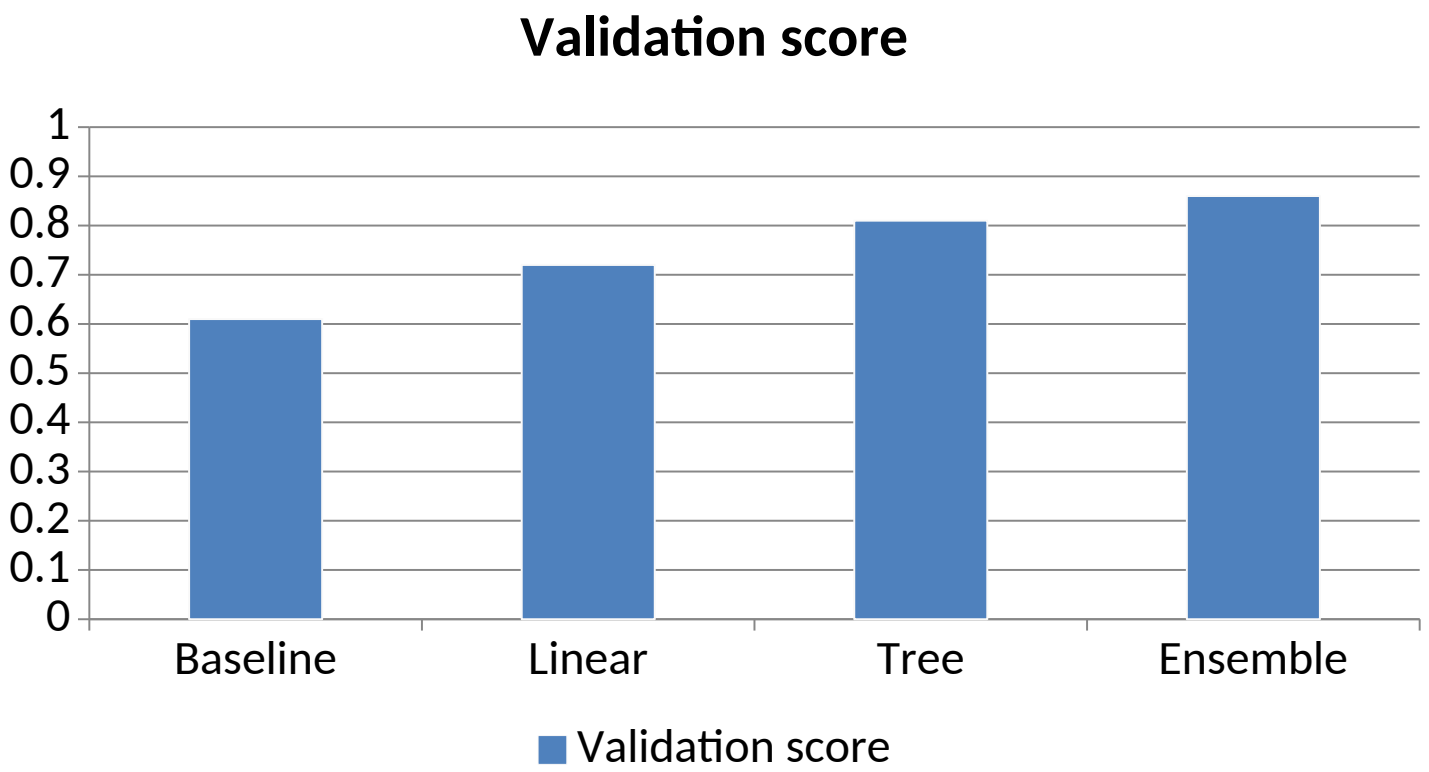
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LEGACY COVERAGE · SLIDE 056

Topic 2 Explore Visual Tools For Machine Learning

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What is Machine Learning



Machine learning is a technique that uses mathematics and statistics to create a model that can predict unknown values.

Identify the input contract for What is Machine Learning.

Trace the processing component and its state change.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Types of Machine Learning

1 INPUT / SIGNAL

There are two general approaches to machine learning: supervised and unsupervised machine learning.

2 MECHANISM

In both approaches, you train a model to make predictions.

3 OUTPUT / EVIDENCE

The supervised machine learning approach requires you to start with a dataset with known label values.

4 CONTROL / VERIFY

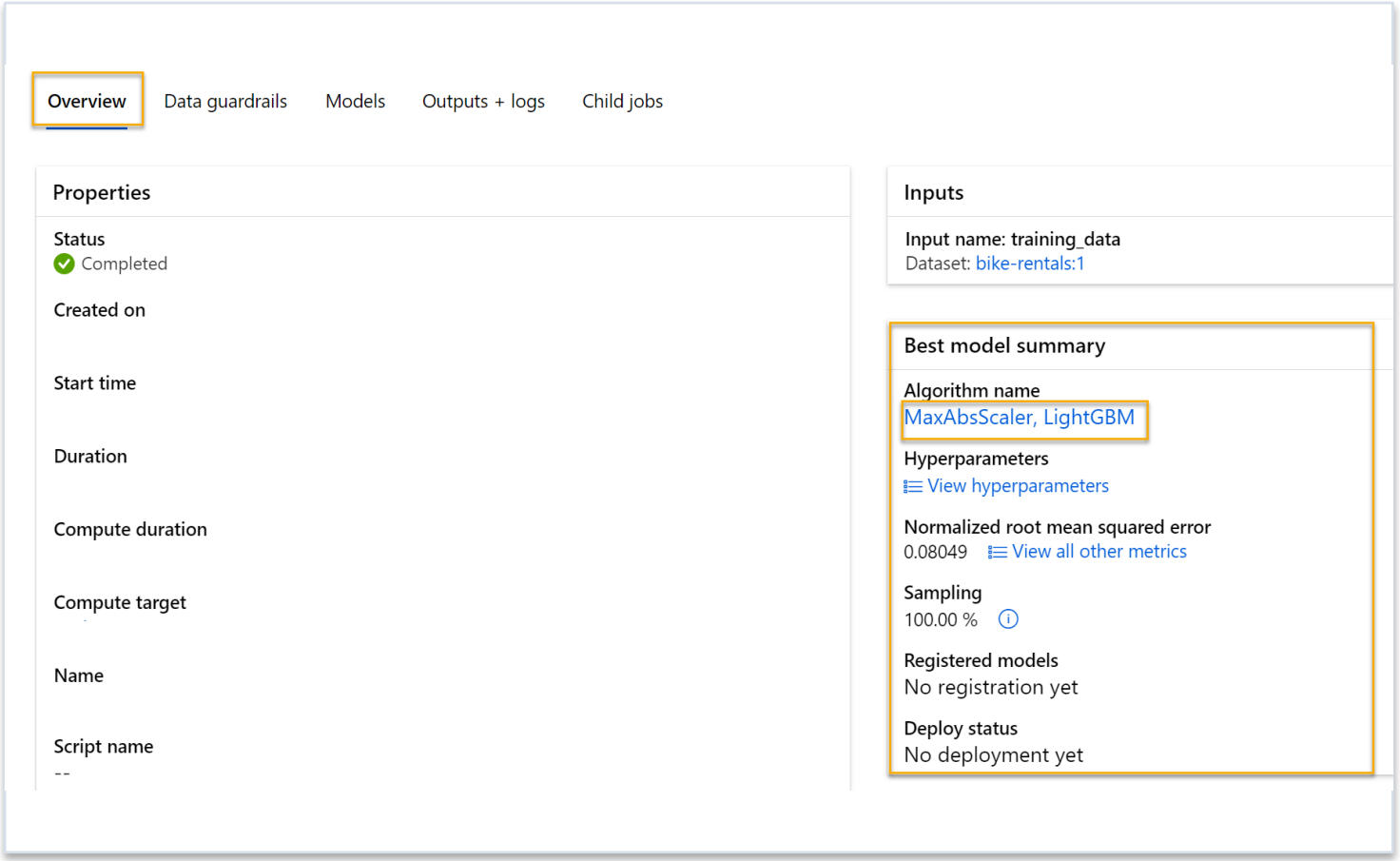
Two types of supervised machine learning tasks include regression and classification.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Azure Machine Learning Studio



1. Azure Machine Learning Studio is the web portal for data scientist developers in Azure Machine Learning.

2. The studio combines no-code and code-first experiences for an inclusive data science platform.

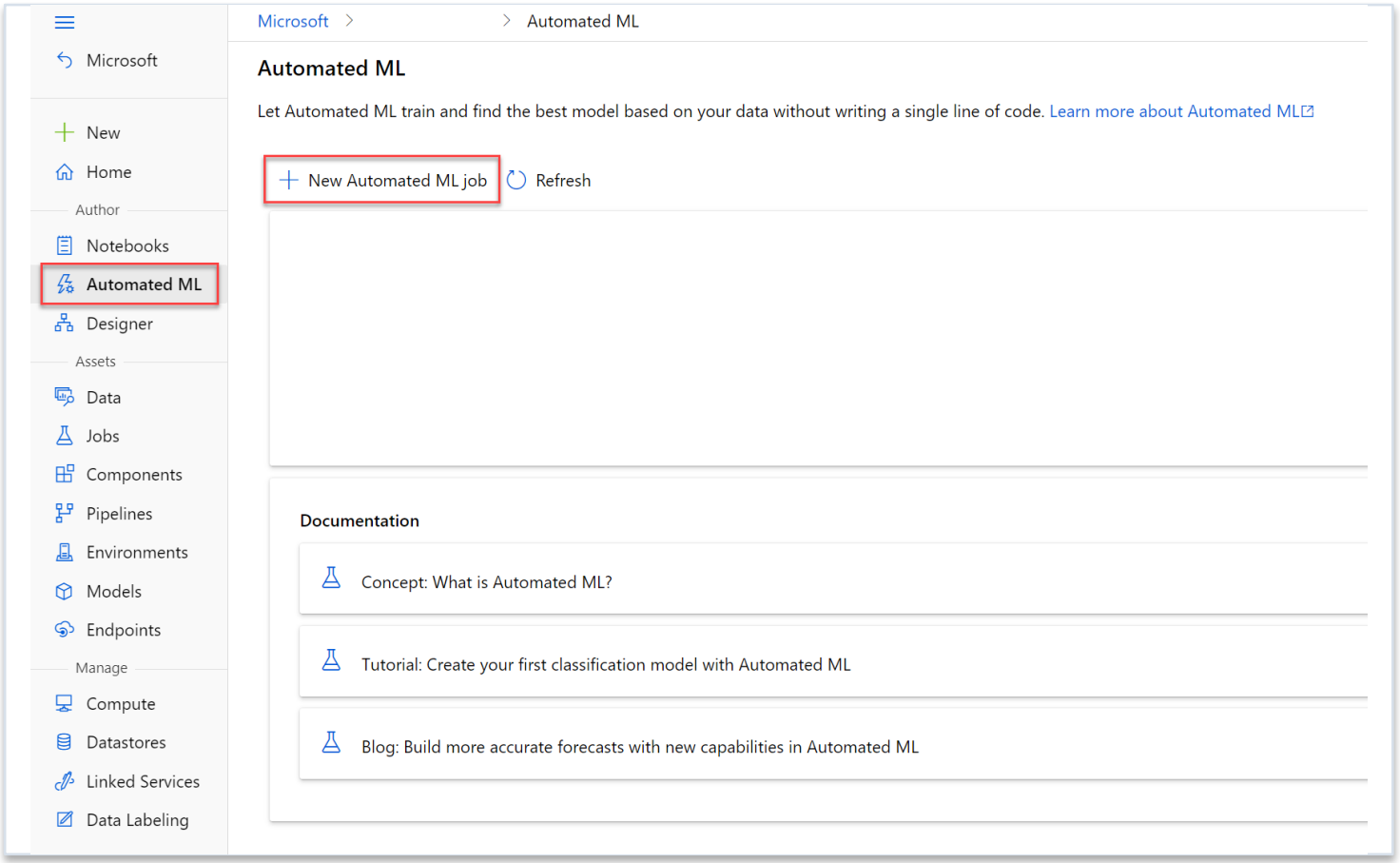
3. The studio offers multiple authoring experiences depending on the type project and the level of user experience.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Azure Machine Learning Studio



1 Identify the input contract for Azure Machine Learning Studio.

2 Trace the processing component and its state change.

3 Capture a named output, metric or observable artifact.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Azure Machine Learning Compute

1

In Azure Machine Learning studio, you can manage the compute targets for your data science activities.

2

Compute Clusters: Scalable clusters of virtual machines for on-demand processing of experiment code.

3

Inference Clusters: Deployment targets for predictive services that use your trained models.

4

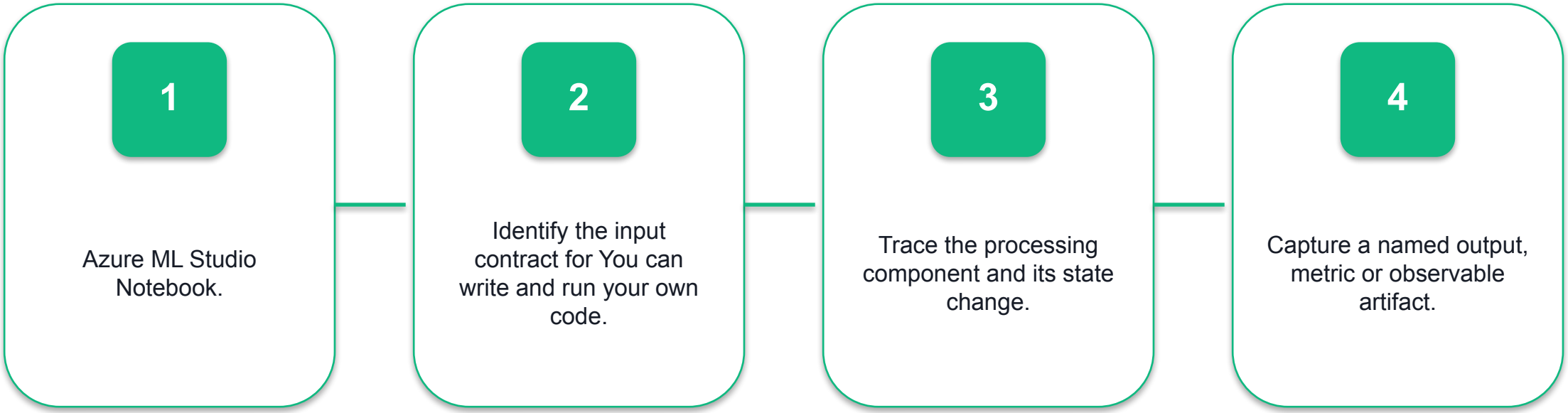
Attached Compute: Links to existing Azure compute resources, such as Virtual Machines or Azure Databricks clusters.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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You can write and run your own code



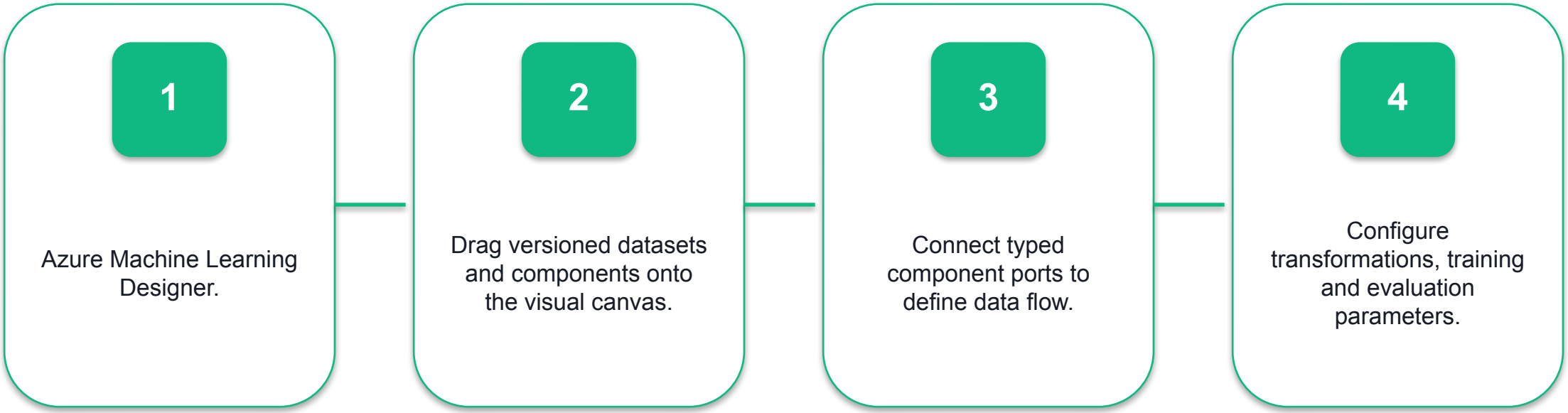
Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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LEGACY SLIDE 063 · EVIDENCE

Train Models with Azure Machine Learning Designer

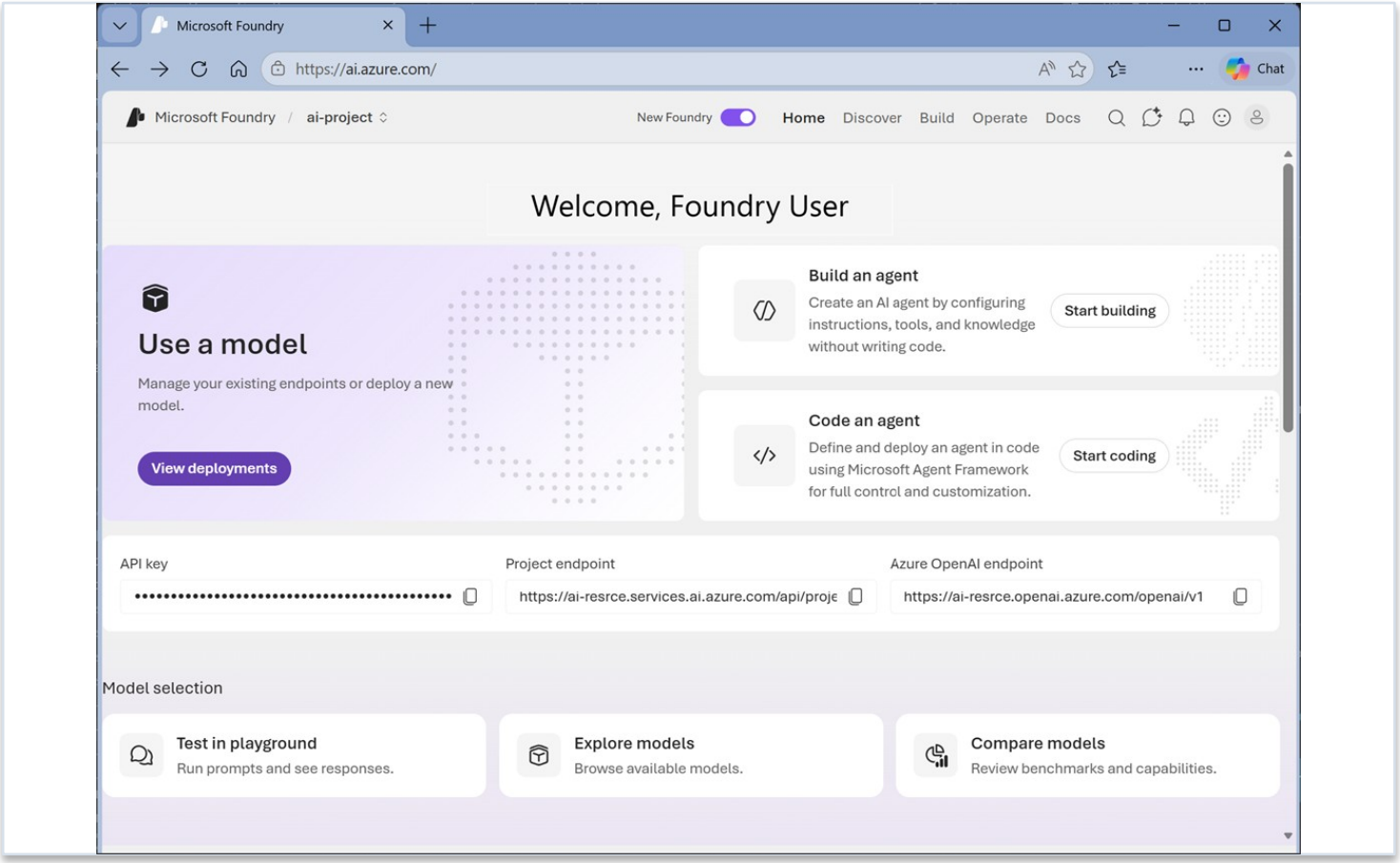


Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Azure Machine Learning Designer



1. Drag versioned datasets and components onto the visual canvas.
2. Connect typed component ports to define data flow.
3. Configure transformations, training and evaluation parameters.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Pipelines

1

Pipelines let you organize, manage, and reuse complex machine learning workflows across projects and users.

2

A pipeline starts with the dataset from which you want to train the model.

3

Each time you run a pipeline, the configuration of the pipeline and its results are stored in your workspace as a pipeline job.

4

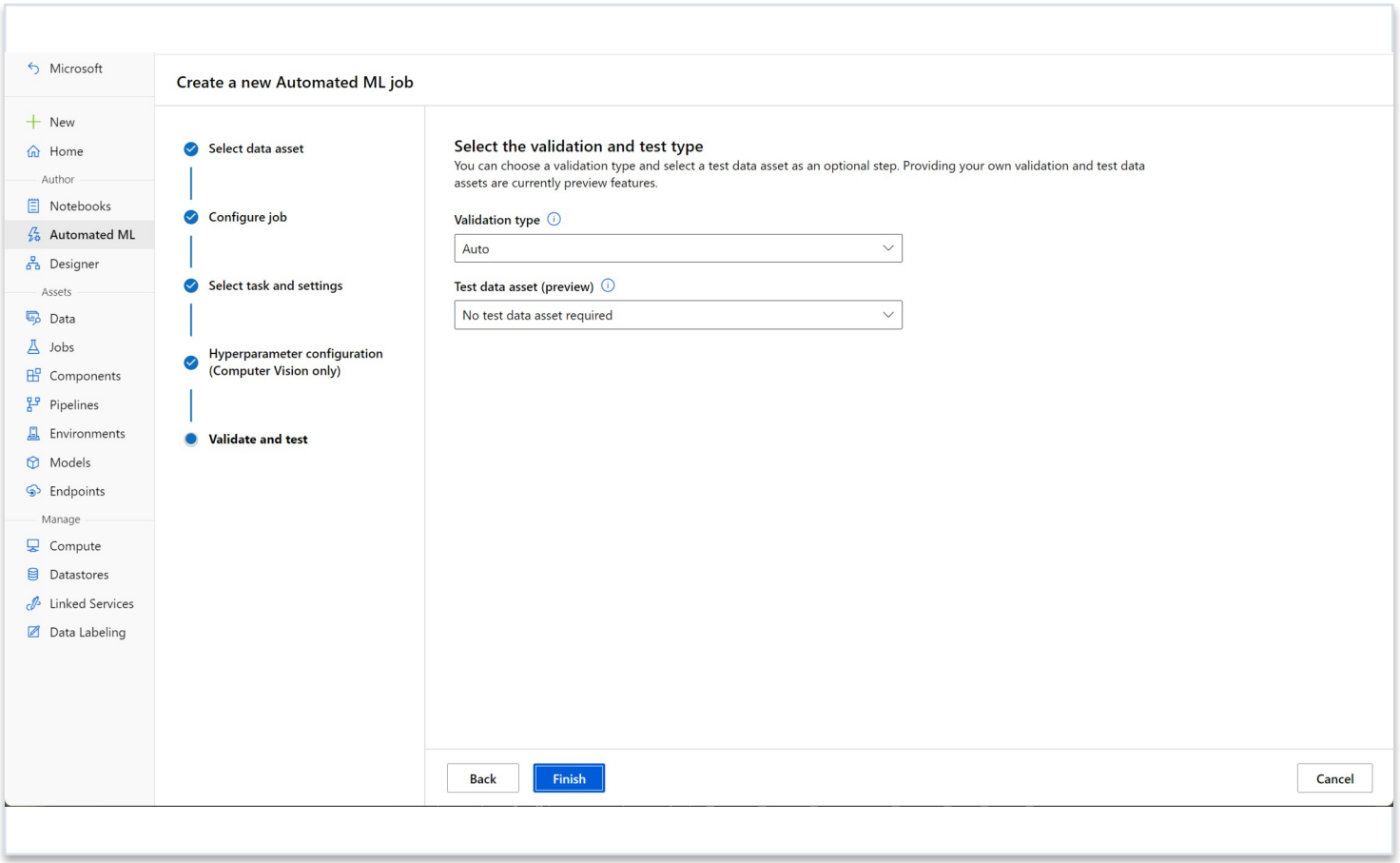
A pipeline connects versioned inputs to transformations and outputs.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Pipelines



1. A pipeline connects versioned inputs to transformations and outputs.
2. Each component exposes a typed boundary and run state.
3. Run history provides repeatability and failure evidence.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Components

1

An Azure Machine Learning component encapsulates one step in a machine learning pipeline.

2

You can think of a component as a programming function and as a building block for Azure Machine Learning pipelines.

3

In a pipeline project, you can access data assets and components from the left panel's Asset Library tab.

4

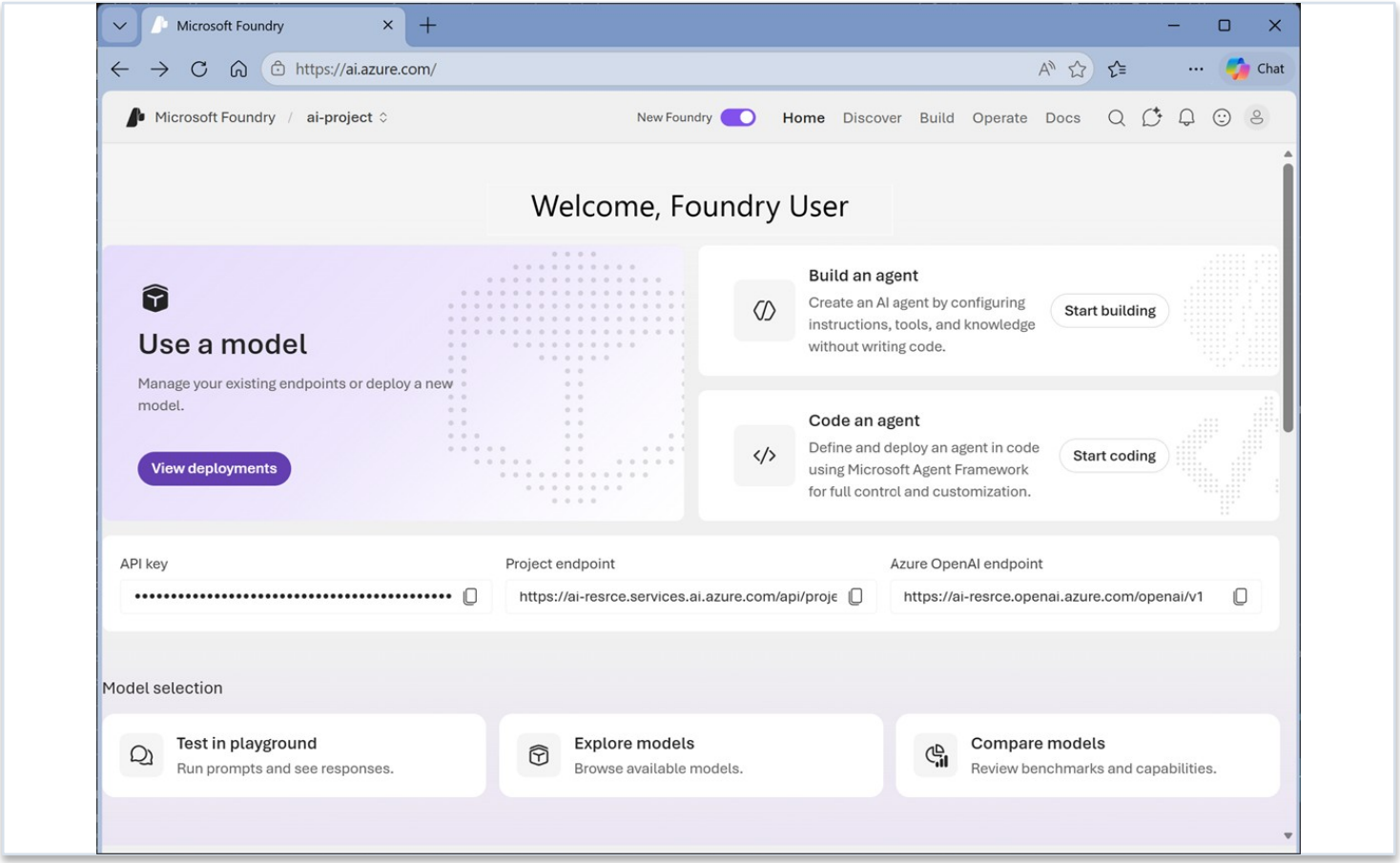
Identify the input contract for Components.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Components



1. Identify the input contract for Components.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.

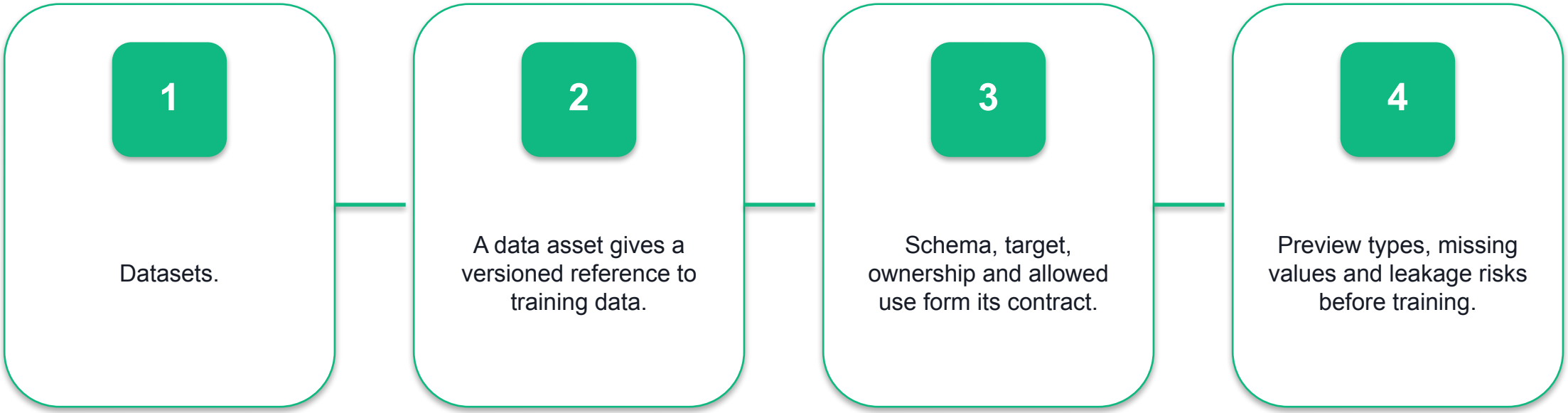
UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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LEGACY SLIDE 069 · MODEL

Create a Machine Learning Data Asset

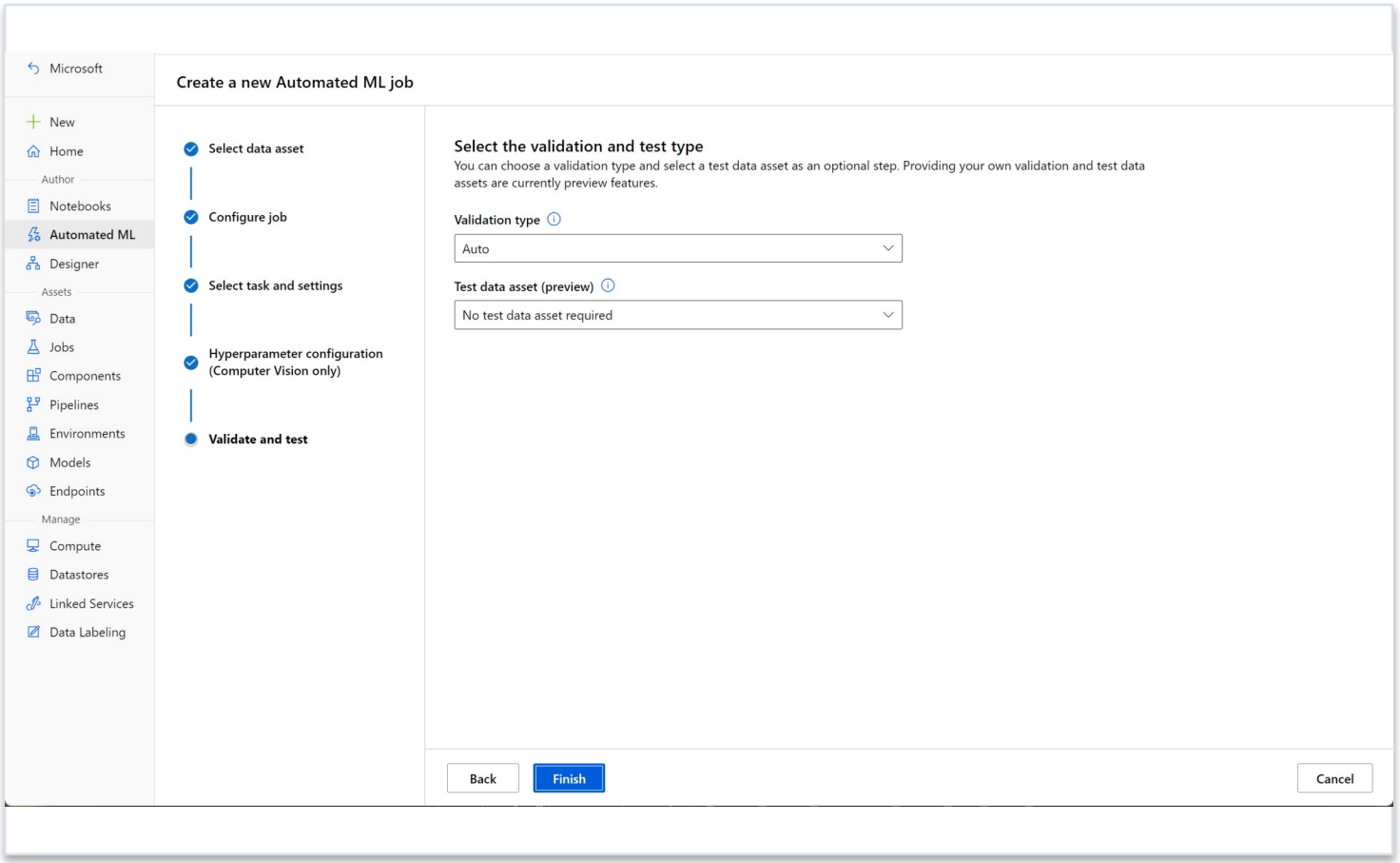


Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Track Experiments with Machine Learning Jobs



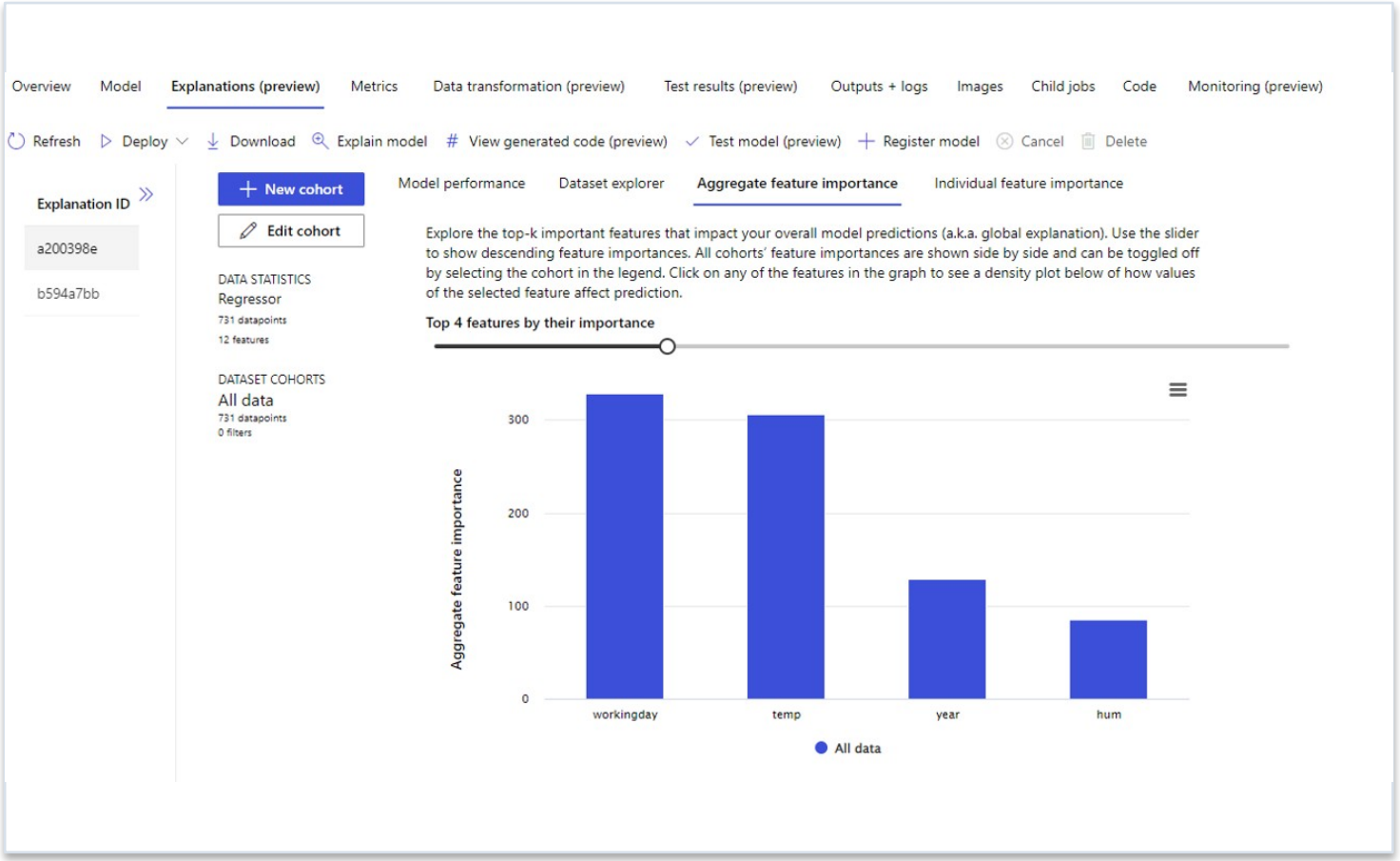
1. Azure Machine Learning Jobs.
2. A job binds code or pipeline, inputs, environment and compute.
3. The run record captures parameters, logs, metrics and outputs.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Azure Machine Learning Jobs



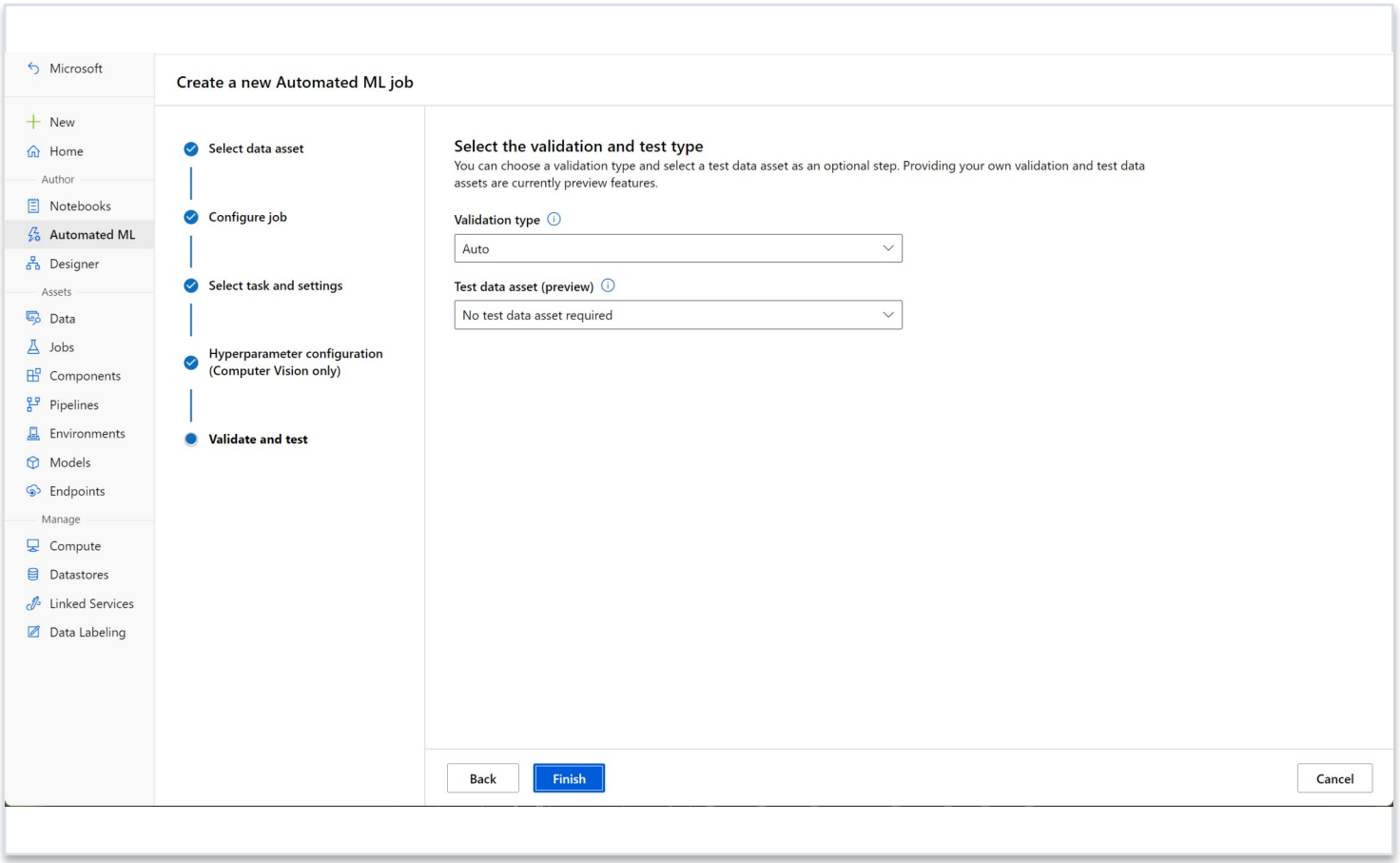
1. A job binds code or pipeline, inputs, environment and compute.
2. The run record captures parameters, logs, metrics and outputs.
3. Version data and environments for repeatability.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Azure Machine Learning Jobs



1. A job binds code or pipeline, inputs, environment and compute.
2. The run record captures parameters, logs, metrics and outputs.
3. Version data and environments for repeatability.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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TOPIC 1 · HANDS-ON BRIDGE

Lab 02 · Machine Learning Fundamentals and Metrics

55 MIN

A bike-share team must choose the correct technique and reject a misleading model before deployment.

SCENARIO

Differentiate regression, classification and clustering and compute evidence-based model metrics.

MECHANISM

Spreadsheet or Python; samples/model-results.csv

EVIDENCE

A technique-selection table and verified MAE, RMSE, precision, recall and confusion-matrix calculations.

VERIFY

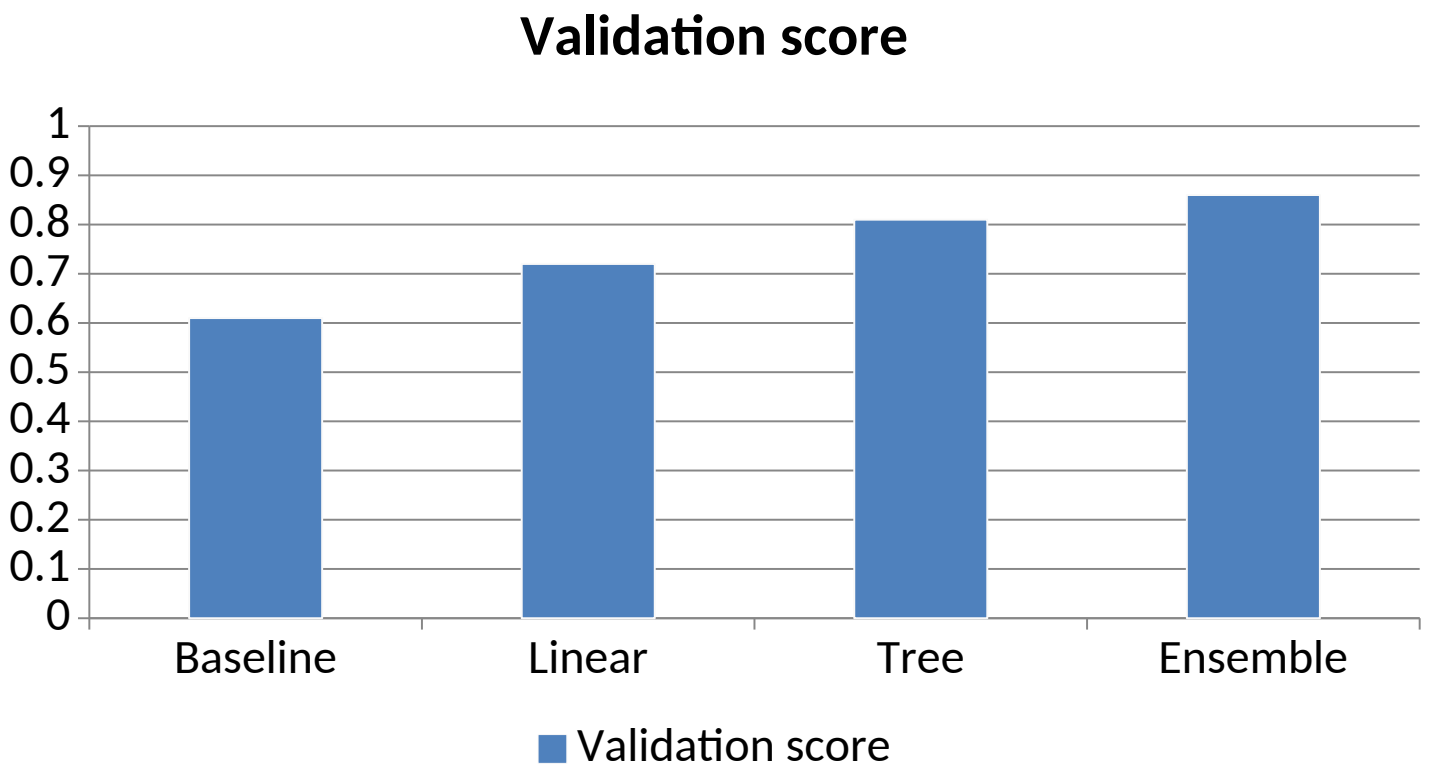
Every scenario has a justified technique · MAE and RMSE are reproducible

Full step-by-step: Learner Guide + labs/lab-02-machine-learning-fundamentals-metrics/README.md

Lab bridge only—no click-by-click procedure on the slide.

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Regression



Regression is a form of machine learning that is used to predict a numeric label based on an item's features.

Features map to a numeric target such as price or demand.

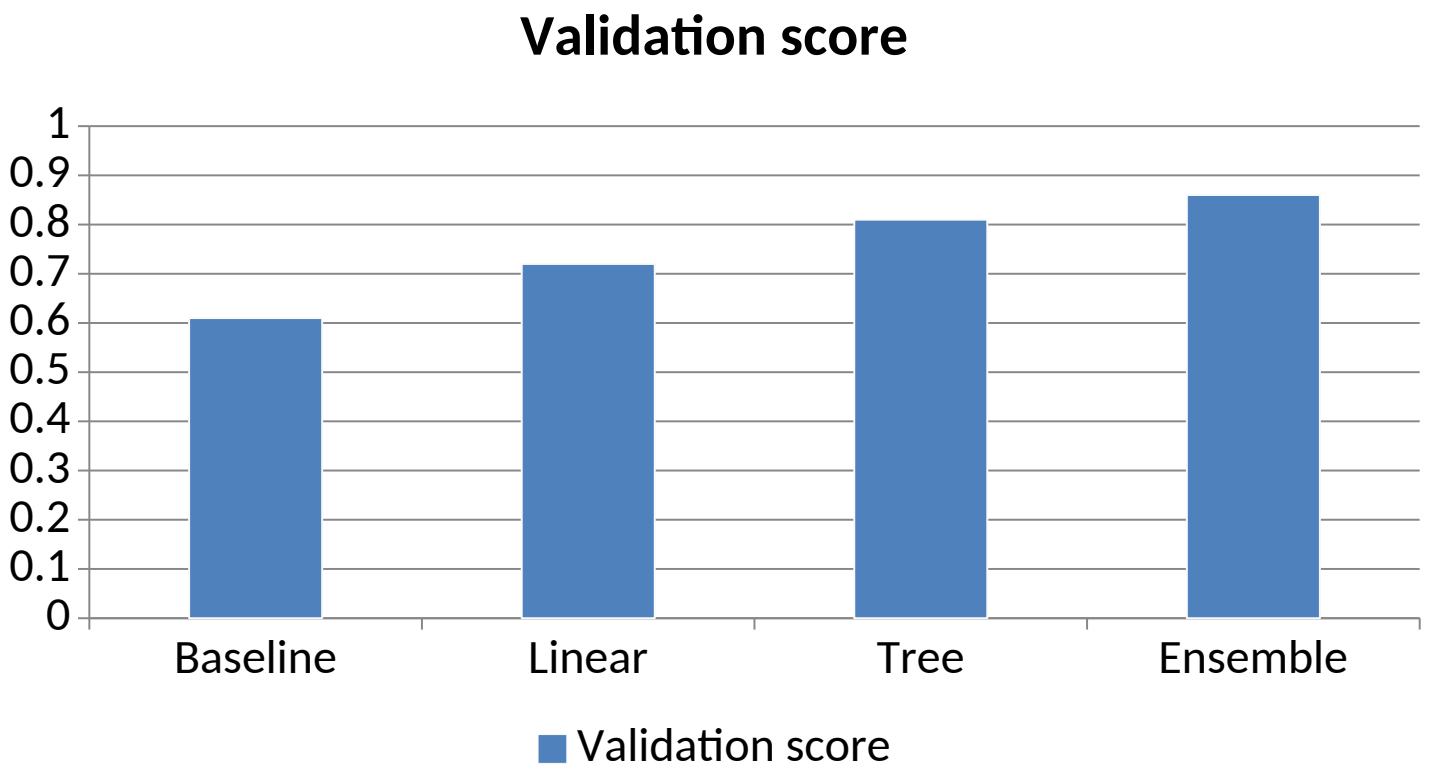
Training fits a function; validation tests unseen examples.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Regression Example



Prepare data: Identify the features and label in a dataset.

Pre-process, or clean and transform, the data as needed.

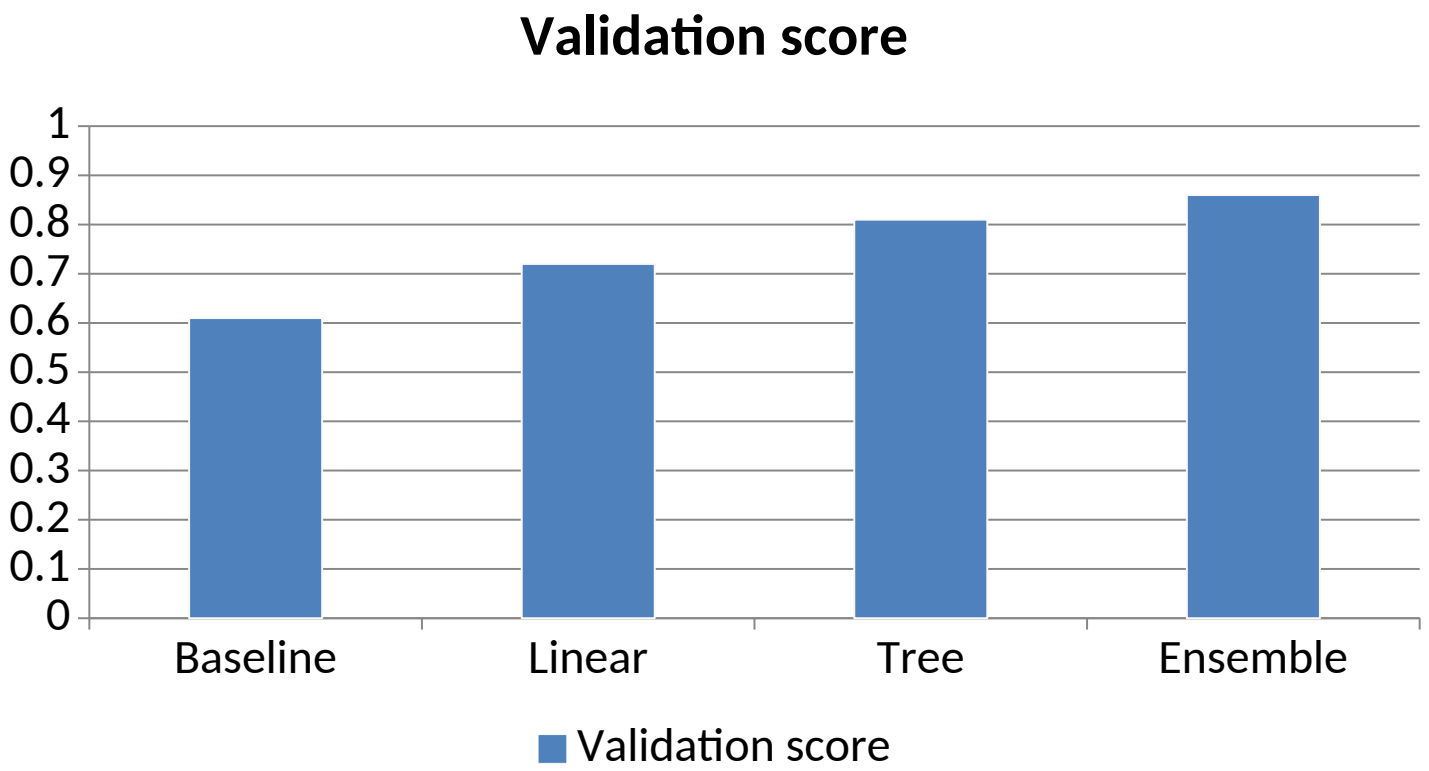
Train model: Split the data into two groups, a training and a validation set.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Regression Example



Features map to a numeric target such as price or demand.

Training fits a function; validation tests unseen examples.

MAE shows average absolute error; RMSE penalizes large errors.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Inference Pipeline

1

A pipeline connects versioned inputs to transformations and outputs.

2

Each component exposes a typed boundary and run state.

3

Run history provides repeatability and failure evidence.

4

Deploy only a validated inference path.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Automatic Price Prediction Regression

1 INPUT / SIGNAL

Features map to a numeric target such as price or demand.

2 MECHANISM

Training fits a function; validation tests unseen examples.

3 OUTPUT / EVIDENCE

MAE shows average absolute error; RMSE penalizes large errors.

4 CONTROL / VERIFY

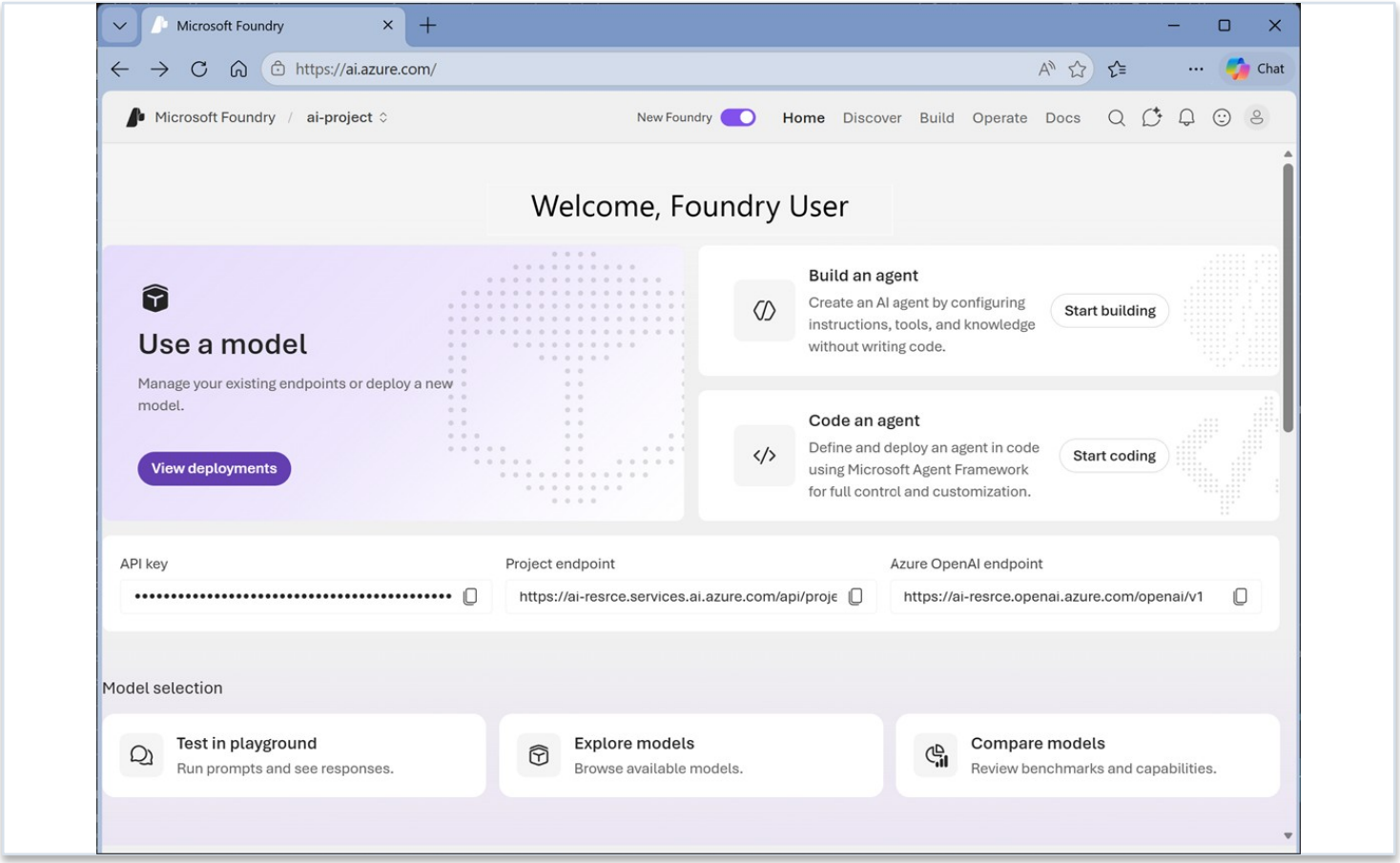
Accept only when the metric and residual pattern meet the target.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Add the Automobile Dataset



1. Add and Explore a Dataset.
2. Identify the input contract for Add the Automobile Dataset.
3. Trace the processing component and its state change.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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LEGACY SLIDE 079 · MODEL

Transform the Automobile Dataset

1. Add Data Transformations.
2. Identify the input contract for Transform the Automobile Dataset.
3. Trace the processing component and its state change.
4. Capture a named output, metric or observable artifact.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Configure the Select Columns Module

1 INPUT / SIGNAL

Select the Select Column.

2 MECHANISM

Identify the input contract for Configure the Select Columns Module.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

Capture a named output, metric or observable artifact.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 081 · MODEL

Clean Missing Data

1

Identify the input contract for Clean Missing Data.

2

Trace the processing component and its state change.

3

Capture a named output, metric or observable artifact.

4

Define the acceptance threshold and human-review route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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LEGACY SLIDE 082 · MODEL

Normalize Data

1

Identify the input contract for Normalize Data.

2

Trace the processing component and its state change.

3

Capture a named output, metric or observable artifact.

4

Define the acceptance threshold and human-review route.

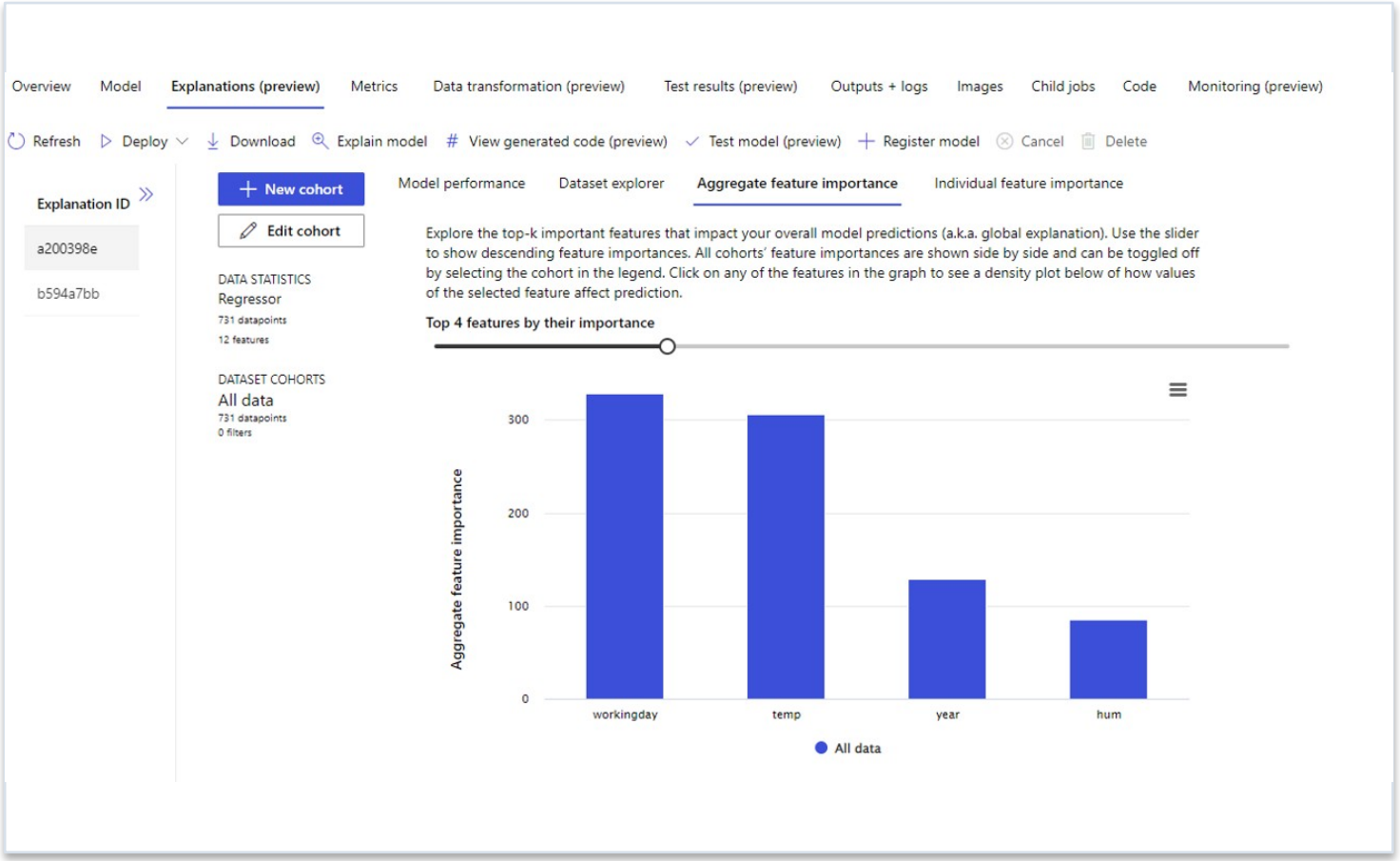
Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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LEGACY SLIDE 083 · AUTHENTIC UI

Run the Pipeline



1. A pipeline connects versioned inputs to transformations and outputs.
2. Each component exposes a typed boundary and run state.
3. Run history provides repeatability and failure evidence.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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LEGACY SLIDE 084 · EVIDENCE

Add Training Modules

1

Identify the input contract for Add Training Modules.

2

Trace the processing component and its state change.

3

Capture a named output, metric or observable artifact.

4

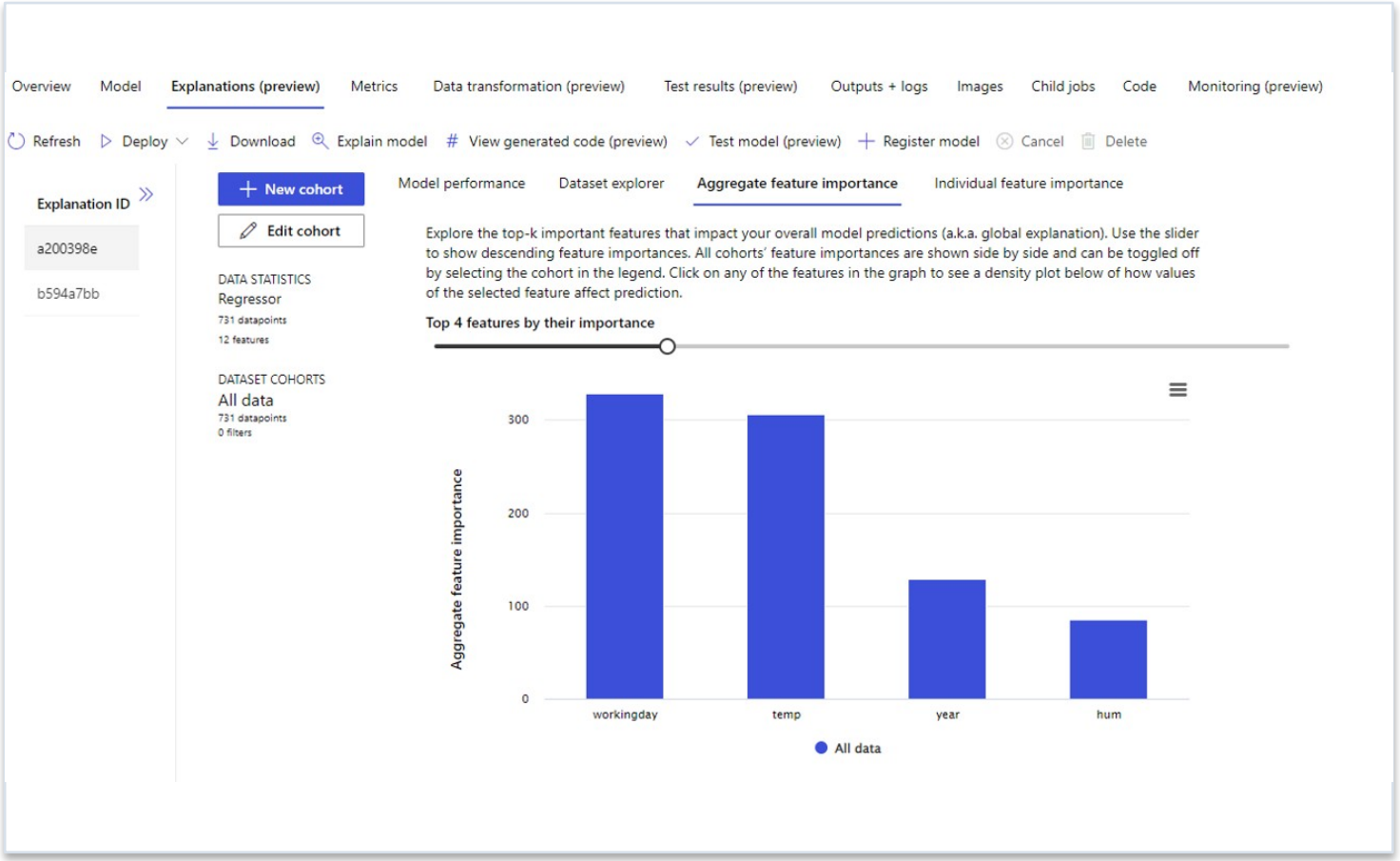
Define the acceptance threshold and human-review route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Run the Training Pipeline



1. A pipeline connects versioned inputs to transformations and outputs.
2. Each component exposes a typed boundary and run state.
3. Run history provides repeatability and failure evidence.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Evaluate a Regression Model

1 INPUT / SIGNAL

Features map to a numeric target such as price or demand.

2 MECHANISM

Training fits a function; validation tests unseen examples.

3 OUTPUT / EVIDENCE

MAE shows average absolute error; RMSE penalizes large errors.

4 CONTROL / VERIFY

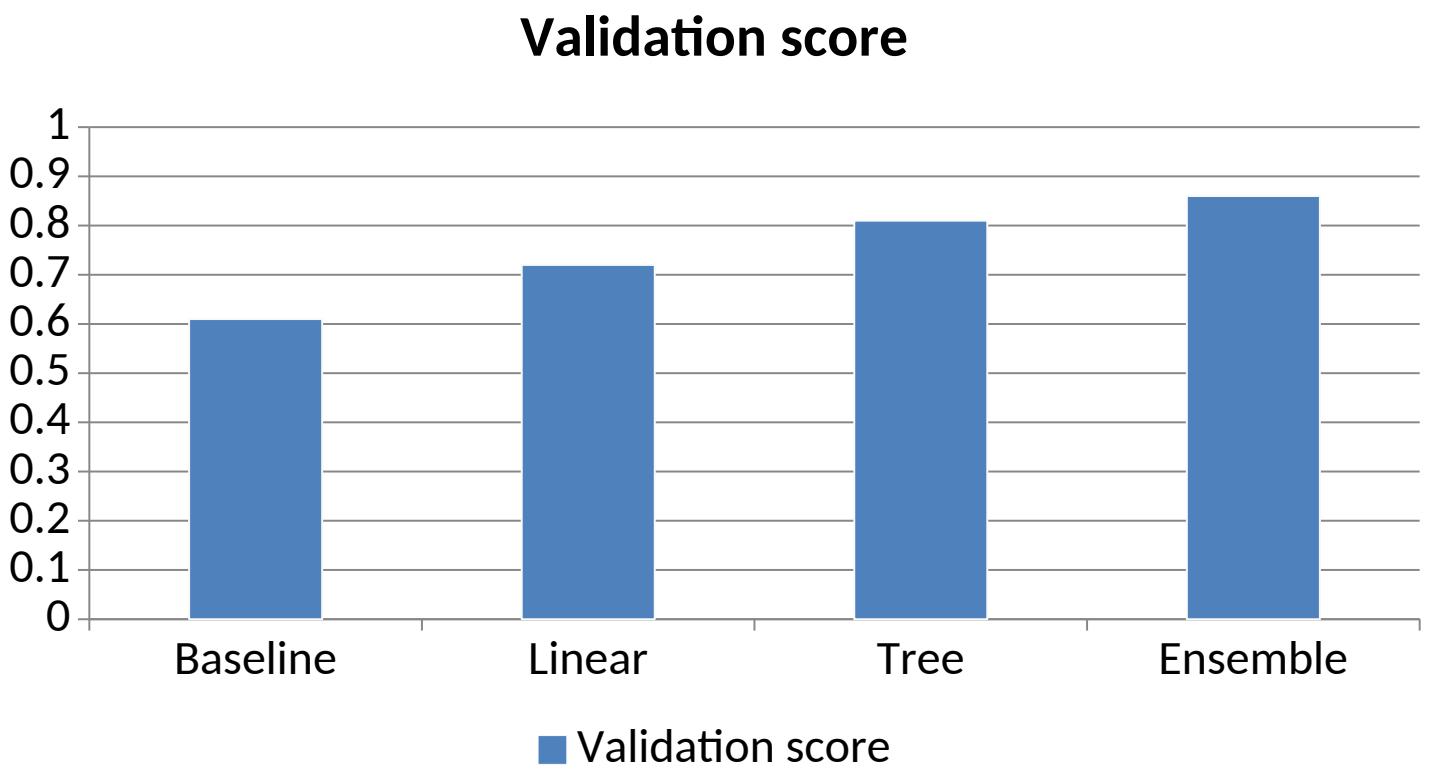
Accept only when the metric and residual pattern meet the target.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Model Performance Metrics



Mean Absolute Error (MAE): The average difference between predicted values and true values.

Root Mean Squared Error (RMSE): The square root of the mean squared difference between predicted and true values.

The closer to 1 this value is, the better the model is performing.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Create an Inference Pipeline

1

A pipeline connects versioned inputs to transformations and outputs.

2

Each component exposes a typed boundary and run state.

3

Run history provides repeatability and failure evidence.

4

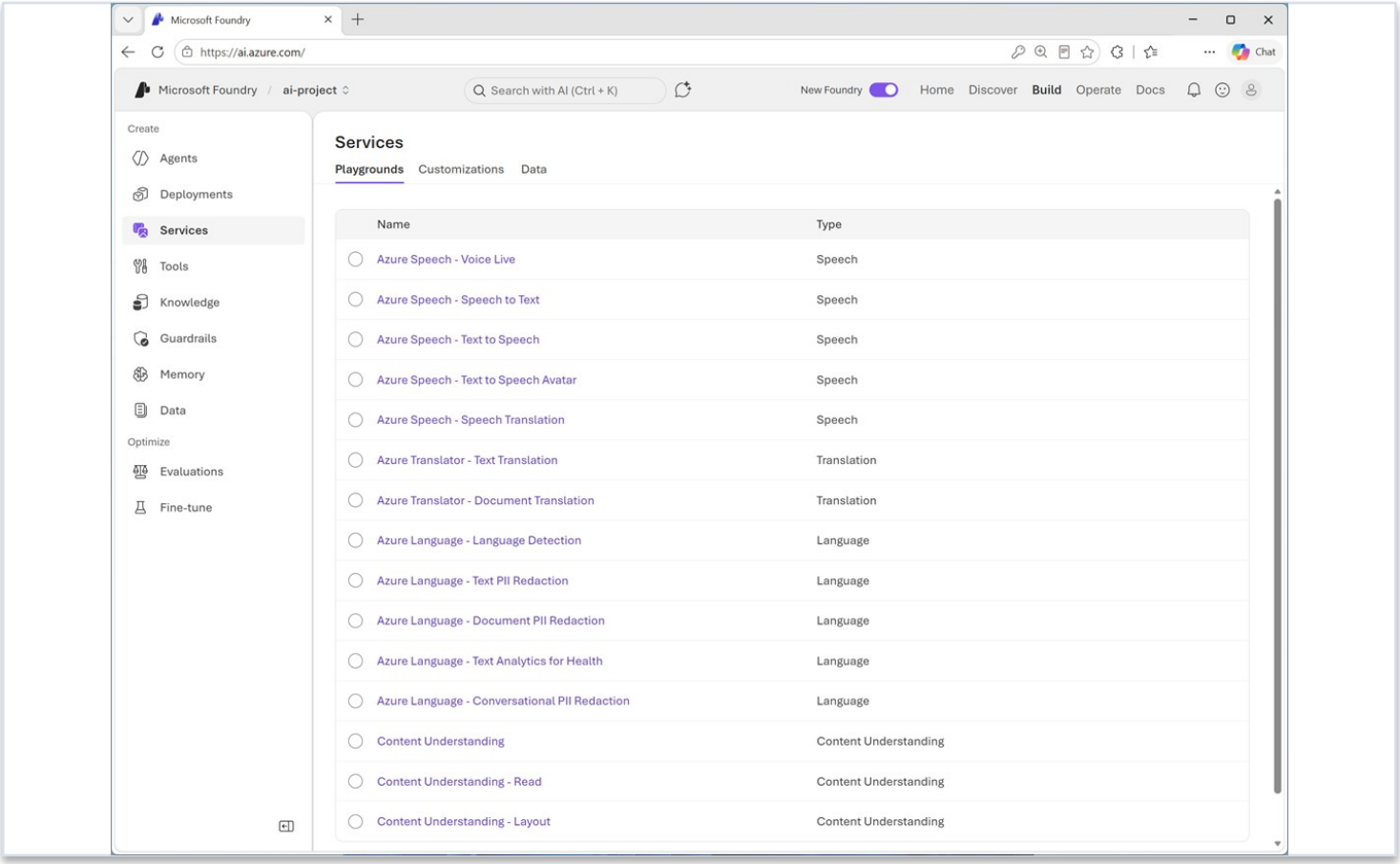
Deploy only a validated inference path.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Deployment



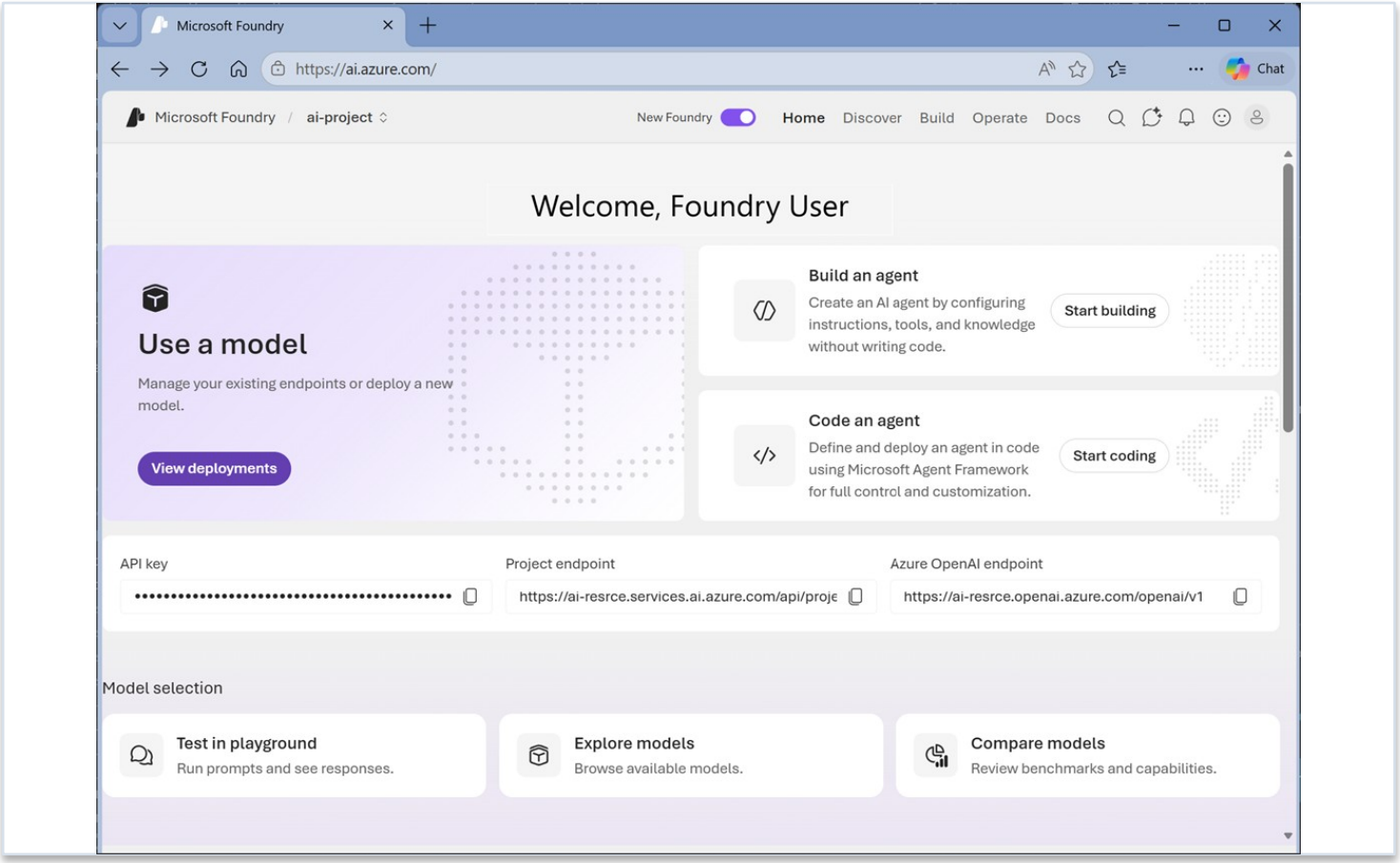
1. A deployment binds a model, environment, code and compute.
2. The endpoint exposes a versioned request and response contract.
3. Authentication and traffic rules control access.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Testing



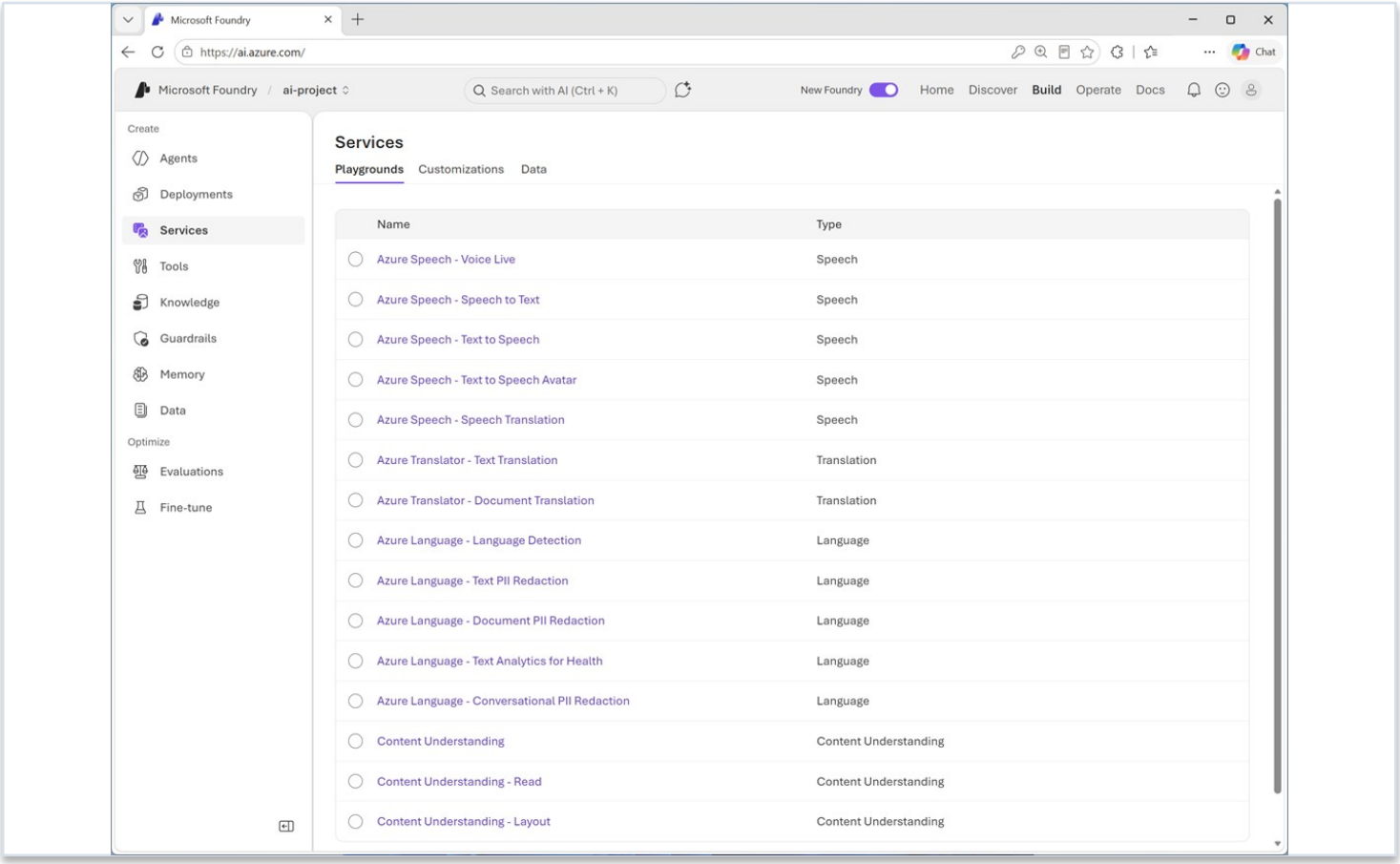
1. Identify the input contract for Testing.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Testing



1. Identify the input contract for Testing.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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TOPIC 1 · HANDS-ON BRIDGE

Lab 03 · Azure Machine Learning AutoML and Model Lifecycle

70 MIN

Contoso Mobility needs a repeatable demand-forecast experiment that its service provider can review.

SCENARIO

Configure an AutoML regression experiment, interpret ranking evidence, and document deployment controls.

MECHANISM

Azure Machine Learning studio; Automated ML; CSV data asset

EVIDENCE

An AutoML experiment record with data, task, target, compute, metric, best-model evidence and deployment decision.

VERIFY

Versioned data asset recorded · Regression target and metric justified

Full step-by-step: Learner Guide + labs/lab-03-azure-machine-learning-automl-lifecycle/README.md

Lab bridge only—no click-by-click procedure on the slide.

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Activity: Regression

1 INPUT / SIGNAL

Features map to a numeric target such as price or demand.

2 MECHANISM

Training fits a function; validation tests unseen examples.

3 OUTPUT / EVIDENCE

MAE shows average absolute error; RMSE penalizes large errors.

4 CONTROL / VERIFY

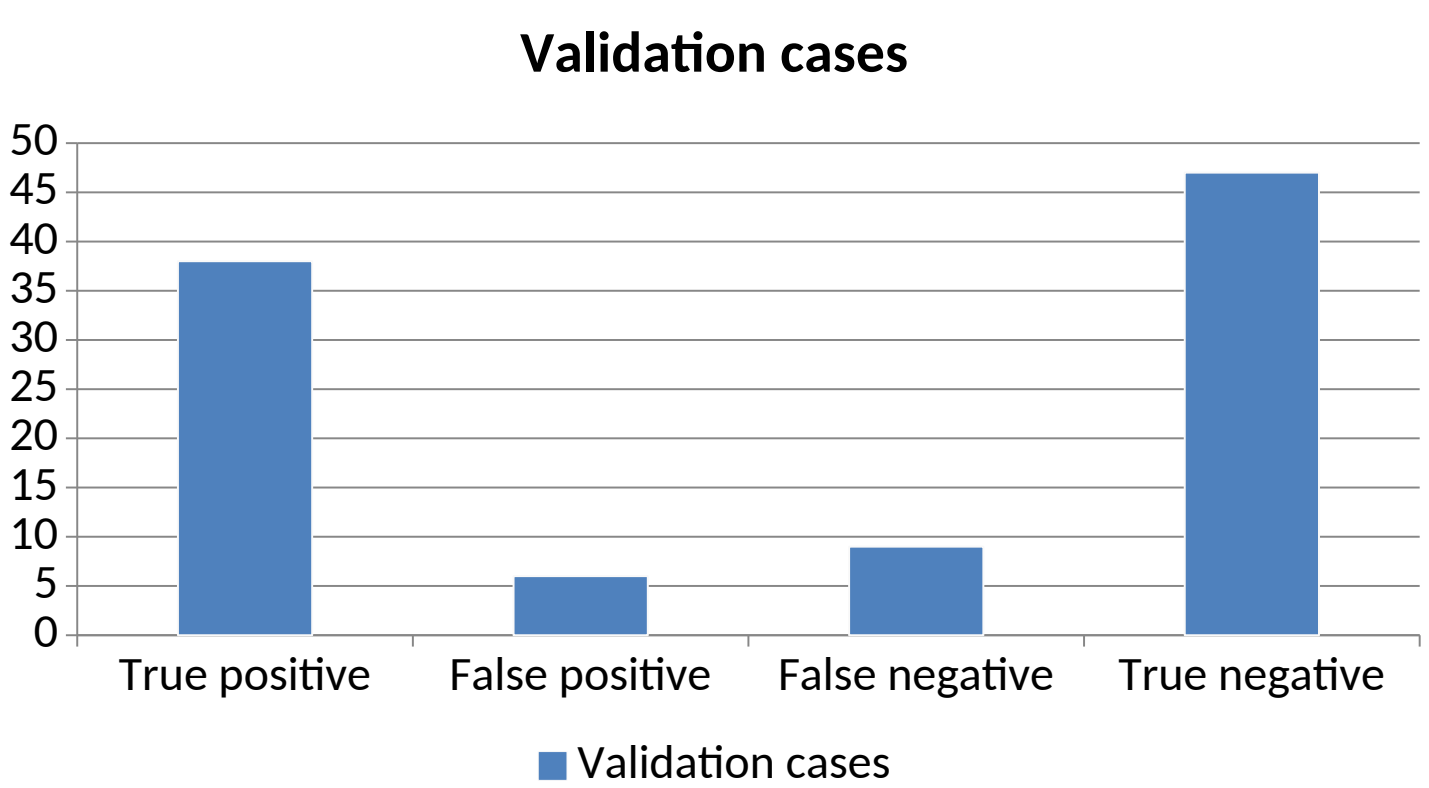
Accept only when the metric and residual pattern meet the target.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Classification Fundamentals



Classification.

Features map to one of two or more categorical labels.

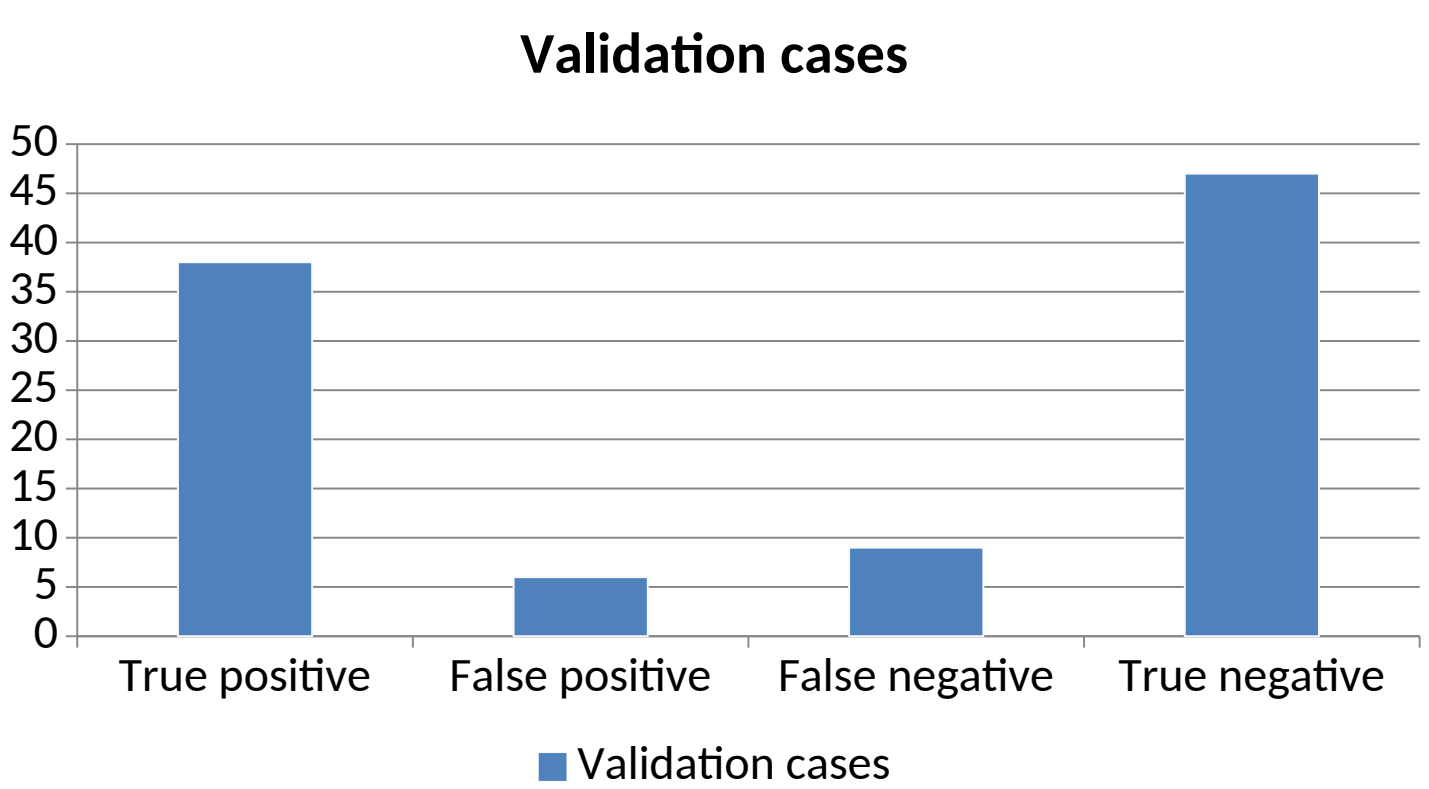
The model emits class scores or probabilities.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Classification Example



Features map to one of two or more categorical labels.

The model emits class scores or probabilities.

A threshold converts scores into predicted labels.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Prepare Classification Data

1. Add Data Transformations.
2. Features map to one of two or more categorical labels.
3. The model emits class scores or probabilities.
4. A threshold converts scores into predicted labels.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Configure Normalization

1 INPUT / SIGNAL

Normalize Data.

2 MECHANISM

Identify the input contract for Configure Normalization.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

Capture a named output, metric or observable artifact.

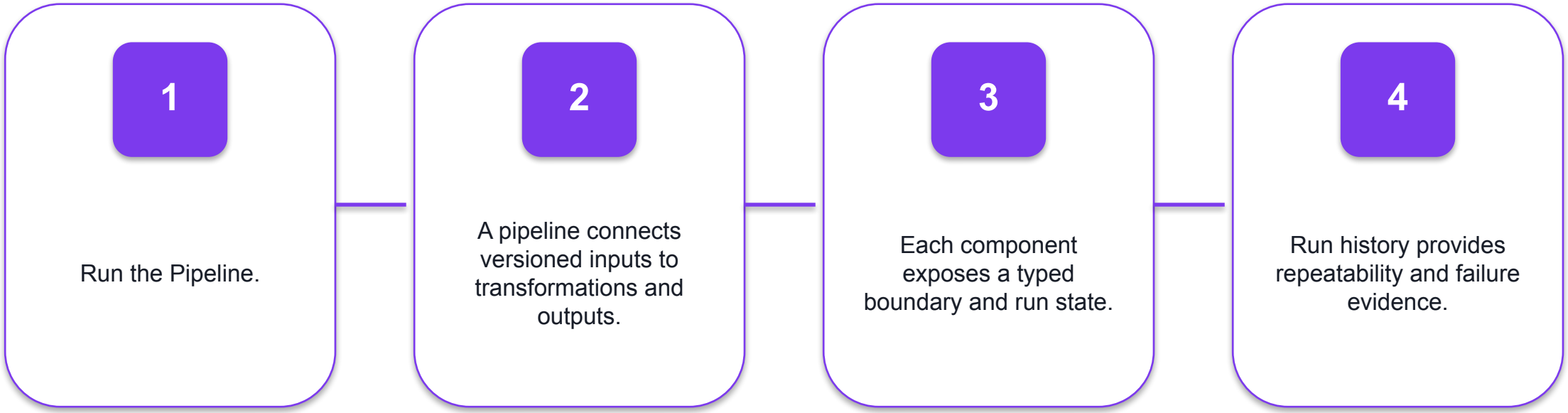
Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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LEGACY SLIDE 097 · UI

Run the Data Transformation Pipeline



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Add Training Modules

1 INPUT / SIGNAL

Identify the input contract for Add Training Modules.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Run the Training Pipeline

1 INPUT / SIGNAL

A pipeline connects versioned inputs to transformations and outputs.

2 MECHANISM

Each component exposes a typed boundary and run state.

3 OUTPUT / EVIDENCE

Run history provides repeatability and failure evidence.

4 CONTROL / VERIFY

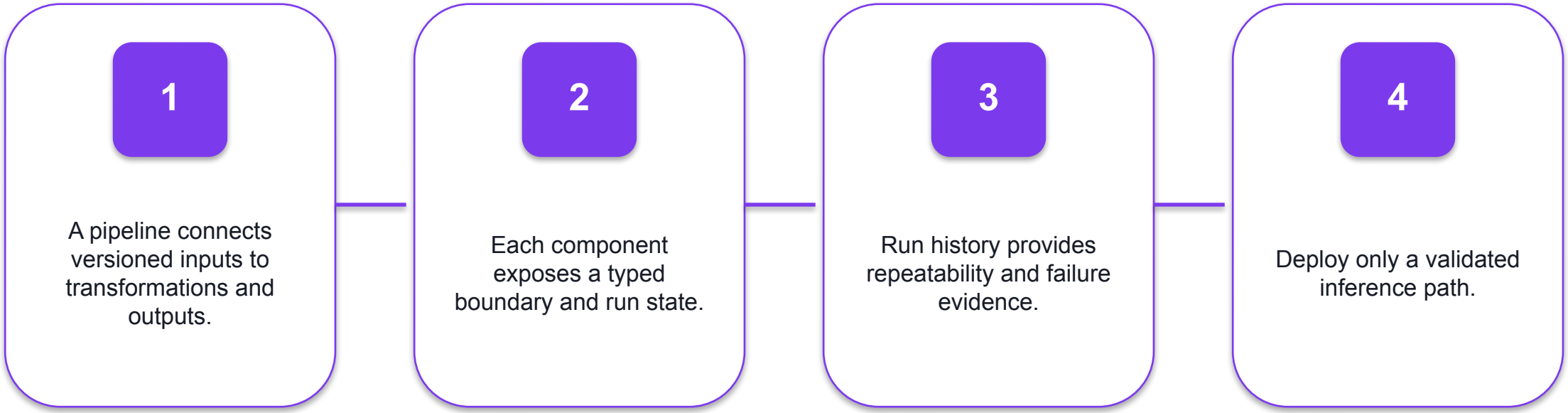
Deploy only a validated inference path.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Create an Inference Pipeline



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Open Evaluation Outputs

1 INPUT / SIGNAL

Confusion Matrix.

2 MECHANISM

Identify the input contract for Open Evaluation Outputs.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

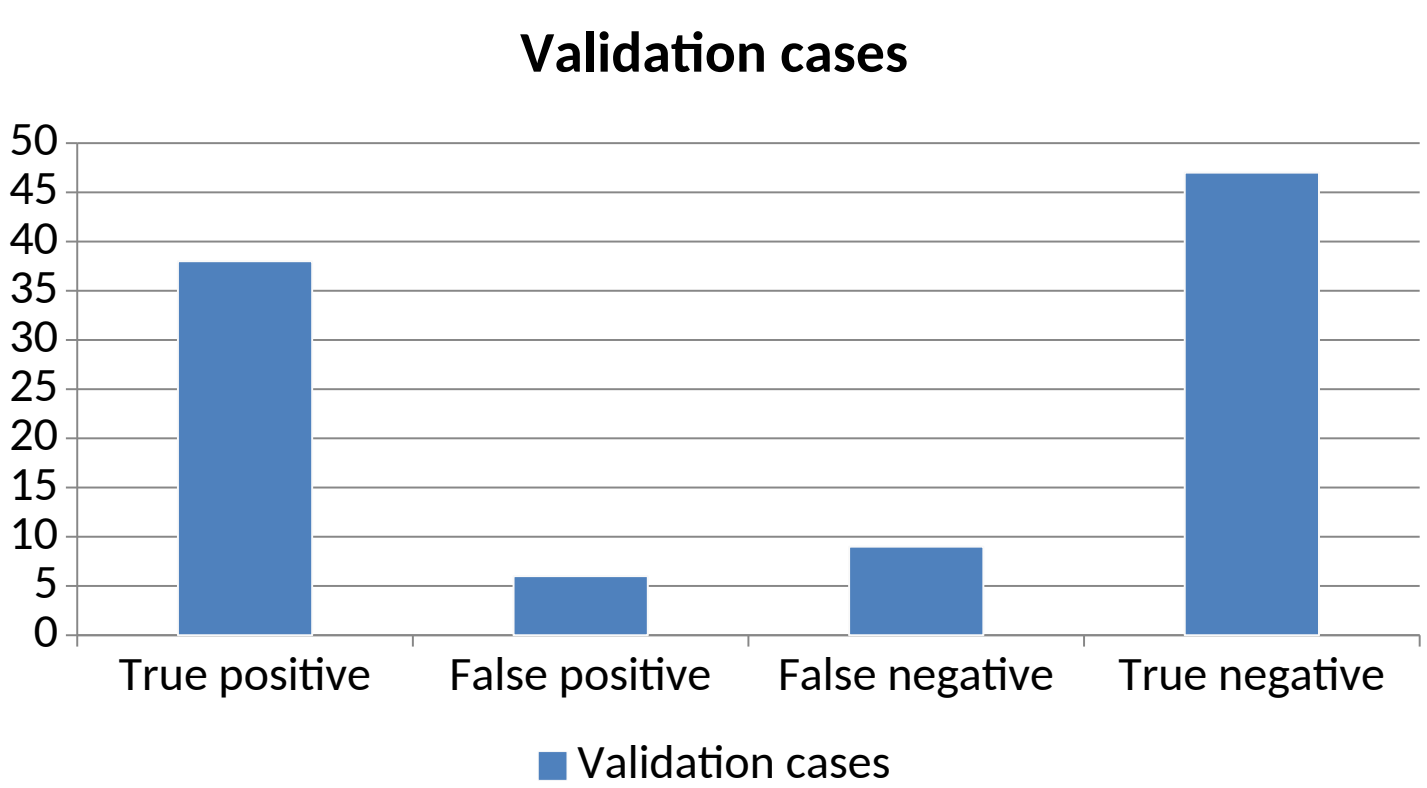
Capture a named output, metric or observable artifact.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Confusion Matrix



The confusion matrix is a tool used to assess the quality of a classification model's predictions.

It compares predicted labels against actual labels.

True Positive: The model predicts the patient has diabetes, and the patient does actually have diabetes.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Read a Confusion Matrix

1

Confusion Matrix.

2

Rows represent actual classes; columns represent predicted classes.

3

True positives and true negatives are correct predictions.

4

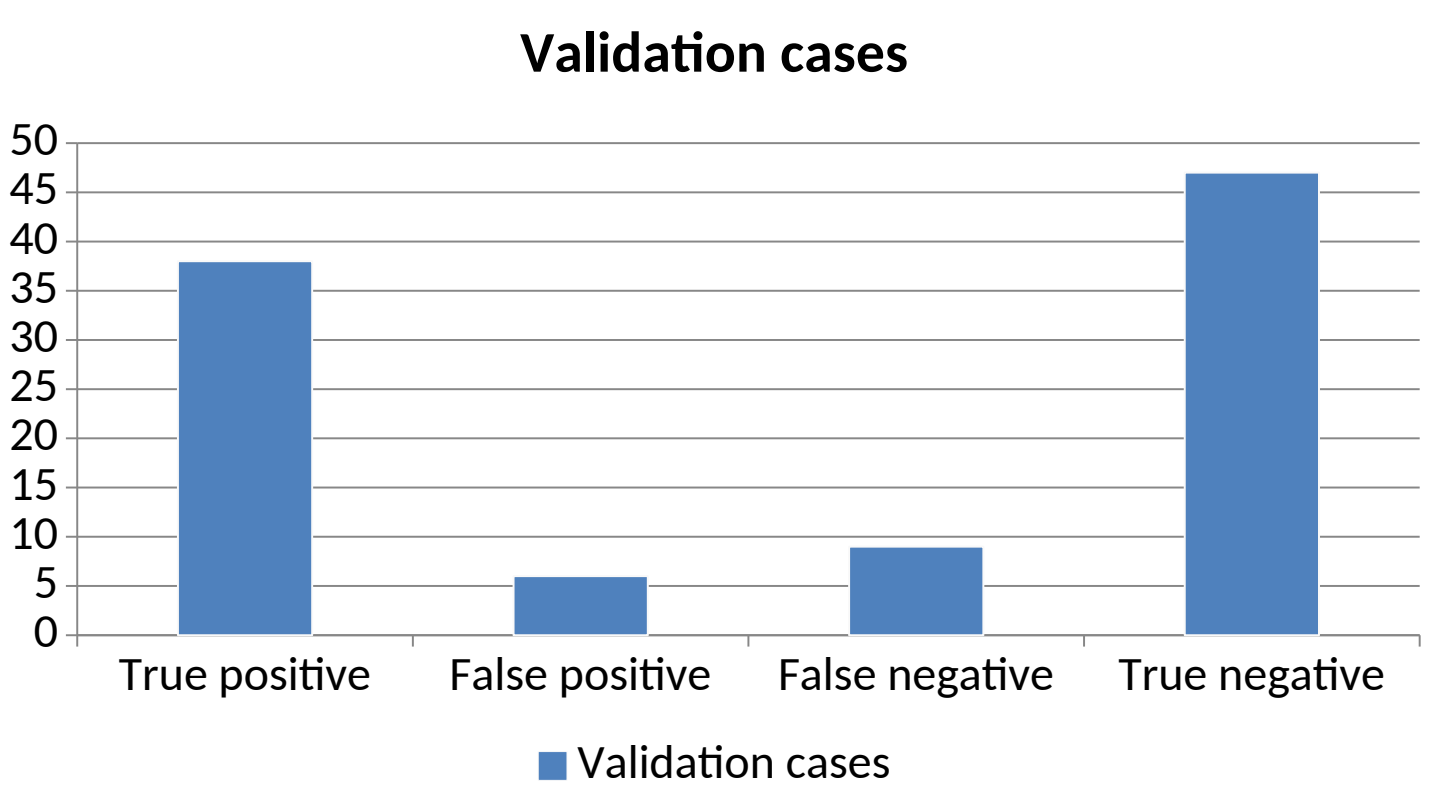
False positives and false negatives are different operational risks.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Confusion Matrix Worked Example



Confusion Matrix Example.

Rows represent actual classes; columns represent predicted classes.

True positives and true negatives are correct predictions.

Interpret the evidence; do not infer causality.

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Metrics

1

Accuracy: The number of correct predictions (true positives + true negatives) divided by the total number of predictions.

2

Recall: The fraction of positive cases correctly identified: the number of true positives divided by (the number of true positives plus false negatives).

3

F1 Score: An overall metric that essentially combines precision and recall.

4

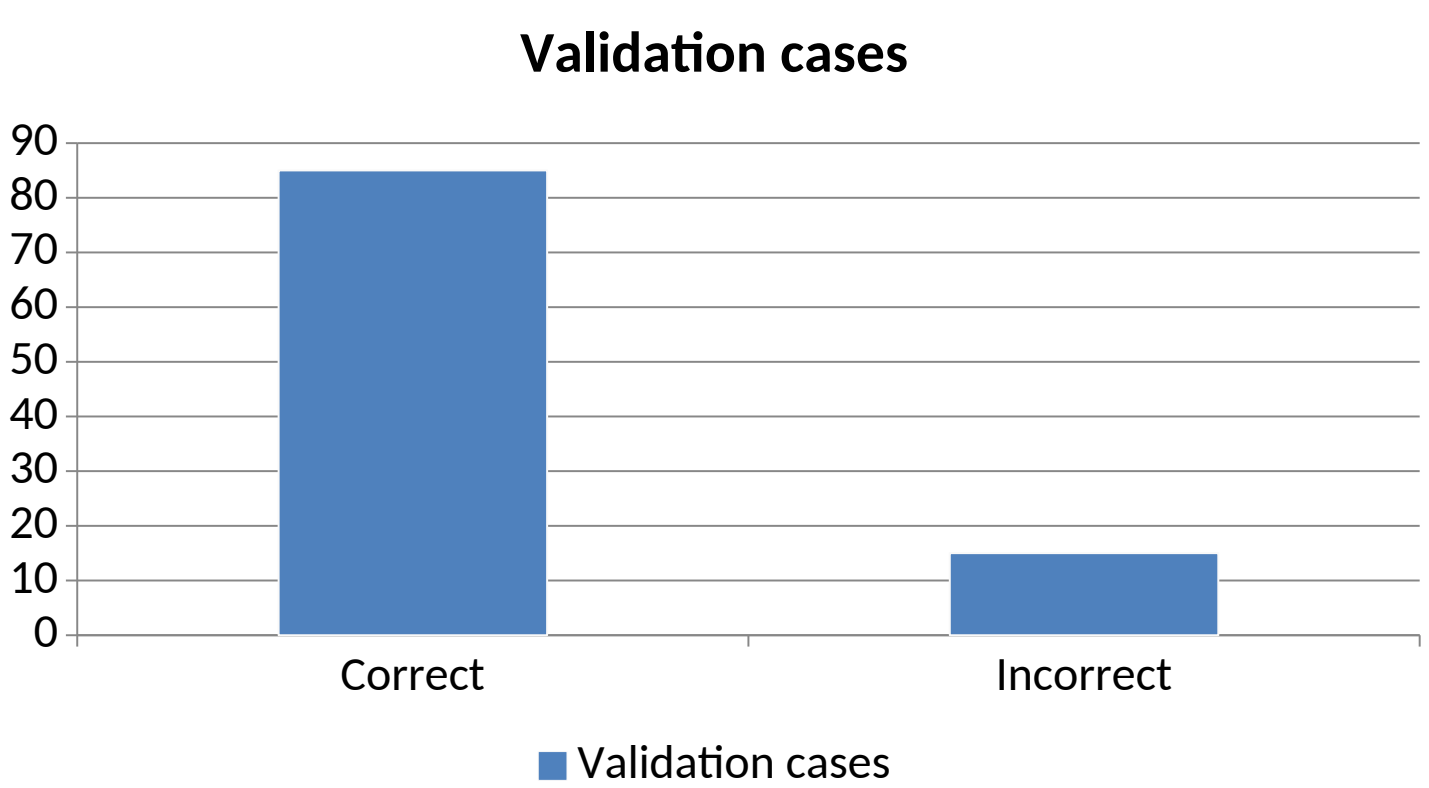
Identify the input contract for Metrics.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Accuracy



Accuracy = (TP + TN) / all cases.

It summarizes the fraction of correct predictions.

Class imbalance can make accuracy misleading.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Precision

1

Precision = TP / (TP + FP).

2

It asks: of predicted positives, how many were correct?

3

Prioritize it when false positives are costly.

4

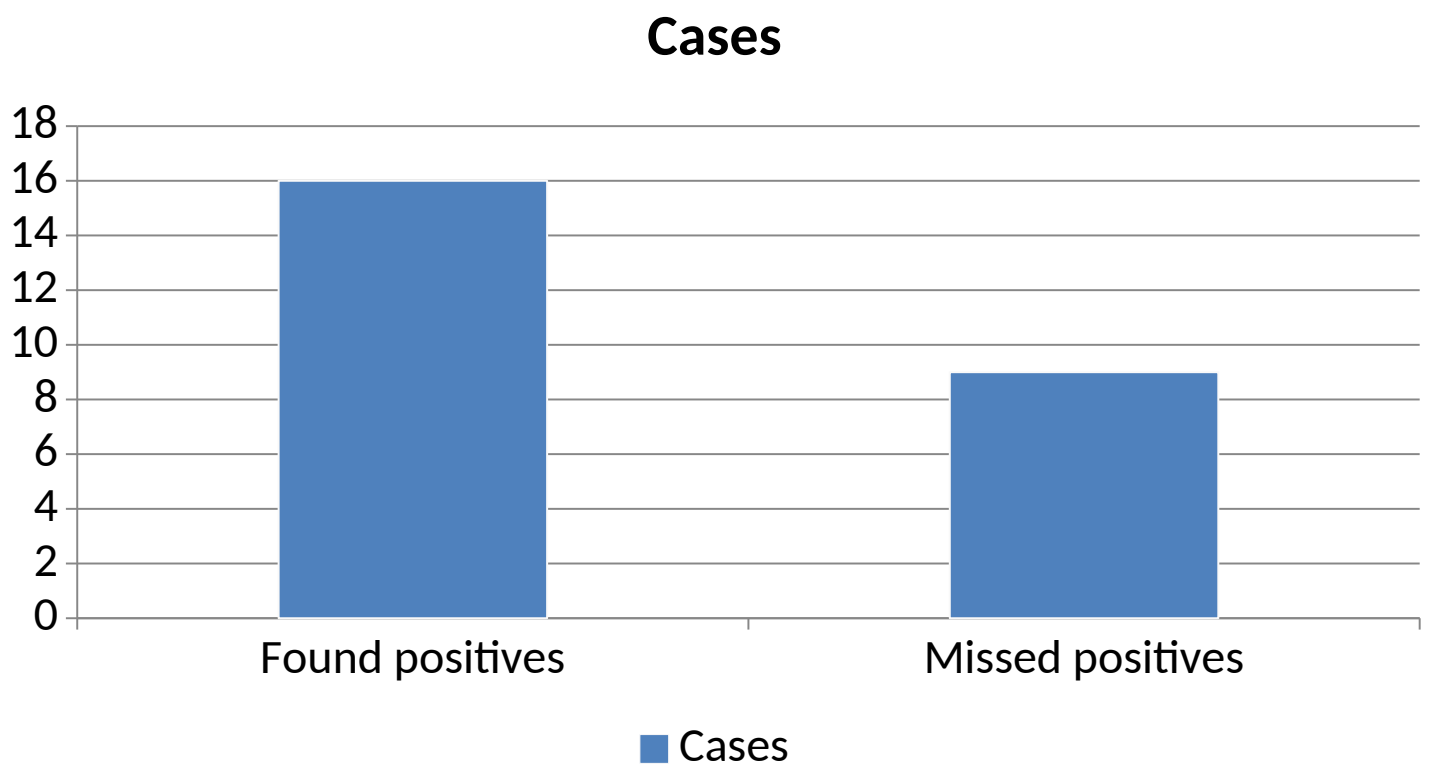
Report the class, threshold and validation population.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Recall or Sensitivity



Recall = TP / (TP + FN).

It asks: of actual positives, how many were found?

Prioritize it when missed positives are costly.

Interpret the evidence; do not infer causality.

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Specificity

1

Specificity = TN / (TN + FP).

2

It measures how well actual negatives are rejected.

3

Low specificity produces excessive false alarms.

4

Validate it with recall and the selected threshold.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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F1 Score

1

F1 = 2 × precision × recall / (precision + recall).

2

It balances precision and recall through a harmonic mean.

3

Use it when both false-positive and false-negative costs matter.

4

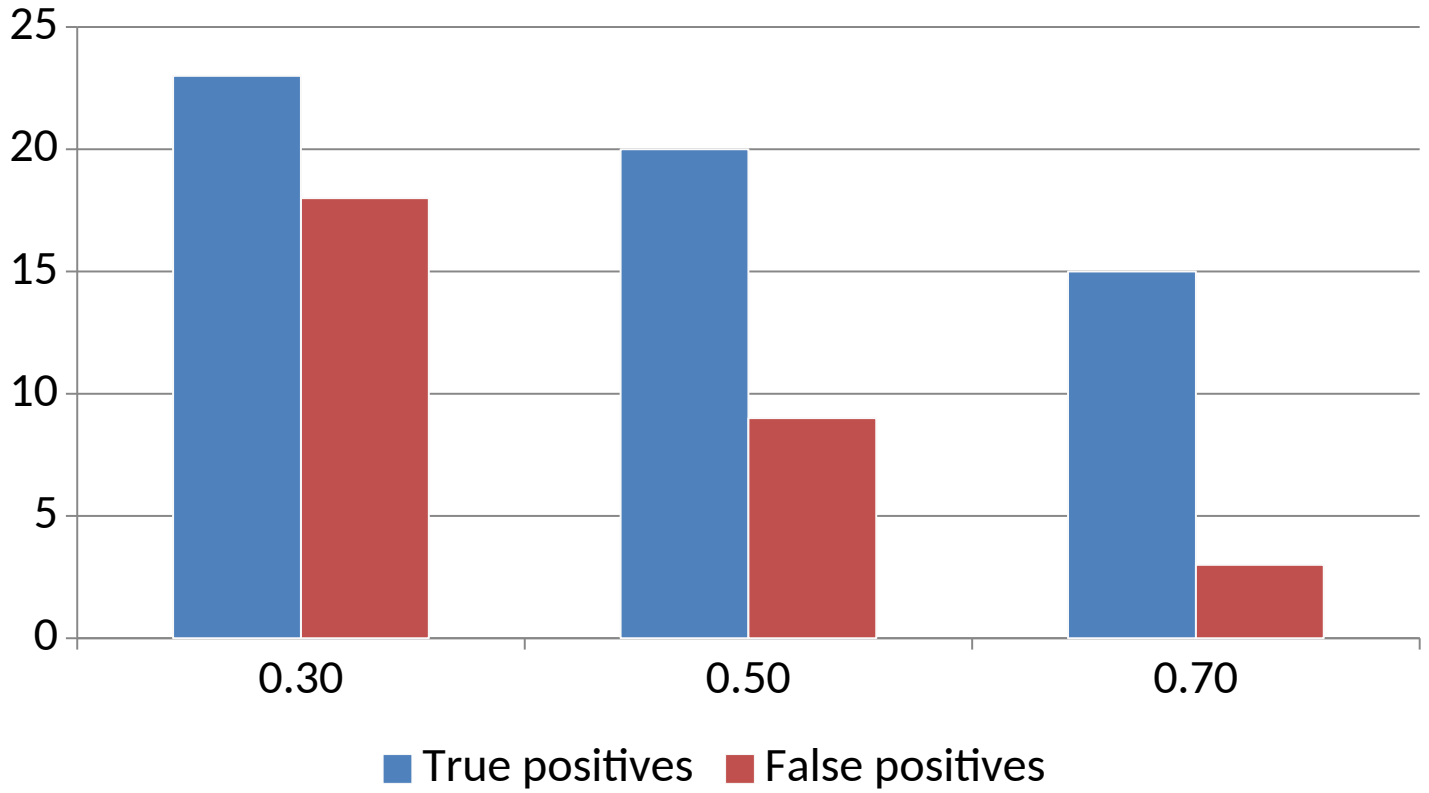
Retain the underlying confusion-matrix counts.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Threshold



Designer has a useful threshold slider for reviewing how the model performance would change depending on the set threshold.

A score threshold converts probability into a class decision.

Lower thresholds usually increase recall and false positives.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Threshold

1

A score threshold converts probability into a class decision.

2

Lower thresholds usually increase recall and false positives.

3

Higher thresholds usually increase precision and false negatives.

4

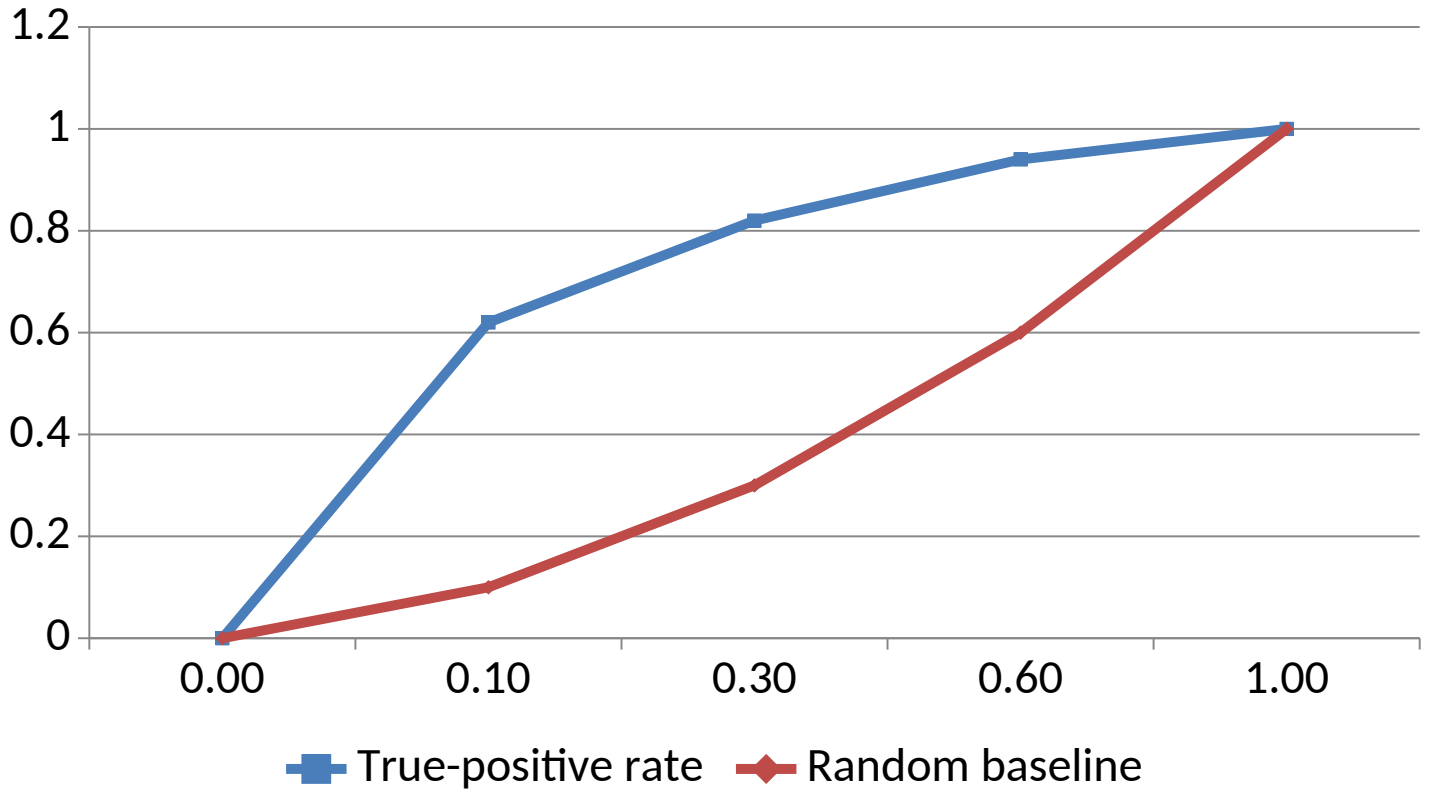
Choose the operating point from business risk and evidence.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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ROC



Plotting these metrics against each other for every possible threshold value between 0 and 1 results in a curve, known as the ROC curve.

ROC plots true-positive rate against false-positive rate.

Each point represents a different classification threshold.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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ROC

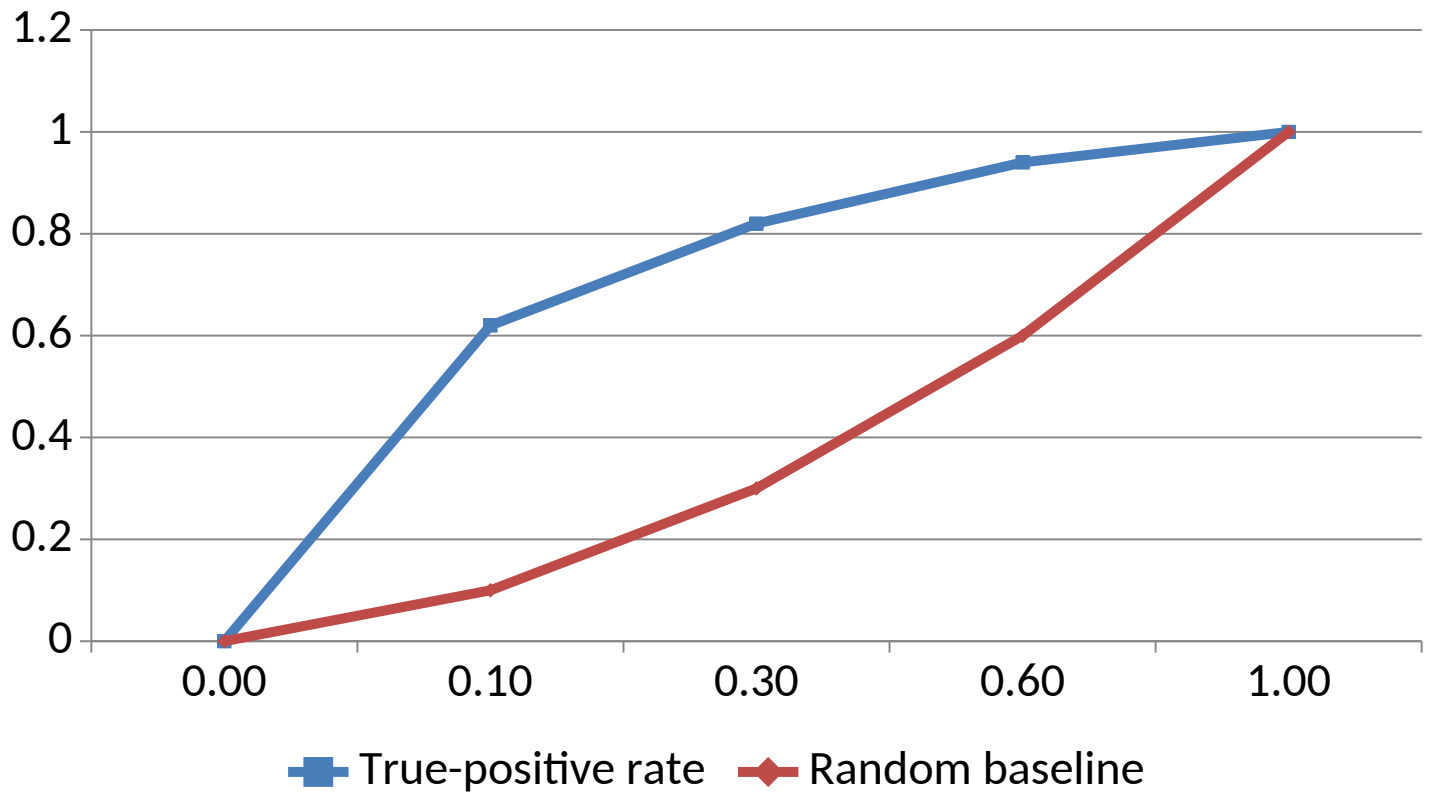
1. ROC plots true-positive rate against false-positive rate.
2. Each point represents a different classification threshold.
3. A curve near the upper-left indicates stronger separation.
4. Choose an operating point; AUC alone does not set the threshold.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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AUC



In an ideal model, the curve would go all the way up the left side and across the top, so that it covers the full area of the chart.

The larger the area under the curve, of AUC metric, (which can be any value from 0 to 1), the better the model is performing.

AUC summarizes ranking performance across thresholds.

Interpret the evidence; do not infer causality.

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Activity: Classification

1. Features map to one of two or more categorical labels.
2. The model emits class scores or probabilities.
3. A threshold converts scores into predicted labels.
4. Validate with a confusion matrix and class-specific metrics.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Iris Flower Dataset

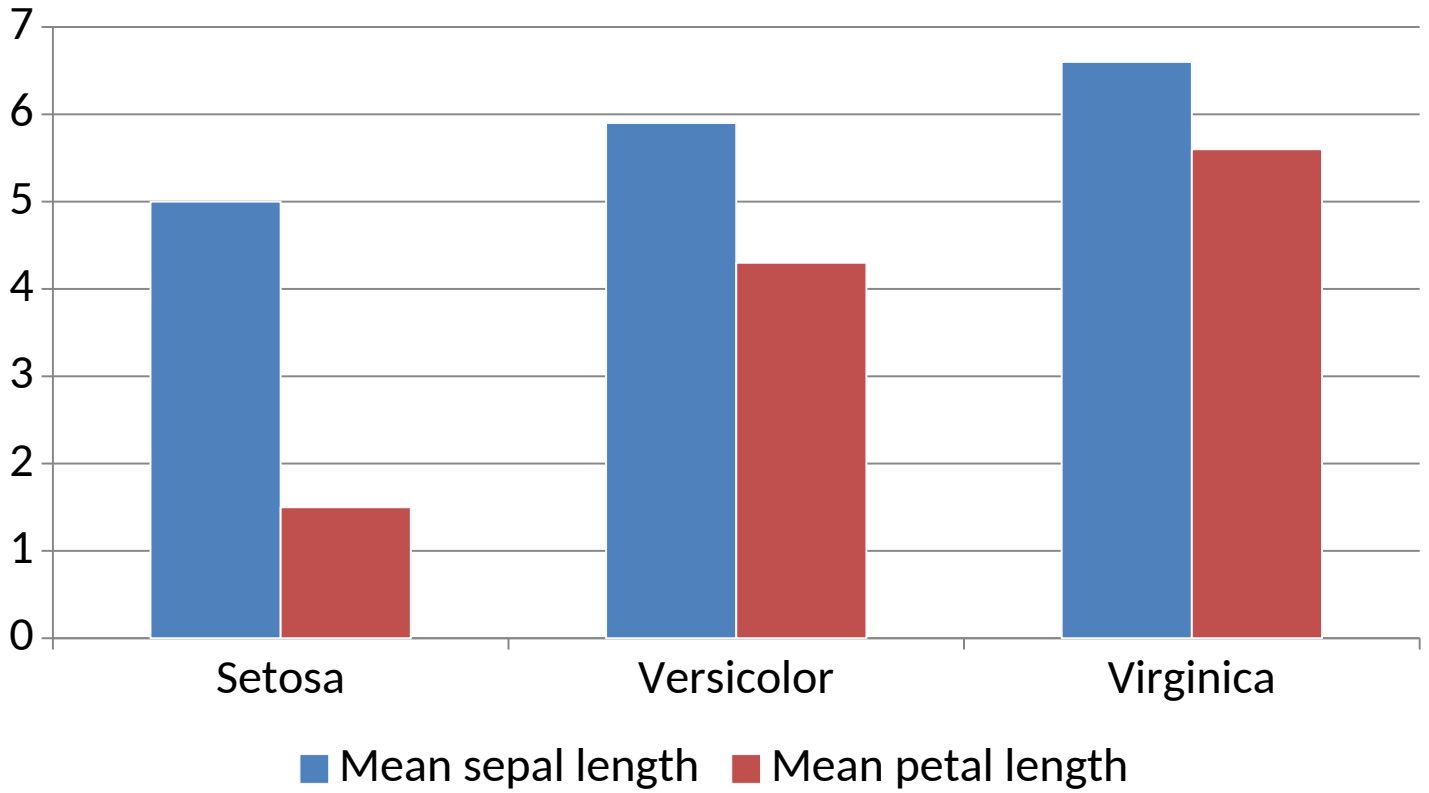


Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Iris Flower Dataset



Identify the input contract for Iris Flower Dataset.

Trace the processing component and its state change.

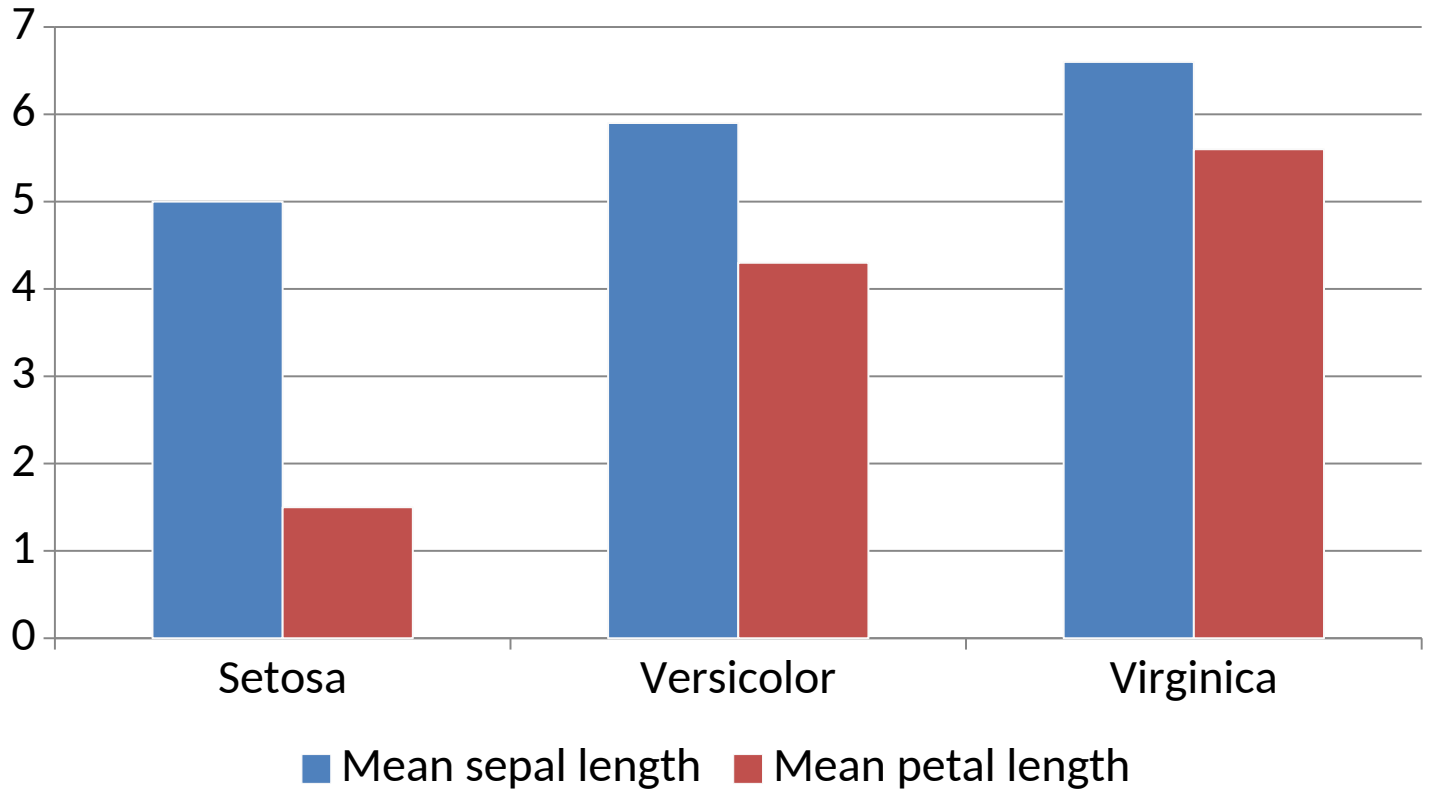
Capture a named output, metric or observable artifact.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Hint for Activity



Identify the input contract for Hint for Activity.

Trace the processing component and its state change.

Capture a named output, metric or observable artifact.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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120

LEGACY COVERAGE · SLIDE 120

Clustering

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Clustering Example

1

Clustering groups unlabeled records by similarity.

2

Feature scaling changes the distance geometry.

3

Cluster IDs are not predefined business labels.

4

Profile and validate each cluster before operational use.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Create a Pipeline

1

A pipeline connects versioned inputs to transformations and outputs.

2

Each component exposes a typed boundary and run state.

3

Run history provides repeatability and failure evidence.

4

Deploy only a validated inference path.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 123 · MODEL

Add Data Transformations

1. Identify the input contract for Add Data Transformations.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.
4. Define the acceptance threshold and human-review route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Select Columns

1 INPUT / SIGNAL

Identify the input contract for Select Columns.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Clean Missing Data

1

Identify the input contract for Clean Missing Data.

2

Trace the processing component and its state change.

3

Capture a named output, metric or observable artifact.

4

Define the acceptance threshold and human-review route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 126 · MODEL

Normalize Data

1 INPUT / SIGNAL

Identify the input contract for Normalize Data.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 127 · UI

Run the Pipeline

1 INPUT / SIGNAL

A pipeline connects versioned inputs to transformations and outputs.

2 MECHANISM

Each component exposes a typed boundary and run state.

3 OUTPUT / EVIDENCE

Run history provides repeatability and failure evidence.

4 CONTROL / VERIFY

Deploy only a validated inference path.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Add Training Modules

1

Identify the input contract for Add Training Modules.

2

Trace the processing component and its state change.

3

Capture a named output, metric or observable artifact.

4

Define the acceptance threshold and human-review route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Run the Training Pipeline

1 INPUT / SIGNAL

A pipeline connects versioned inputs to transformations and outputs.

2 MECHANISM

Each component exposes a typed boundary and run state.

3 OUTPUT / EVIDENCE

Run history provides repeatability and failure evidence.

4 CONTROL / VERIFY

Deploy only a validated inference path.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Evaluate a Clustering Model

1 INPUT / SIGNAL

Clustering groups unlabeled records by similarity.

2 MECHANISM

Feature scaling changes the distance geometry.

3 OUTPUT / EVIDENCE

Cluster IDs are not predefined business labels.

4 CONTROL / VERIFY

Profile and validate each cluster before operational use.

Control: define the threshold or failure that routes to human review.

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Create an Inference Pipeline

1

A pipeline connects versioned inputs to transformations and outputs.

2

Each component exposes a typed boundary and run state.

3

Run history provides repeatability and failure evidence.

4

Deploy only a validated inference path.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 132 · ARCH

Test the Service

1 INPUT / SIGNAL

Identify the input contract for Test the Service.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

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What is Azure Automated Machine Learning?

1 INPUT / SIGNAL

These automated capabilities use the power of cloud compute to find the best performing supervised machine learning model for your data.

2 MECHANISM

Identify the input contract for What is Azure Automated Machine Learning?.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

Capture a named output, metric or observable artifact.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Understand the AutoML process

1

Select a supported ML task, versioned data asset and target column.

2

Set the primary metric, compute budget, timeout and validation method.

3

AutoML evaluates preprocessing and algorithm pipelines within those bounds.

4

Use leaderboard metrics and model details to justify the selected candidate.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Understand the AutoML process

1 INPUT / SIGNAL

Select a supported ML task, versioned data asset and target column.

2 MECHANISM

Set the primary metric, compute budget, timeout and validation method.

3 OUTPUT / EVIDENCE

AutoML evaluates preprocessing and algorithm pipelines within those bounds.

4 CONTROL / VERIFY

Use leaderboard metrics and model details to justify the selected candidate.

Control: define the threshold or failure that routes to human review.

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Azure Machine Learning studio

1 INPUT / SIGNAL

Identify the input contract for Azure Machine Learning studio.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

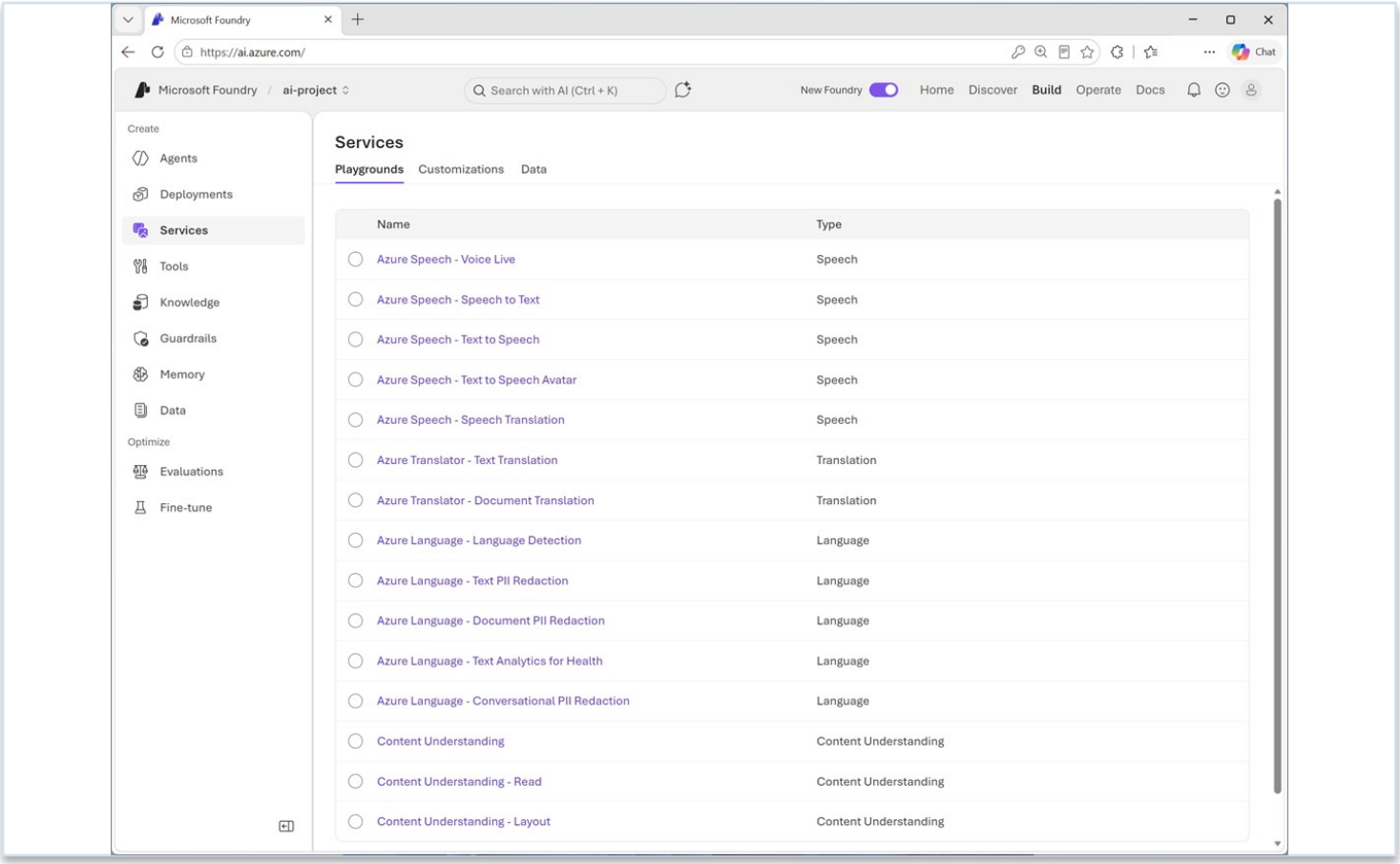
Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

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Prepare data



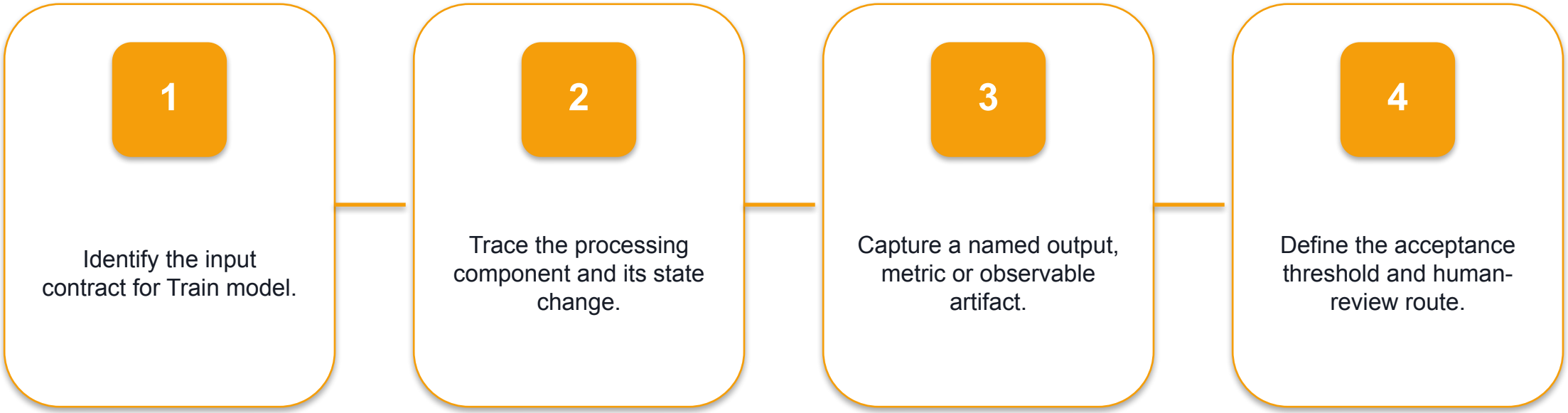
1. Identify the input contract for Prepare data.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Train model



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Configuration

1 INPUT / SIGNAL

Identify the input contract for Configuration.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Additional Configuration

1 INPUT / SIGNAL

Identify the input contract for Additional Configuration.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

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Additional Configuration

1 INPUT / SIGNAL

Identify the input contract for Additional Configuration.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Auto ML Experiment

1. Identify the input contract for Auto ML Experiment.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.
4. Define the acceptance threshold and human-review route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Auto ML Experiment

1 INPUT / SIGNAL

Identify the input contract for Auto ML Experiment.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Evaluate Performance

1 **INPUT / SIGNAL**

Identify the input contract for Evaluate Performance.

2 **MECHANISM**

Trace the processing component and its state change.

3 **OUTPUT / EVIDENCE**

Capture a named output, metric or observable artifact.

4 **CONTROL / VERIFY**

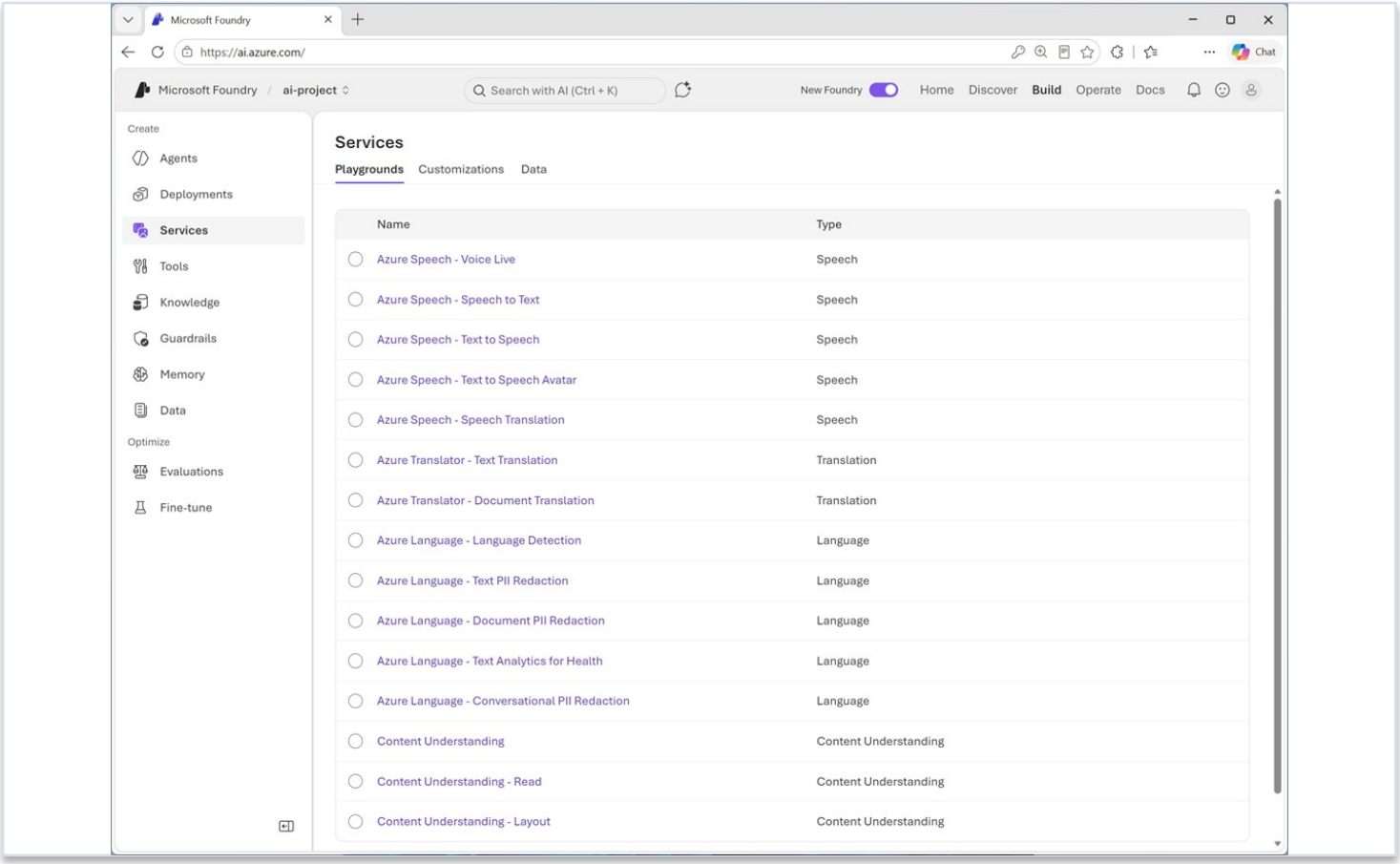
Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Review Result



1. Identify the input contract for Review Result.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Model Summary

1. Identify the input contract for Model Summary.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.
4. Define the acceptance threshold and human-review route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Review the Best Model

1 INPUT / SIGNAL

Identify the input contract for Review the Best Model.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Deploy a Predictive Service

1 INPUT / SIGNAL

In Azure Machine Learning, you can deploy a service as an Azure Container Instances (ACI) or to an Azure Kubernetes Service (AKS) cluster.

2 MECHANISM

For production scenarios, an AKS deployment is recommended, for which you must create an inference cluster compute target.

3 OUTPUT / EVIDENCE

In this exercise, you'll use an ACI service, which is a suitable deployment target for testing, and does not require you to create an inference cluster.

4 CONTROL / VERIFY

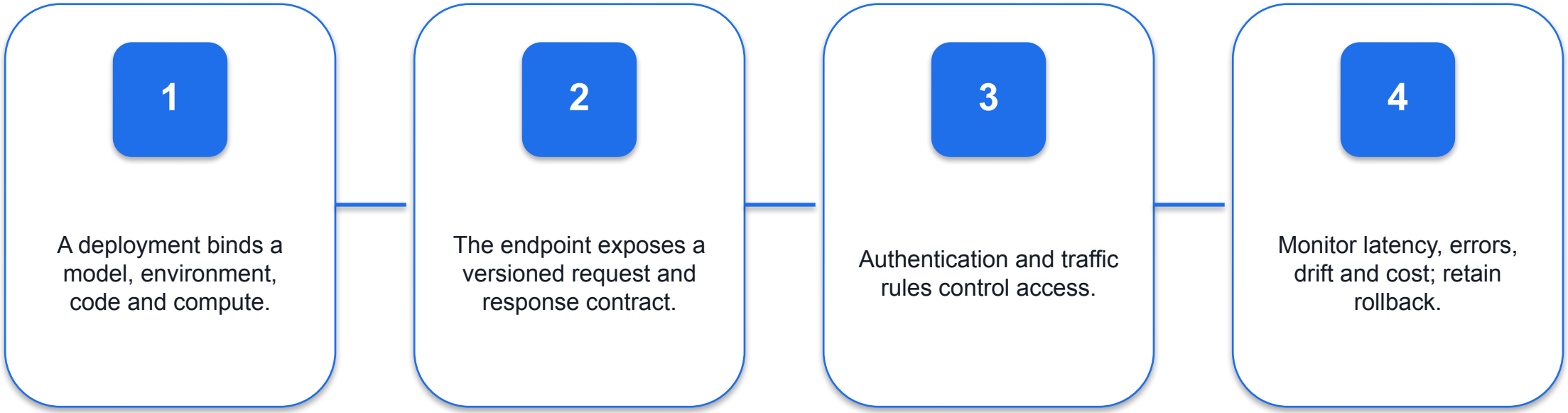
A deployment binds a model, environment, code and compute.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Deploy a Model as a Service



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Deploy a Model as a Service

1 INPUT / SIGNAL

A deployment binds a model, environment, code and compute.

2 MECHANISM

The endpoint exposes a versioned request and response contract.

3 OUTPUT / EVIDENCE

Authentication and traffic rules control access.

4 CONTROL / VERIFY

Monitor latency, errors, drift and cost; retain rollback.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Topic 3 Explore Computer Vision

1 INPUT / SIGNAL

Identify the input contract for Topic 3 Explore Computer Vision.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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What is Computer Vision

1 INPUT / SIGNAL

Identify the input contract for What is Computer Vision.

2 MECHANISM

Trace the processing component and its state change.

3 OUTPUT / EVIDENCE

Capture a named output, metric or observable artifact.

4 CONTROL / VERIFY

Define the acceptance threshold and human-review route.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Image Classification

1. Features map to one of two or more categorical labels.
2. The model emits class scores or probabilities.
3. A threshold converts scores into predicted labels.
4. Validate with a confusion matrix and class-specific metrics.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Azure Resources for Computer Vision

1 INPUT / SIGNAL

To use the Computer Vision service, you need to create a resource for it in your Azure subscription.

2 MECHANISM

Cognitive Services: A general cognitive services resource that includes Custom Vision along with many other cognitive services.

3 OUTPUT / EVIDENCE

You can use this type of resource for training, prediction, or both.

4 CONTROL / VERIFY

Identify the input contract for Azure Resources for Computer Vision.

Control: define the threshold or failure that routes to human review.

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LEGACY SLIDE 155 · FORMULA

Model Evaluation

1 INPUT / SIGNAL

Precision: What percentage of the class predictions made by the model were correct?

2 MECHANISM

For example, if the model predicted that 10 images are oranges, of which eight were actually oranges, then the precision is 0.8 (80%).

3 OUTPUT / EVIDENCE

Recall: What percentage of class predictions did the model correctly identify?

4 CONTROL / VERIFY

For example, if there are 10 images of apples, and the model found 7 of them, then the recall is 0.7 (70%).

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Using the model for prediction

1

Project ID: The unique ID of the Custom Vision project you created to train the model.

2

Model name: The name you assigned to the model during publishing.

3

Prediction endpoint: The HTTP address of the endpoints for the prediction resource to which you published the model (not the training resource).

4

Prediction key: The authentication key for the prediction resource to which you published the model (not the training resource).

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Object Detection Outputs

1 INPUT / SIGNAL

Object Detection.

2 MECHANISM

The class of each object identified in the image.

3 OUTPUT / EVIDENCE

The model returns a label, confidence and bounding box per object.

4 CONTROL / VERIFY

Multiple objects can appear in one image.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Face Detection Bounding Boxes

1 INPUT / SIGNAL

Face Detection.

2 MECHANISM

Detection returns a face region; identification asserts identity.

3 OUTPUT / EVIDENCE

Consent, purpose and legal authority must be explicit.

4 CONTROL / VERIFY

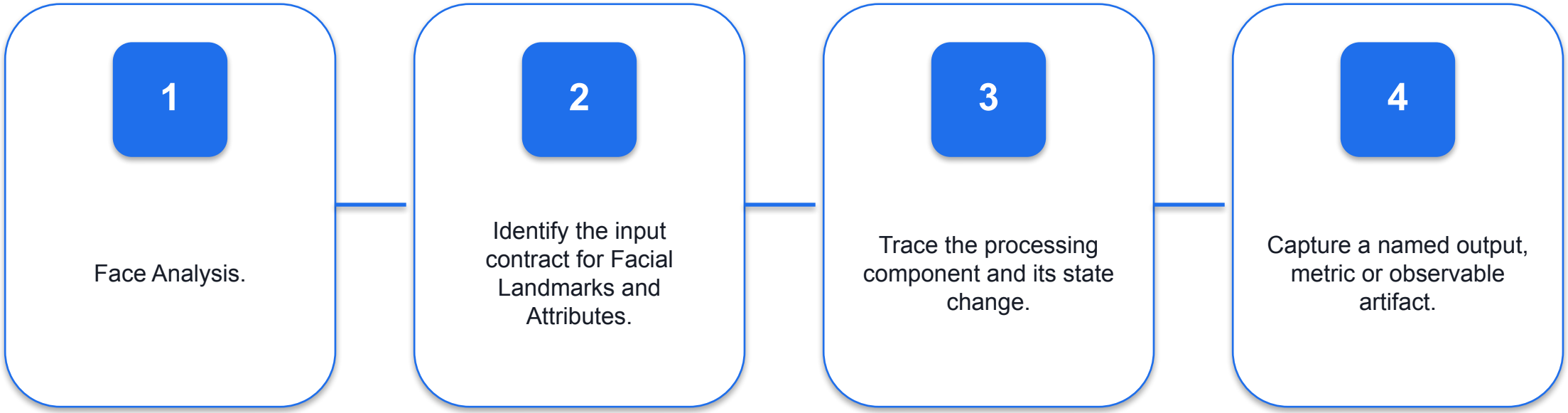
Do not infer sensitive traits from facial appearance.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Facial Landmarks and Attributes



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Face Identification and Responsible Use

1 INPUT / SIGNAL

Face Recognition.

2 MECHANISM

Detection returns a face region; identification asserts identity.

3 OUTPUT / EVIDENCE

Consent, purpose and legal authority must be explicit.

4 CONTROL / VERIFY

Do not infer sensitive traits from facial appearance.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Face Analysis in Azure

1 INPUT / SIGNAL

Computer Vision, which offers face detection and some basic face analysis, such as returning the bounding box coordinates around an image.

2 MECHANISM

Video Indexer, which you can use to detect and identify faces in a video.

3 OUTPUT / EVIDENCE

Face, which offers pre-built algorithms that can detect, recognize, and analyze faces.

4 CONTROL / VERIFY

Detection returns a face region; identification asserts identity.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Legacy Face Lab Migration

1 INPUT / SIGNAL

Activity: Face Analysis.

2 MECHANISM

Detection returns a face region; identification asserts identity.

3 OUTPUT / EVIDENCE

Consent, purpose and legal authority must be explicit.

4 CONTROL / VERIFY

Do not infer sensitive traits from facial appearance.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Read API

1

The Read API uses the latest recognition models and is optimized for images that have a significant amount of text or has considerable visual noise.

2

The Read API can handle scanned documents that have a lot of text.

3

OCR detects printed or handwritten text and geometry.

4

The response links text spans to pages and regions.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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LEGACY SLIDE 164 · CASE

Legacy OCR Lab Migration

1 INPUT / SIGNAL

Activity: OCR.

2 MECHANISM

OCR detects printed or handwritten text and geometry.

3 OUTPUT / EVIDENCE

The response links text spans to pages and regions.

4 CONTROL / VERIFY

Compare extracted text with the source image.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Azure AI Document Intelligence

1 INPUT / SIGNAL

Custom models are trained using your own data, which helps to tailor this model to your specific forms.

2 MECHANISM

Starting with only five samples of your forms, you can train the custom model.

3 OUTPUT / EVIDENCE

A prebuilt or custom model extracts typed document fields.

4 CONTROL / VERIFY

Results include values, tables, geometry and confidence.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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TOPIC 2 · HANDS-ON BRIDGE

Lab 04 · Computer Vision, Image Analysis, and OCR

65 MIN

A product-quality team needs visual inspection and text extraction without treating model output as ground truth.

SCENARIO

Analyze an image, compare vision outputs with source evidence, and report accuracy or privacy issues.

MECHANISM

Microsoft Foundry; vision-capable model; Content Understanding OCR

EVIDENCE

An image-analysis result, OCR extraction, confidence review and issue report.

VERIFY

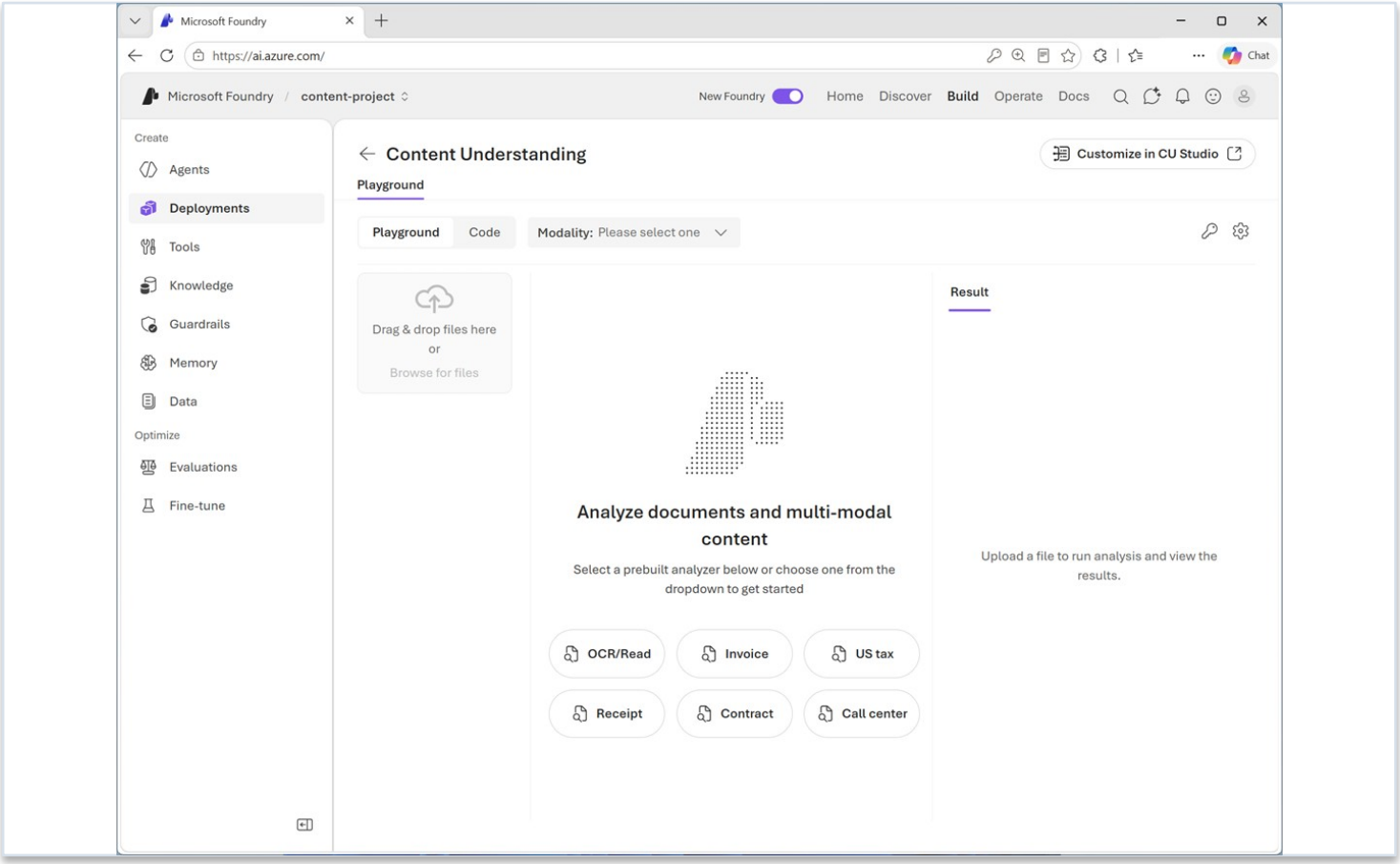
Image input and model recorded · Output compared with source image

Full step-by-step: Learner Guide + labs/lab-04-computer-vision-image-analysis-ocr/README.md

Lab bridge only—no click-by-click procedure on the slide.

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Pre-built Receipt Model



1. A prebuilt or custom model extracts typed document fields.
2. Results include values, tables, geometry and confidence.
3. Validate critical fields against the original document.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Knowledge Quiz - Fundamentals of Computer Vision

1 INPUT / SIGNAL

Distinguish classification, detection, segmentation and OCR.

2 MECHANISM

Read labels, confidence scores and bounding regions.

3 OUTPUT / EVIDENCE

Choose evidence that proves the visual result.

4 CONTROL / VERIFY

Identify a privacy or fairness risk in image analysis.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Knowledge Quiz - Fundamentals of Facial Recognition

1 INPUT / SIGNAL

Separate face detection from identification.

2 MECHANISM

Interpret a face bounding box and confidence score.

3 OUTPUT / EVIDENCE

Apply consent, access and data-retention controls.

4 CONTROL / VERIFY

Route uncertain or high-impact results to human review.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

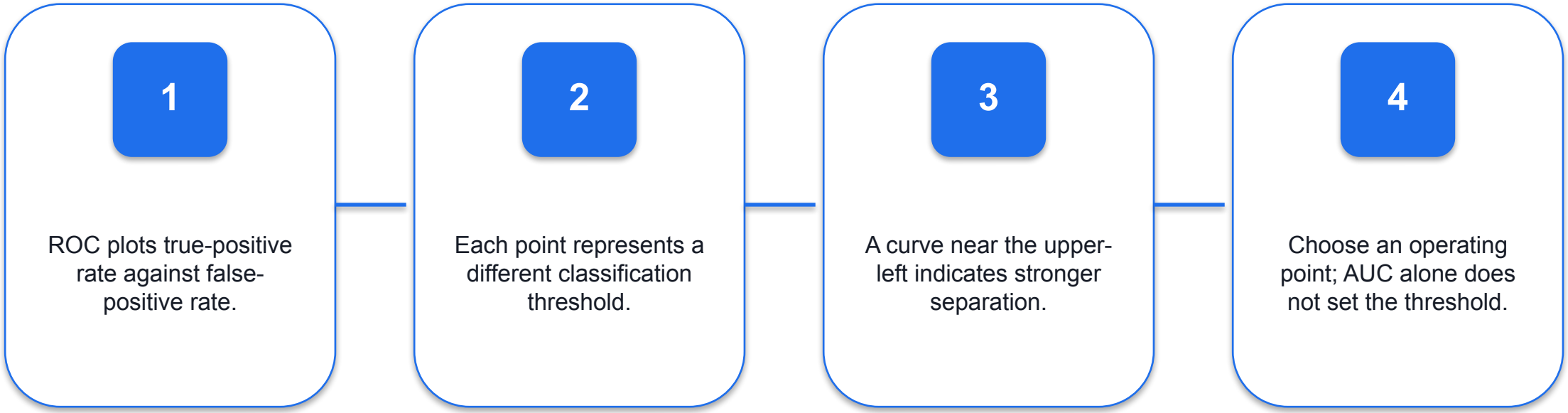
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LEGACY COVERAGE · SLIDE 169

Knowledge Quiz - Fundamentals of optical character recognition

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Topic 4 Explore Natural Language Processing



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Natural Language Processing

1

Natural language processing supports applications that can see, hear, speak with, and understand users.

2

ROC plots true-positive rate against false-positive rate.

3

Each point represents a different classification threshold.

4

A curve near the upper-left indicates stronger separation.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Language Detection



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Sentiment Analysis

1

The text analytics capabilities in the Language service can evaluate text and return sentiment scores and labels for each sentence.

2

This capability is useful for detecting positive and negative sentiment in social media, customer reviews, discussion forums and more.

3

Scores that are close to the middle of the range (0.5) are considered neutral or indeterminate.

4

Sentiment returns positive, neutral or negative confidence.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Key Phrase Extraction

1

table friendly hostess dinner ambiance place.

2

attentive service great food birthday celebration fantastic experience.

3

Key phrase extraction is the concept of evaluating the text of a document, or documents, and then identifying the main talking points of the document(s).

4

Key phrase extraction can provide some context to this review by extracting the following phrases:.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Named-Entity Recognition

1

Entity Recognition.

2

Entity recognition returns typed spans and offsets.

3

Linked entities may include a knowledge-base identifier.

4

Offsets must survive preprocessing and encoding.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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LEGACY SLIDE 176 · CASE

Text Analysis Lab

1

Activity: Text Analytics.

2

Identify the input contract for Text Analysis Lab.

3

Trace the processing component and its state change.

4

Capture a named output, metric or observable artifact.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Speech Recognition

1

The spoken words can be in the form of a recorded voice in an audio file, or live audio from a microphone.

2

Speech patterns are analyzed in the audio to determine recognizable patterns that are mapped to words.

3

Speech recognition converts audio into timestamped text.

4

Speech synthesis converts text into encoded audio.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Speech Synthesis

1

Speech synthesis is in many respects the reverse of speech recognition.

2

It is concerned with vocalizing data, usually by converting text to speech.

3

A speech synthesis solution typically requires the following information: The text to be spoken.

4

The voice to be used to vocalize the speech.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Speech Service

1

Text-to-speech - used to generate spoken audio from a text source.

2

Speech Translation - used to translate speech in one language to text or speech in another.

3

Speech recognition converts audio into timestamped text.

4

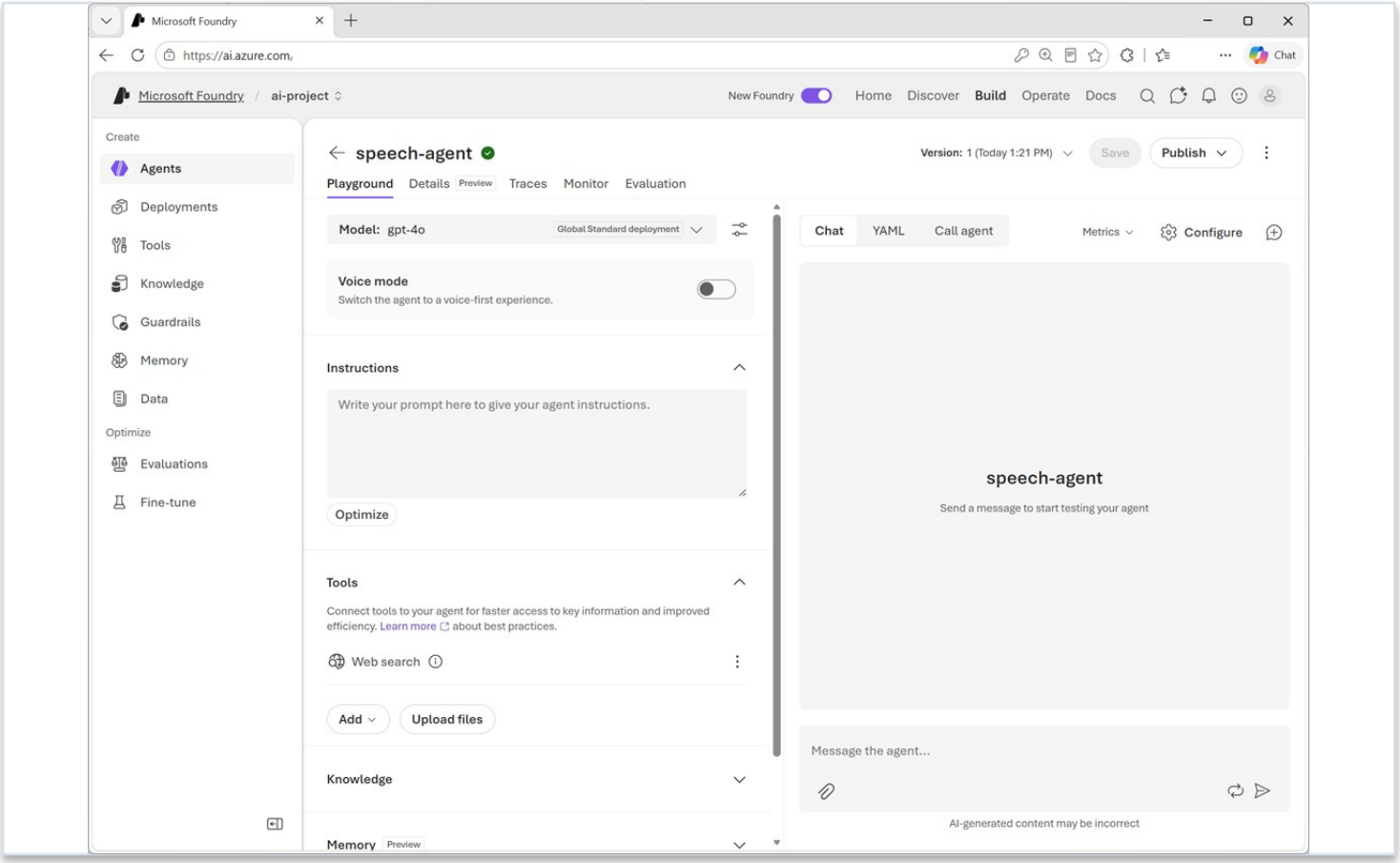
Speech synthesis converts text into encoded audio.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Speech-to-Text API



UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

1. You can use the speech-to-text API to perform real-time or batch transcription of audio into a text format.

2. The audio source for transcription can be a real-time audio stream from a microphone or an audio file.

3. The model that is used by the speech-to-text API, is based on the Universal Language Model that was trained by Microsoft.

UI may evolve; verify the named state and output.

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TOPIC 2 · HANDS-ON BRIDGE

Lab 05 · Language: Text Analysis, PII, and Clean Inputs

60 MIN

A service desk wants useful themes from feedback while preventing personal data from entering reports.

SCENARIO

Analyze text, detect language and PII, clean malformed input, and verify the output contract.

MECHANISM

Microsoft Foundry; Azure AI Language tool; JSON

EVIDENCE

A before/after clean-data record, text-analysis output, PII-redaction evidence and schema validation.

VERIFY

Clean dataset preserves row IDs · Summary cites source themes

Full step-by-step: Learner Guide + labs/lab-05-language-text-analysis-pii/README.md

Lab bridge only—no click-by-click procedure on the slide.

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Text-to-Speech API

1

When you use the text-to-speech API, you can specify the voice to be used to vocalize the text.

2

This capability offers you the flexibility to personalize your speech synthesis solution and give it a specific character.

3

Speech recognition converts audio into timestamped text.

4

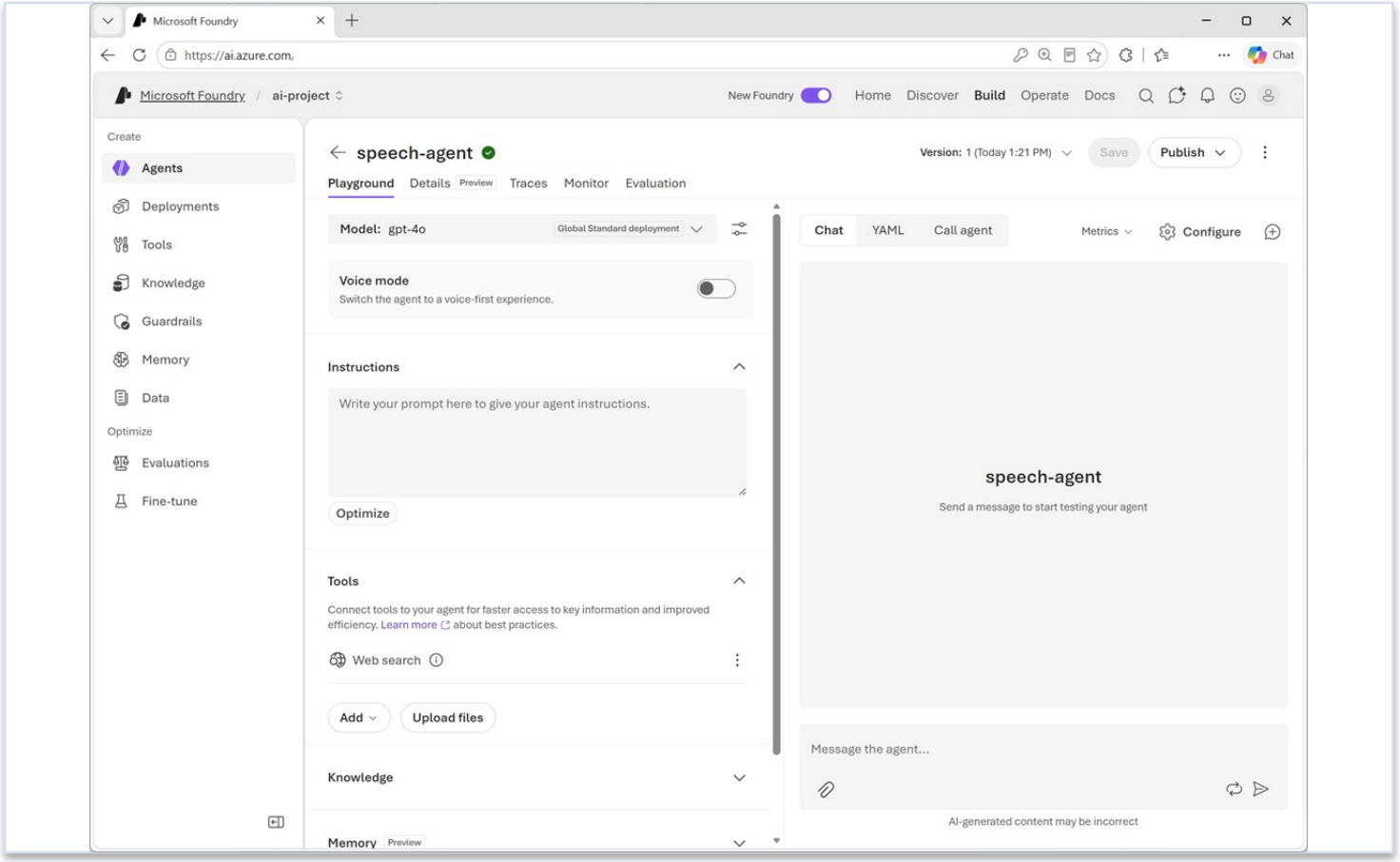
Speech synthesis converts text into encoded audio.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Speech Lab



1 Activity: Speech.

2 response_text = 'Turning the light on.'.

3 response_text = 'Turning the light off.'.

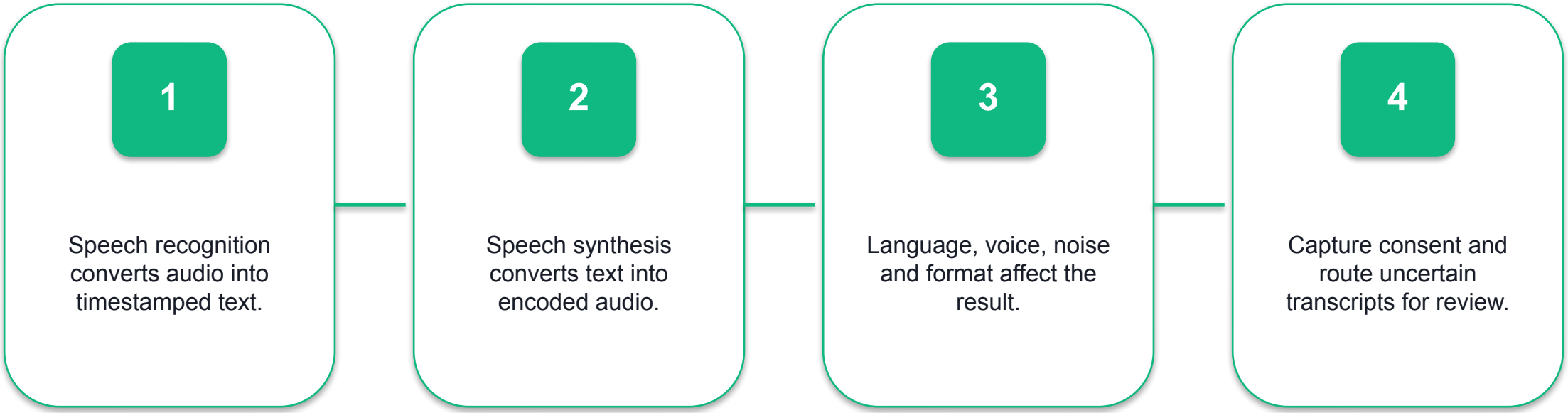
UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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LEGACY SLIDE 183 · EVIDENCE

Text and Speech Translation



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Conversational Language Lab

1 INPUT / SIGNAL

Activity: Text and Speech Translation.

2 MECHANISM

An utterance is mapped to an intent and extracted entities.

3 OUTPUT / EVIDENCE

Confidence and competing intents determine the route.

4 CONTROL / VERIFY

Fallback handles unsupported or ambiguous requests.

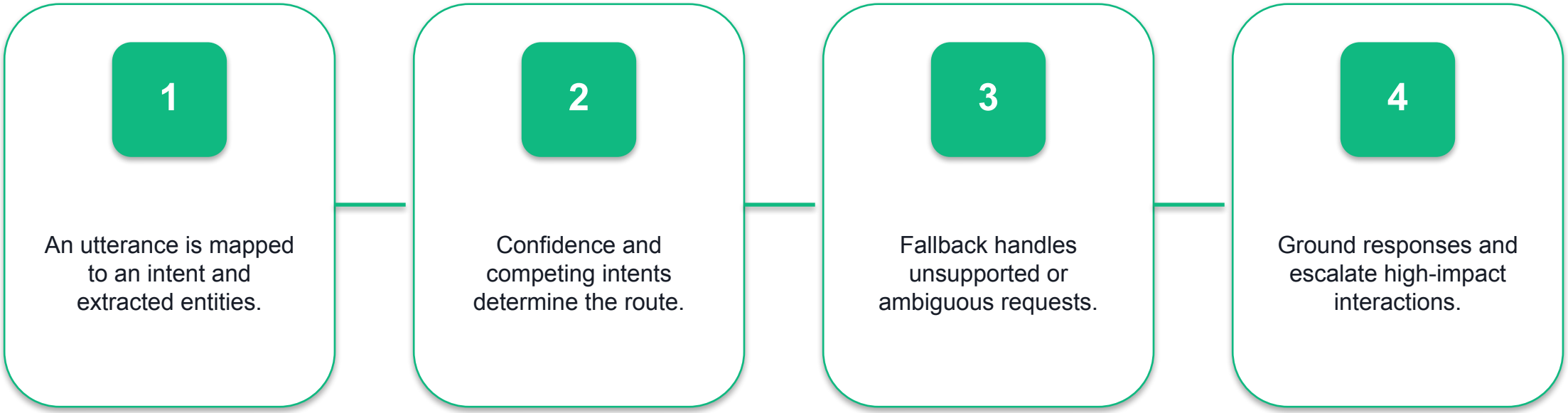
Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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LEGACY SLIDE 185 · ARCH

Conversational language understanding



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Legacy LUIS Lab Migration

1 INPUT / SIGNAL

Activity: Language Understanding.

2 MECHANISM

Identify the input contract for Legacy LUIS Lab Migration.

3 OUTPUT / EVIDENCE

Trace the processing component and its state change.

4 CONTROL / VERIFY

Capture a named output, metric or observable artifact.

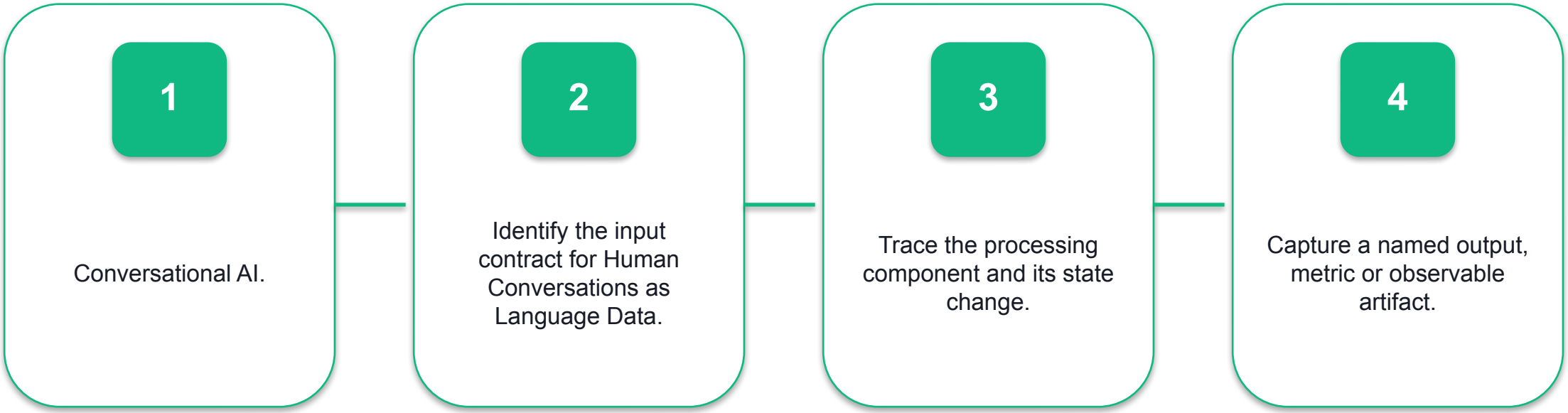
Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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LEGACY SLIDE 187 · MODEL

Human Conversations as Language Data



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 188 · CASE

Conversational AI

1 **INPUT / SIGNAL**

Bots are designed to interact with users in a conversational manner, as shown in this example of a chat interface:.

2 **MECHANISM**

An utterance is mapped to an intent and extracted entities.

3 **OUTPUT / EVIDENCE**

Confidence and competing intents determine the route.

4 **CONTROL / VERIFY**

Fallback handles unsupported or ambiguous requests.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Question Answering and Azure Bot Service

1

You can easily create a user support bot solution on Microsoft Azure using a combination of two core technologies: Custom question answering.

2

This cognitive service enables you to create and publish a knowledge base with built-in natural language processing capabilities.

3

Azure Bot Service.

4

This service provides a framework for developing, publishing, and managing bots on Azure.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Azure Custom question answering

1 INPUT / SIGNAL

Approved documents become a bounded knowledge source.

2 MECHANISM

Retrieval finds candidate passages for an answer.

3 OUTPUT / EVIDENCE

Citations provide traceability to source evidence.

4 CONTROL / VERIFY

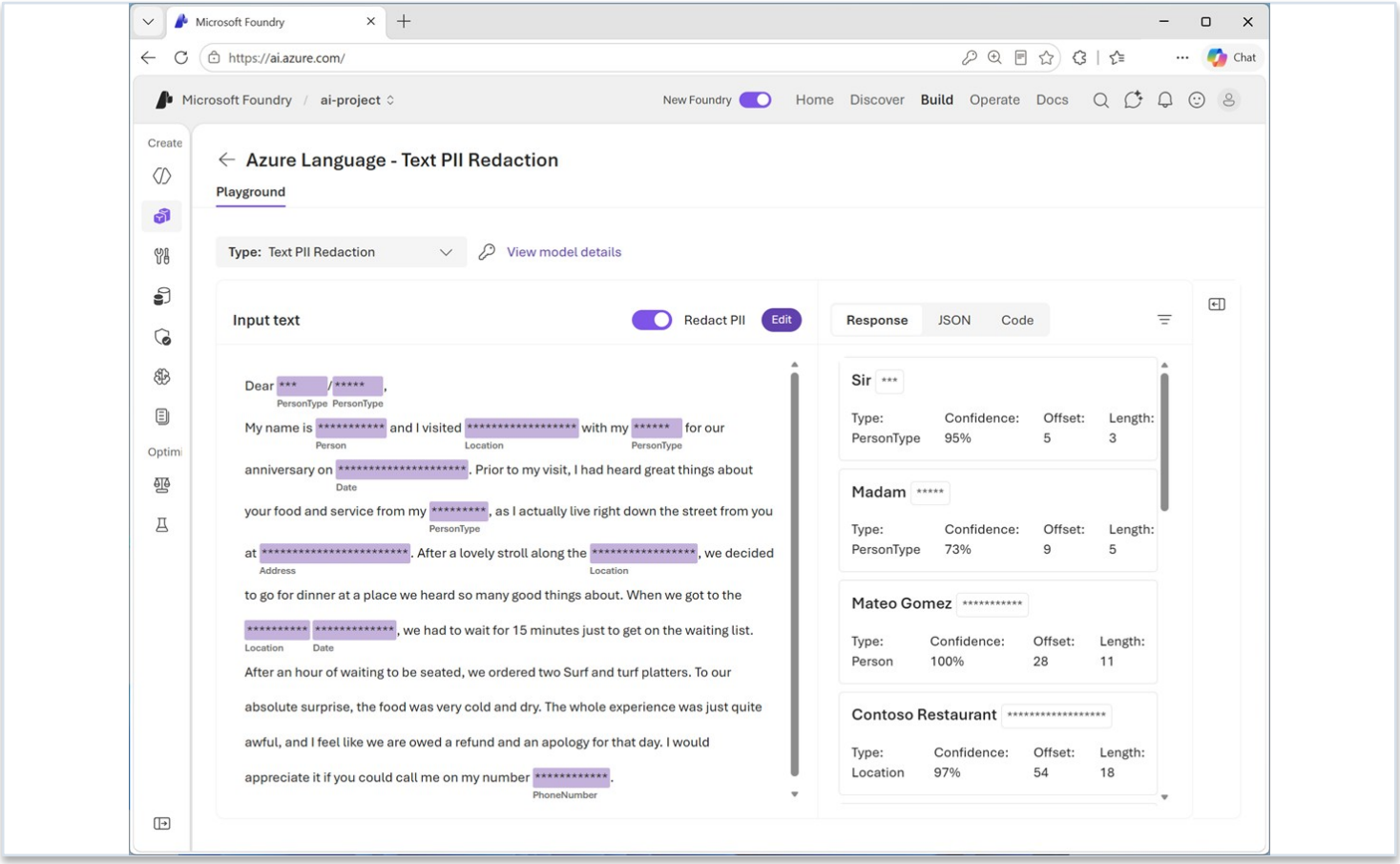
Return no answer or escalate when evidence is absent.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Legacy Language Understanding Lab



1. Activity: Custom question answering.

2. Identify the input contract for Legacy Language Understanding Lab.

3. Trace the processing component and its state change.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Define questions and answers

1

These questions and answers can be: Generated from an existing FAQ document or web page.

2

Entered and edited manually.

3

Identify the input contract for Define questions and answers.

4

Trace the processing component and its state change.

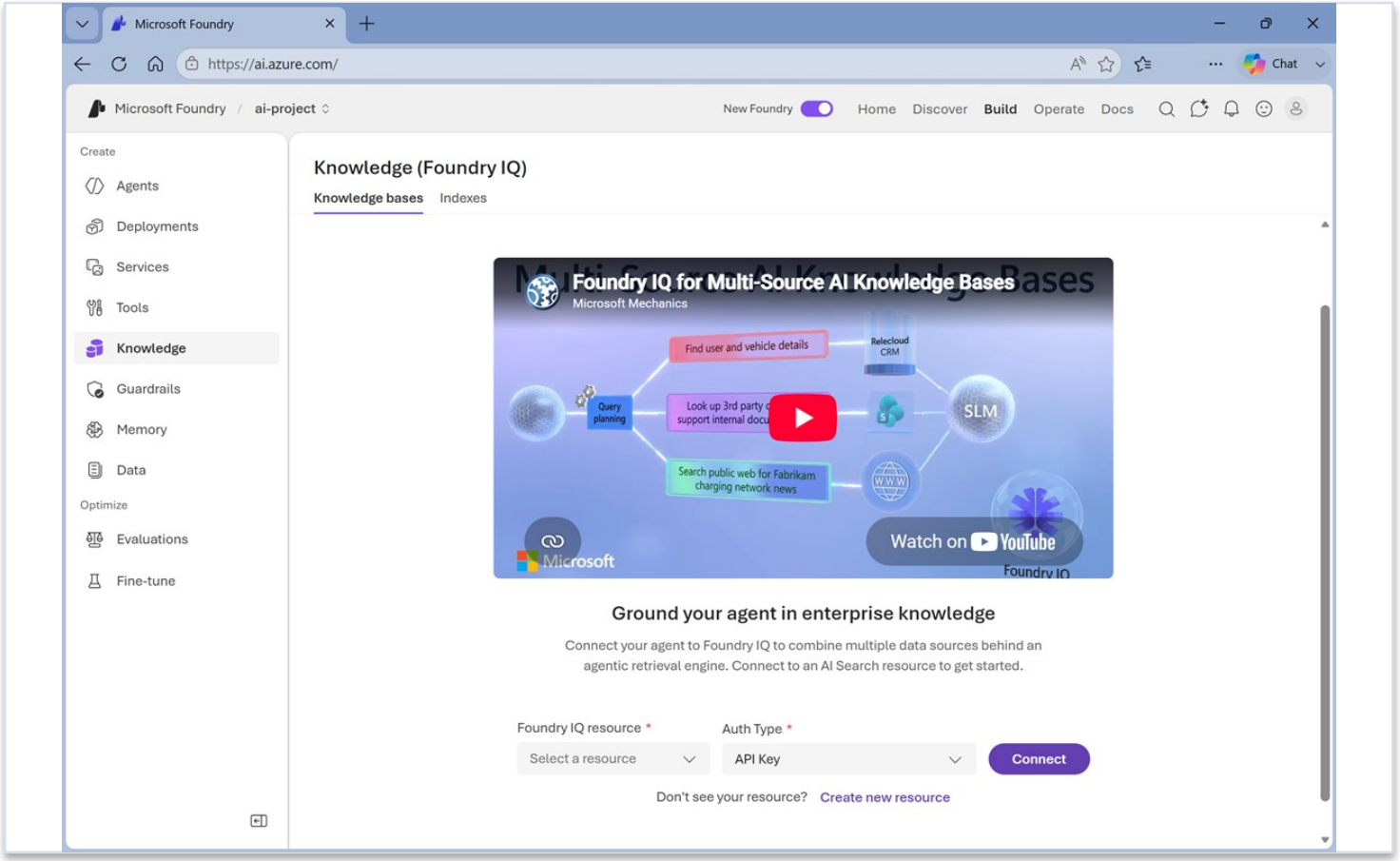
Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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LEGACY SLIDE 193 · AUTHENTIC UI

Test the Knowledge Base



1 After creating a set of question-and-answer pairs, you must save it.

2 Approved documents become a bounded knowledge source.

3 Retrieval finds candidate passages for an answer.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Use the knowledge base

1

When you're satisfied with your knowledge base, deploy it.

2

Then you can use it over its REST interface.

3

To access the knowledge base, client applications require: The knowledge base ID The knowledge base endpoint The knowledge base authorization key.

4

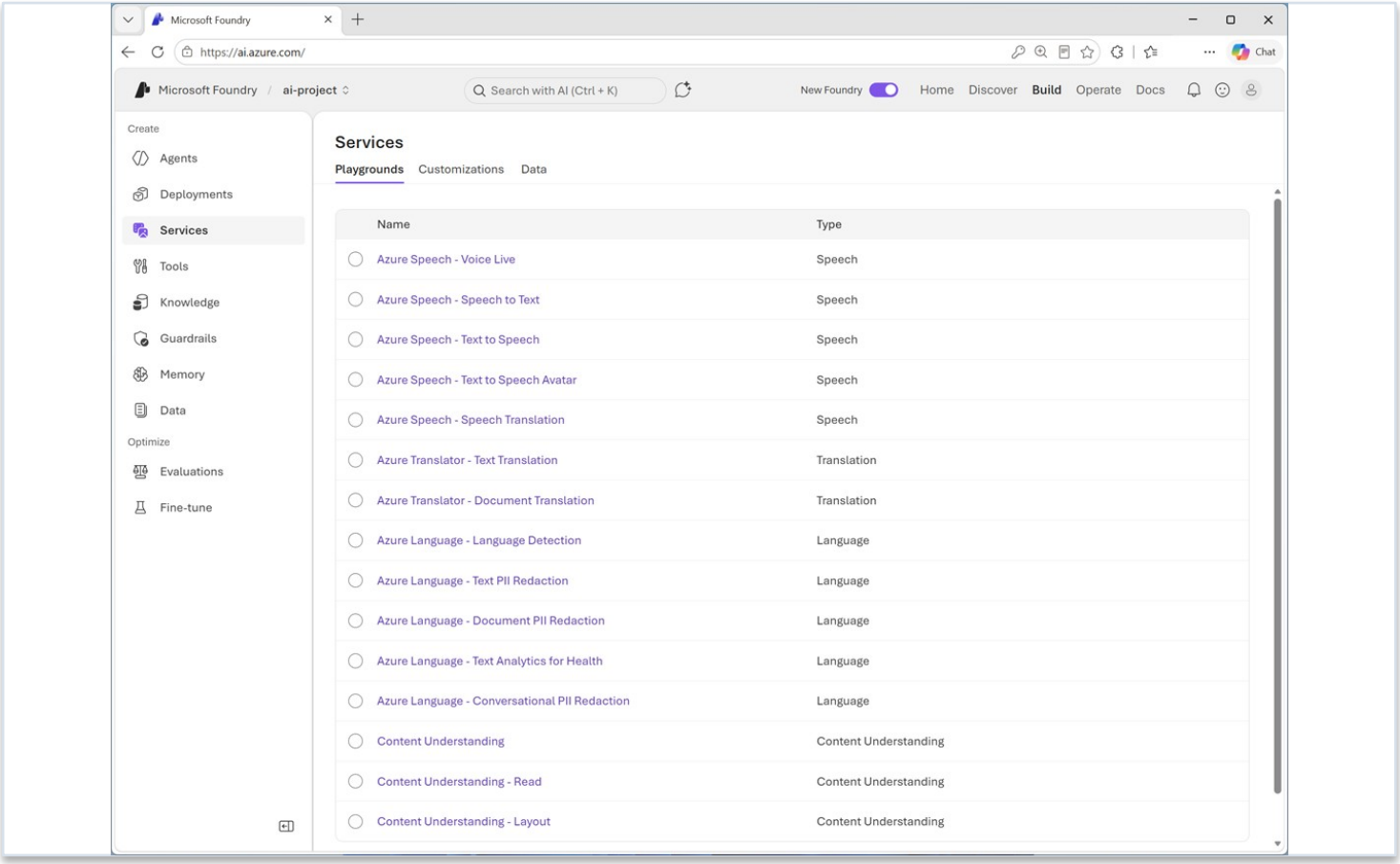
Approved documents become a bounded knowledge source.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Legacy QnA Bot Lab Migration



1. Activity: Custom question answering.

2. Identify the input contract for Legacy QnA Bot Lab Migration.

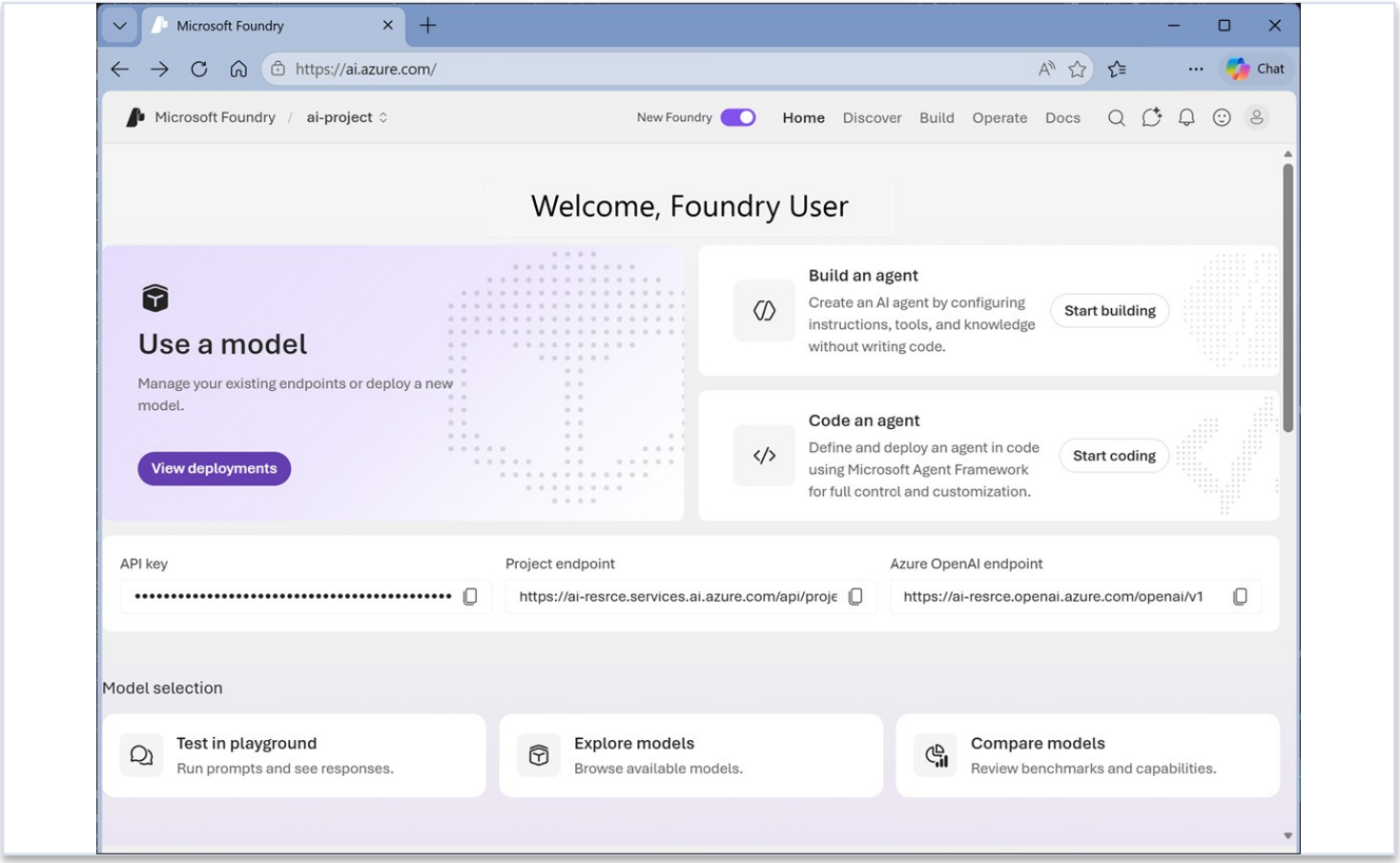
3. Trace the processing component and its state change.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Build a bot with the Azure Bot Service



1. Identify the input contract for Build a bot with the Azure Bot Service.
2. Trace the processing component and its state change.
3. Capture a named output, metric or observable artifact.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Conversational Language Understanding

1

On Microsoft Azure, conversational language understanding is supported through the Language Service.

2

To work with Conversational Language Understanding, you need to take into account three core concepts: utterances entities intents.

3

An utterance is mapped to an intent and extracted entities.

4

Confidence and competing intents determine the route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Utterances

1

An utterance is mapped to an intent and extracted entities.

2

Confidence and competing intents determine the route.

3

Fallback handles unsupported or ambiguous requests.

4

Ground responses and escalate high-impact interactions.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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Entities

1

Identify the input contract for Entities.

2

Trace the processing component and its state change.

3

Capture a named output, metric or observable artifact.

4

Define the acceptance threshold and human-review route.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Intents

1

An utterance is mapped to an intent and extracted entities.

2

Confidence and competing intents determine the route.

3

Fallback handles unsupported or ambiguous requests.

4

Ground responses and escalate high-impact interactions.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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TOPIC 2 · HANDS-ON BRIDGE

Lab 06 · Speech, Translation, and Conversational AI

60 MIN

A multilingual support team needs a voice interaction that fails safely in noisy conditions.

SCENARIO

Use speech input/output, design a translation contract, and define a human handoff for conversational AI.

MECHANISM

Microsoft Foundry; Azure Speech Voice Live or Speech; Translator design worksheet

EVIDENCE

A transcript/output evidence pack, translation message contract and escalation decision table.

VERIFY

Speech input and output both evidenced · Noise/accent difference discussed

Full step-by-step: Learner Guide + labs/lab-06-speech-translation-conversational-ai/README.md

Lab bridge only—no click-by-click procedure on the slide.

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Azure AI Language Capability

1 INPUT / SIGNAL

Identify language, sentiment, key phrases, entities and PII.

2 MECHANISM

Select the correct input and response schema.

3 OUTPUT / EVIDENCE

Validate offsets, confidence and redacted spans.

4 CONTROL / VERIFY

Route ambiguous or sensitive outputs for review.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Azure AI Language Capability

1 **INPUT / SIGNAL**

Define a grounded knowledge source and answer boundary.

2 **MECHANISM**

Distinguish retrieval evidence from generated wording.

3 **OUTPUT / EVIDENCE**

Test supported, unsupported and adversarial questions.

4 **CONTROL / VERIFY**

Escalate when no approved source supports an answer.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Knowledge Quiz - Fundamentals of conversational language understanding

1 INPUT / SIGNAL

Distinguish utterance, intent and entity.

2 MECHANISM

Write representative training and test utterances.

3 OUTPUT / EVIDENCE

Interpret confidence and competing intent evidence.

4 CONTROL / VERIFY

Use a fallback or human route below threshold.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Knowledge Quiz - Fundamentals of Azure AI Speech

1 INPUT / SIGNAL

Distinguish speech recognition from synthesis.

2 MECHANISM

Set language, voice and audio format deliberately.

3 OUTPUT / EVIDENCE

Validate transcript or audio output against the input.

4 CONTROL / VERIFY

Handle noise, consent and uncertain recognition safely.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Knowledge Quiz - Fundamentals of language translation

1 INPUT / SIGNAL

Choose text or speech translation for the channel.

2 MECHANISM

Preserve language codes and encoding across services.

3 OUTPUT / EVIDENCE

Validate meaning, named terms and unsupported language behavior.

4 CONTROL / VERIFY

Retain a human review route for consequential content.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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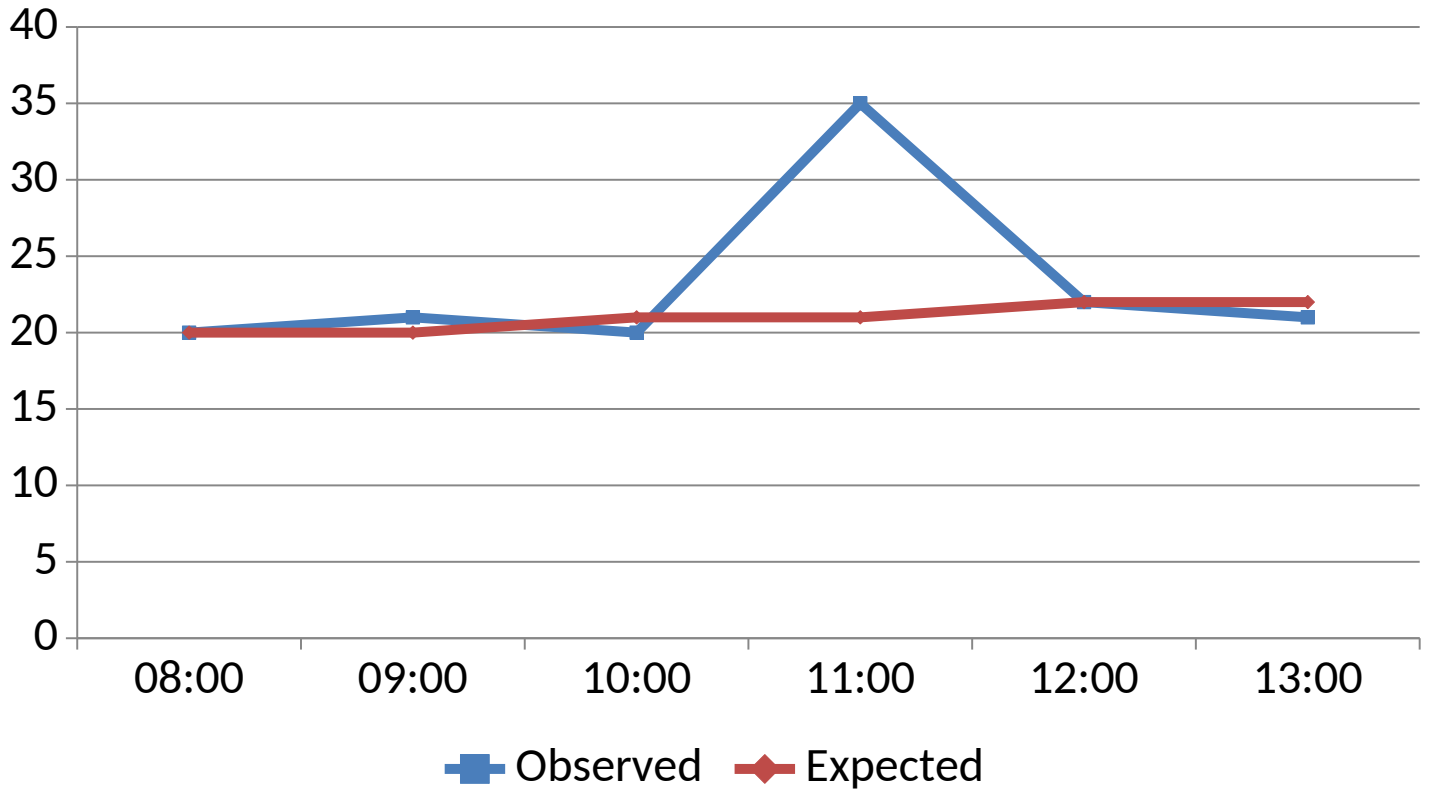
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LEGACY COVERAGE · SLIDE 206

Topic 5 Explore Decision Support

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Anomaly Detection Boundaries



The Anomaly Detector service identifies anomalies that exist outside the scope of a boundary.

The boundary is set using a sensitivity value.

By default, the upper and lower boundaries for anomaly detection are calculated using concepts known as expectedValue, upperMargin, and lowerMargin.

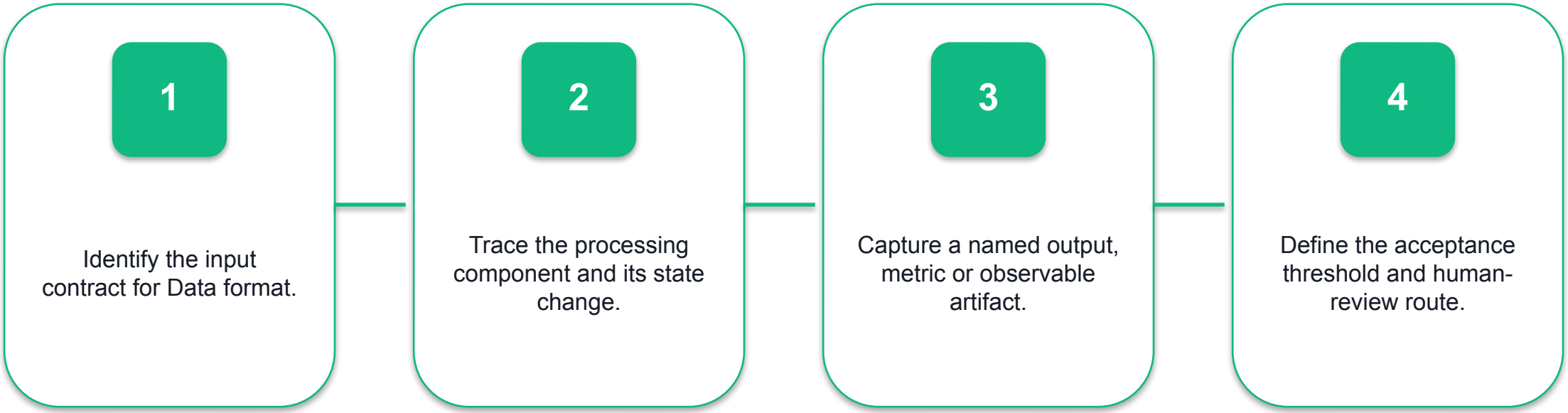
Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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LEGACY SLIDE 208 · MODEL

Data format

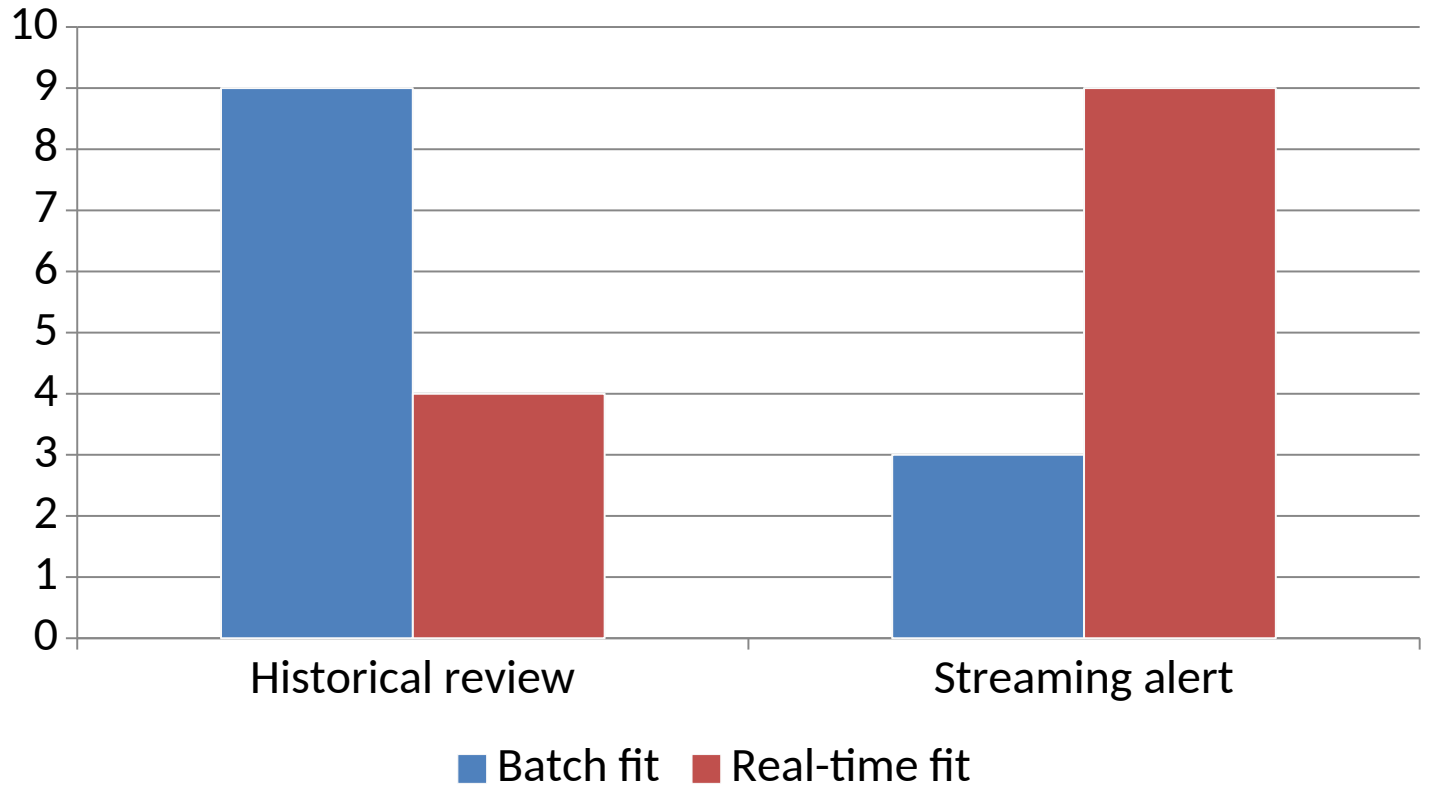


Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Choose Batch or Real-Time Detection



The Anomaly Detector service supports batch processing of time series data and last-point anomaly detection for real-time data.

Identify the input contract for Choose Batch or Real-Time Detection.

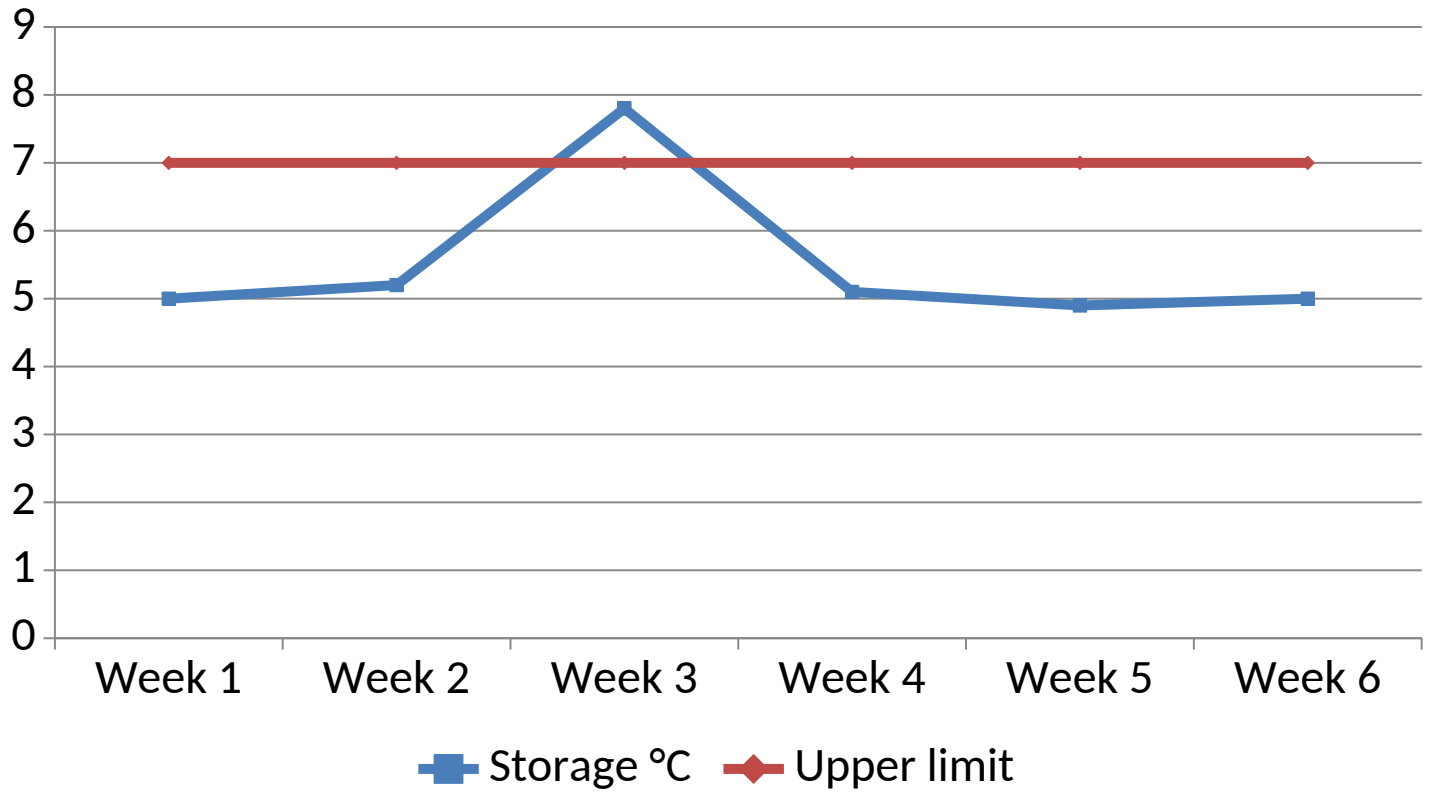
Trace the processing component and its state change.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Batch Detection



Identify the input contract for Batch Detection.

Trace the processing component and its state change.

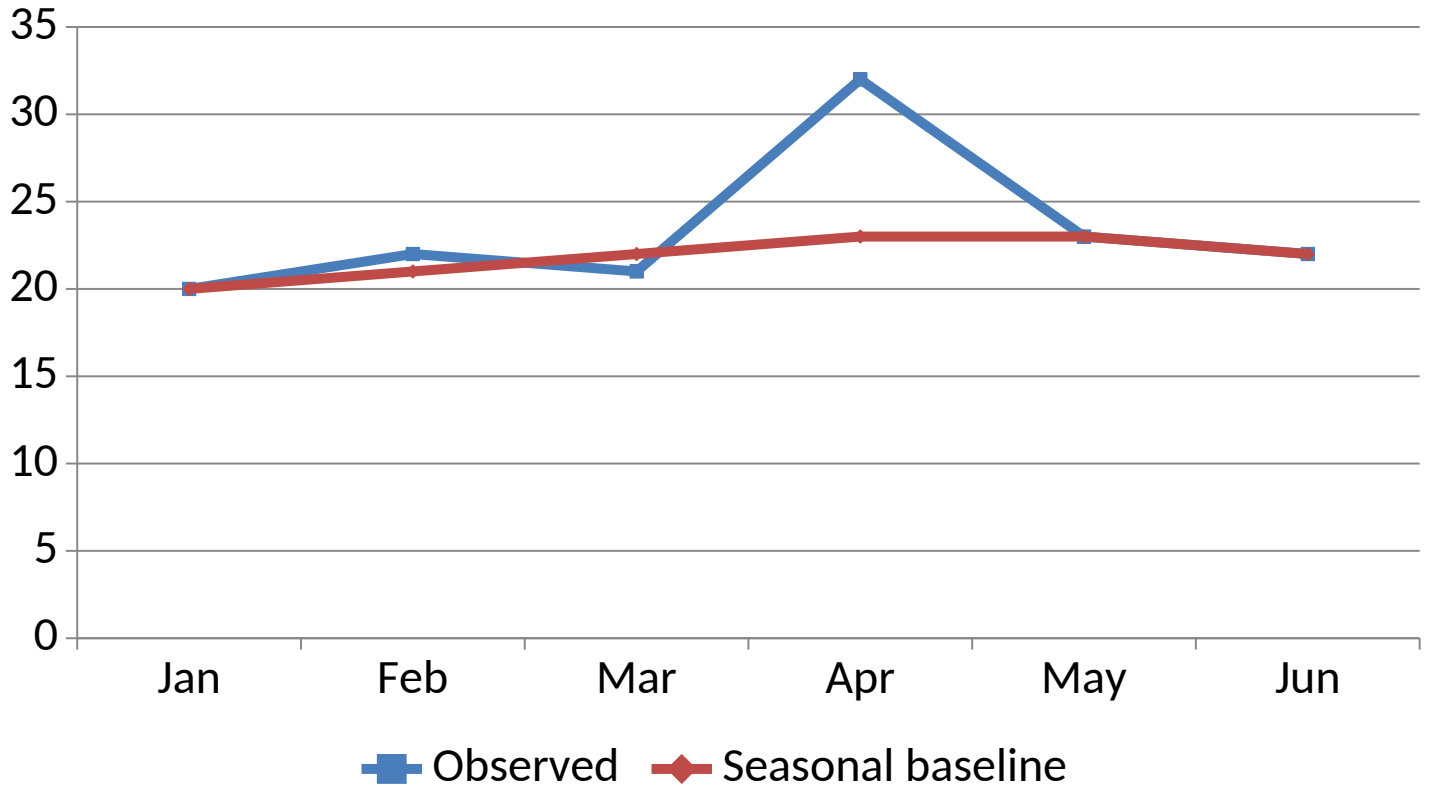
Capture a named output, metric or observable artifact.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Batch Detection Example



Batch detection involves applying the algorithm to an entire data series at one time.

The concept of time series data involves evaluation of a data set as a batch.

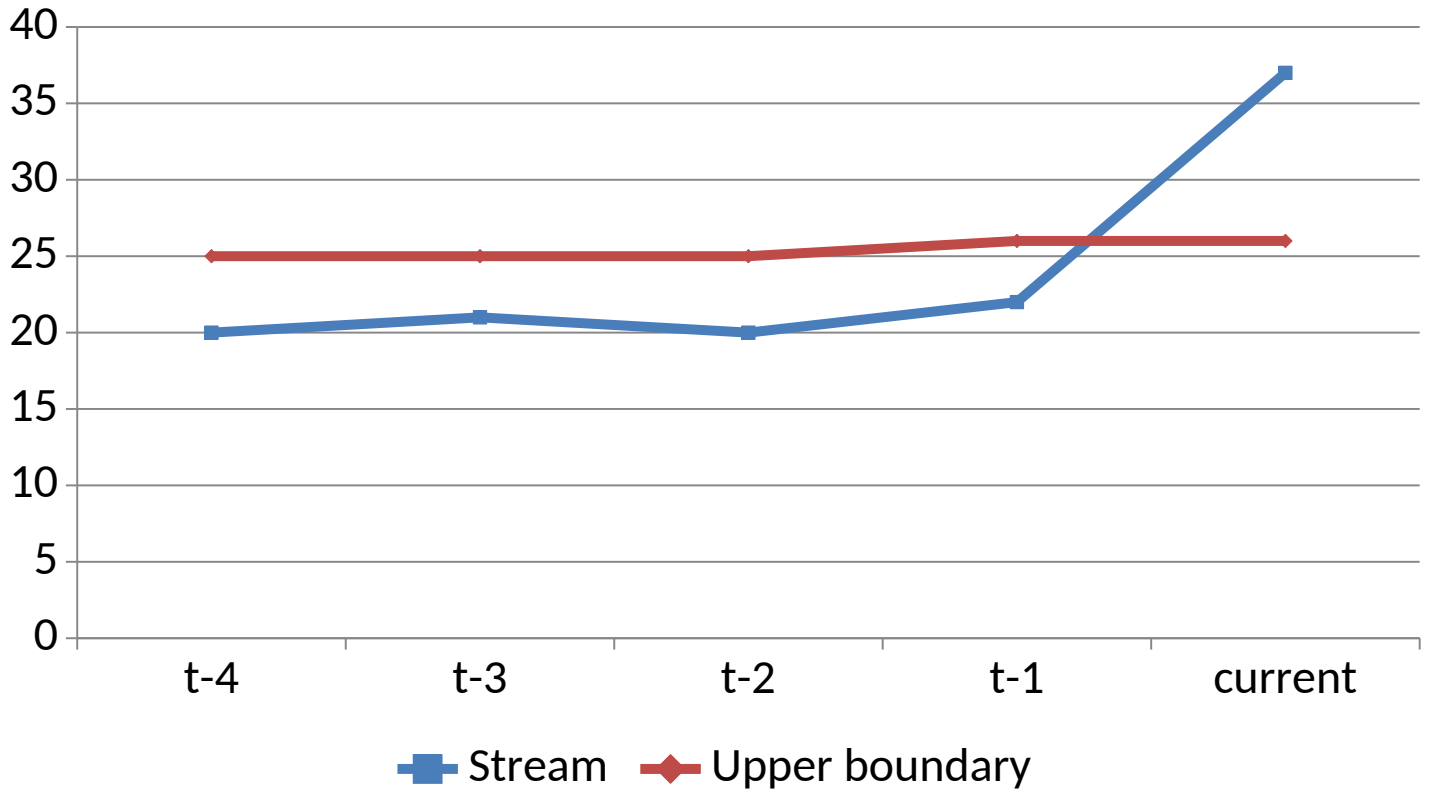
Use your time series to detect any anomalies that might exist throughout your data.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Real-time Detection



Real-time detection uses streaming data by comparing previously seen data points to the last data point to determine if your latest one is an anomaly.

This operation generates a model using the data points you send, and determines if the target (current) point is an anomaly.

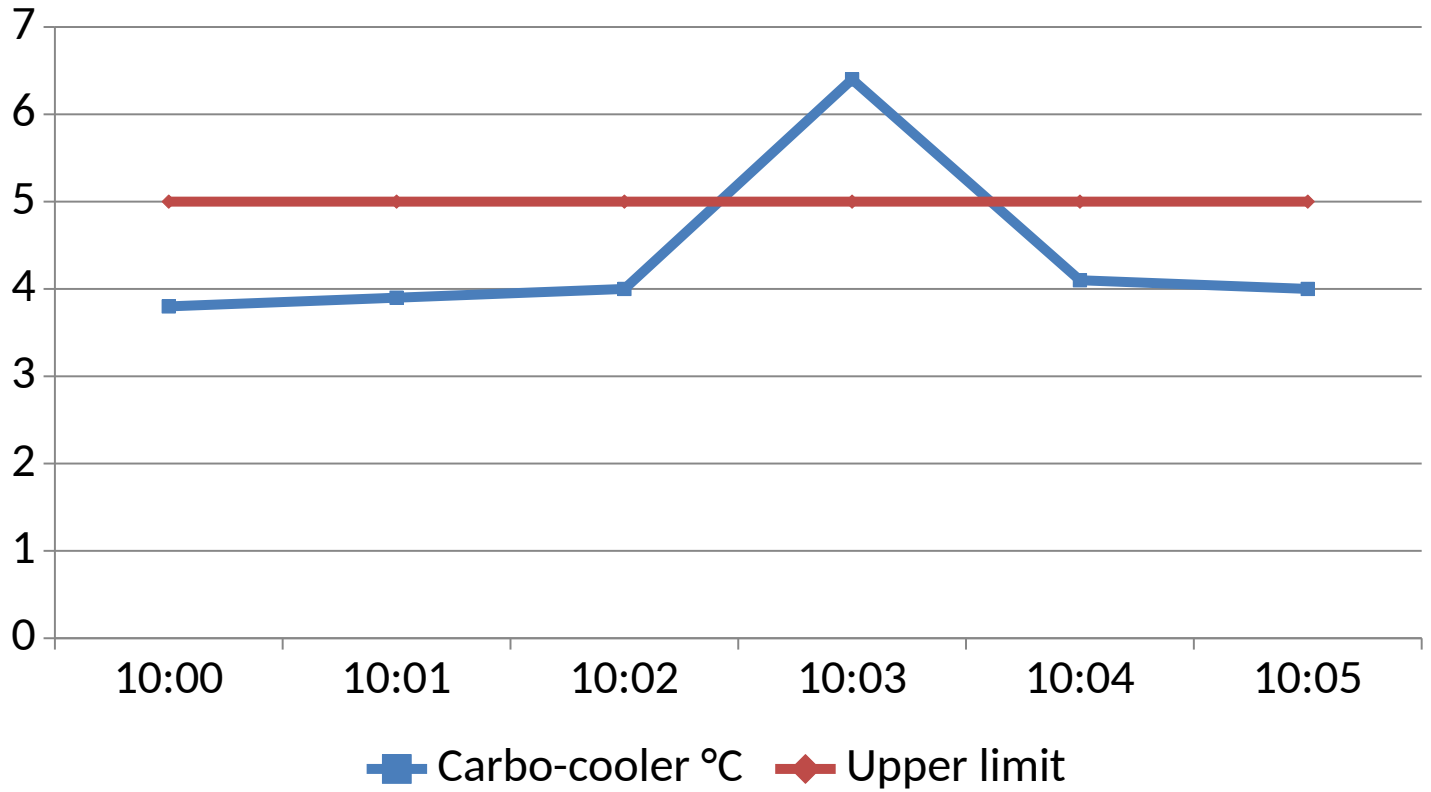
By calling the service with each new data point you generate, you can monitor your data as it's created.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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Real-time Detection Example



Consider a scenario in the carbonated beverage industry where real-time anomaly detection may be useful.

The carbon dioxide added to soft drinks during the bottling or canning process needs to stay in a specific temperature range.

Bottling systems use a device known as a carbo-cooler to achieve the refrigeration of the product for this process.

Interpret the evidence; do not infer causality.

Editable native PowerPoint chart; example values align to Labs 2 and 9.

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TOPIC 3 · HANDS-ON BRIDGE

Lab 09 · Anomaly Evidence, Content Safety, Security, and Governance

60 MIN

Operations must replace a retired managed anomaly service without losing detection evidence or escalation controls.

SCENARIO

Compute anomaly evidence, test content-safety categories, and produce an actionable issue/governance report.

MECHANISM

Spreadsheet or Python; Microsoft Foundry safety guardrails; Azure governance worksheet

EVIDENCE

A reproducible anomaly table, safety-severity results, issue report, access model and cost/cleanup checklist.

VERIFY

Anomaly rule is reproducible · Retirement/migration risk stated

Full step-by-step: Learner Guide + labs/lab-09-anomaly-safety-governance/README.md

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Topic 6 Explore knowledge mining

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LEGACY SLIDE 215 · MODEL

Azure AI Search

1

A key field uniquely identifies each indexed document.

2

Searchable, filterable and retrievable flags shape queries.

3

An indexer moves and enriches content from a data source.

4

Validate freshness, permissions and query evidence.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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LEGACY SLIDE 216 · MODEL

Understand Indexes

1

An Azure AI Search index can be thought of as a container of searchable documents.

2

Conceptually you can think of an index as a table and each row in the table represents a document.

3

Tables have columns, and the columns can be thought of as equivalent to the fields in a document.

4

Columns have data types, just as the fields do on the documents.

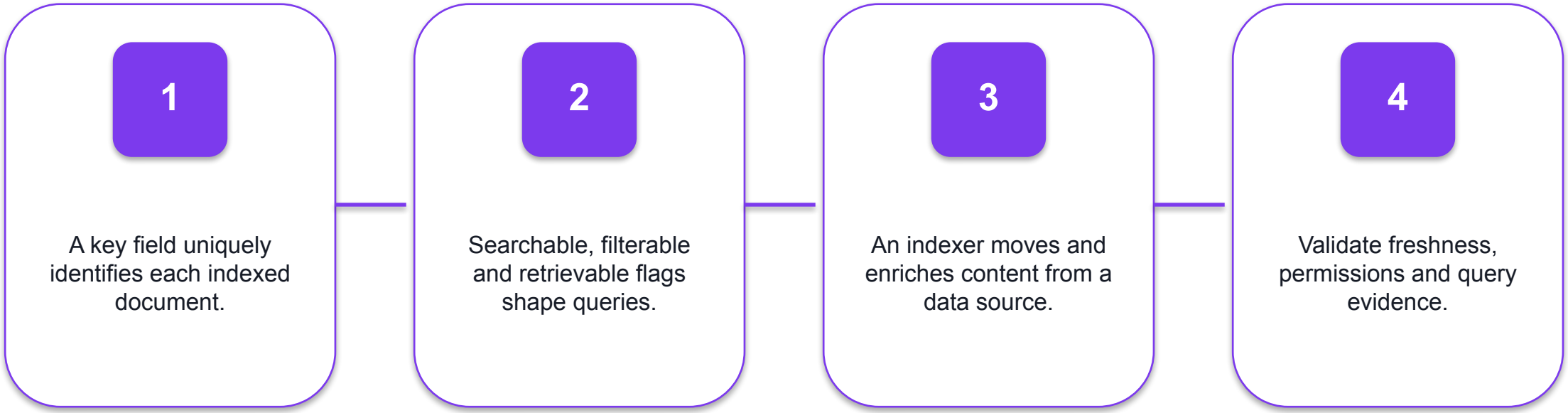
Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 217 · UI

Index Schema



Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

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LEGACY SLIDE 218 · MODEL

Index Attributes

1

A key field uniquely identifies each indexed document.

2

Searchable, filterable and retrievable flags shape queries.

3

An indexer moves and enriches content from a data source.

4

Validate freshness, permissions and query evidence.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Use an indexer to build an index

1

Pushing data has the most flexibility as it has no restrictions on the data source type, location, or frequency of execution.

2

A key field uniquely identifies each indexed document.

3

Searchable, filterable and retrievable flags shape queries.

4

An indexer moves and enriches content from a data source.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Knowledge Store

1

A knowledge store is persistent storage of enriched content.

2

The purpose of a knowledge store is to store the data generated from AI enrichment in a container.

3

For example, you may want to save the results of an AI skillset that generates captions from images.

4

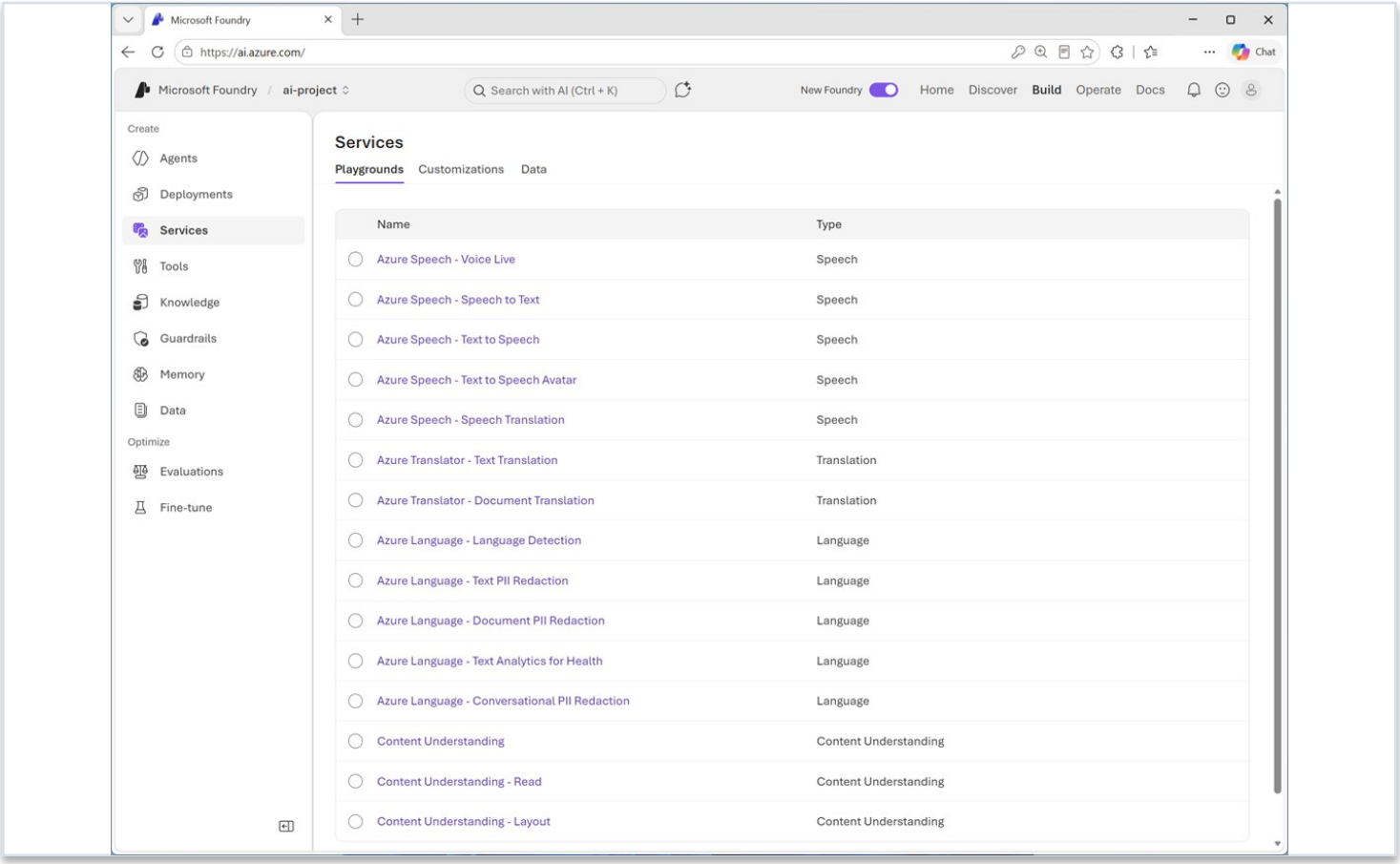
Identify the input contract for Knowledge Store.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Create an index in the Azure portal



1. A key field uniquely identifies each indexed document.

2. Searchable, filterable and retrievable flags shape queries.

3. An indexer moves and enriches content from a data source.

UI may evolve; verify the named state and output.

UI source: MicrosoftLearning repositories; screenshot pinned to the source register.

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Query an Azure AI Search Index

1

Index and query design are closely linked.

2

After we build the index, we can perform queries.

3

A crucial component to understand is that the schema of the index determines what queries can be answered.

4

Azure AI Search queries can be submitted as an HTTP or REST API request, with the response coming back as JSON.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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Simple query requests

1

A query request is a list or words (search terms) and query operators (simple or full) of what you would like to see returned in a result set.

2

Let's look what components make up a search query.

3

Consider this simple search example coffee (-"busy" + "wifi").

4

Identify the input contract for Simple query requests.

Trace the named input, transformation, output and review decision. Full click path: Learner Guide and aligned lab folder.

Mechanism redraw from legacy v9; editable PowerPoint shapes.

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TOPIC 4 · HANDS-ON BRIDGE

Lab 07 · Document Intelligence and Knowledge Mining

70 MIN

Accounts payable needs receipt data that can move safely from extraction into search and reporting systems.

SCENARIO

Extract document fields, normalize them to a data contract, and prepare searchable knowledge records.

MECHANISM

Microsoft Foundry Content Understanding; Azure AI Search design worksheet; JSON Schema

EVIDENCE

A receipt extraction, cleaned interoperable JSON document, field-quality report and search-index design.

VERIFY

Raw and structured outputs compared · Normalized JSON validates

Full step-by-step: Learner Guide + labs/lab-07-document-intelligence-knowledge-mining/README.md

Lab bridge only—no click-by-click procedure on the slide.

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Knowledge Quiz - Fundamentals of Azure AI Document Intelligence

1 INPUT / SIGNAL

Compare prebuilt and custom document extraction.

2 MECHANISM

Validate fields, tables, geometry and confidence.

3 OUTPUT / EVIDENCE

Preserve the original document for traceability.

4 CONTROL / VERIFY

Redact sensitive fields and document review decisions.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Knowledge Check · Knowledge Mining and Azure AI Search

1 INPUT / SIGNAL

Define key, searchable, filterable and retrievable fields.

2 MECHANISM

Trace ingestion from data source through indexer to index.

3 OUTPUT / EVIDENCE

Test simple, filtered and semantic retrieval behavior.

4 CONTROL / VERIFY

Monitor freshness, access scope and failed enrichment.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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TOPIC 4 · HANDS-ON BRIDGE

Lab 10 · AI-900 Capstone: End-to-End Azure AI Solution Review

80 MIN

Contoso must approve one cross-service Azure AI solution for product feedback, receipts and policy support.

SCENARIO

Integrate service selection, workflow deployment, issue reporting, interoperability, cleaning and governance into one evidence-backed design.

MECHANISM

Architecture worksheet; prior lab evidence; Microsoft Foundry; Azure AI Search design

EVIDENCE

A final architecture, data contract, evidence matrix, issue register, validation plan and resource-cleanup record.

VERIFY

Architecture names inputs, outputs and stores · Canonical schema and cleaning rules included

Full step-by-step: Learner Guide + labs/lab-10-ai900-capstone-course-review/README.md

Lab bridge only—no click-by-click procedure on the slide.

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Summary Q&A

1 INPUT / SIGNAL

Connect each workload to its input, model/service and output.

2 MECHANISM

Explain one metric and the decision it supports.

3 OUTPUT / EVIDENCE

Name one Responsible AI control and its evidence.

4 CONTROL / VERIFY

Resolve remaining questions before the assessment briefing.

Control: define the threshold or failure that routes to human review.

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Course Feedback and TRAQOM

1 INPUT / SIGNAL

Complete the course feedback in the LMS-TMS.

2 MECHANISM

Complete the mandatory TRAQOM survey.

3 OUTPUT / EVIDENCE

Use the correct course run and enrolled learner identity.

4 CONTROL / VERIFY

Show the submitted confirmation state to the trainer.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Digital Attendance

1 INPUT / SIGNAL

Scan the assessment attendance QR code shown by the trainer.

2 MECHANISM

Submit with the enrolled learner identity.

3 OUTPUT / EVIDENCE

Confirm the correct course run and session.

4 CONTROL / VERIFY

Show the confirmation state before starting assessment.

Control: define the threshold or failure that routes to human review.

Source: legacy v9 deck; terminology refreshed against MicrosoftLearning current exercises (commit d1a6668).

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Final Assessment

1 INPUT / SIGNAL

Complete WA-SAQ and PP within the stated durations.

2 MECHANISM

Use only approved materials and your own evidence.

3 OUTPUT / EVIDENCE

Check that every response and practical artifact is submitted.

4 CONTROL / VERIFY

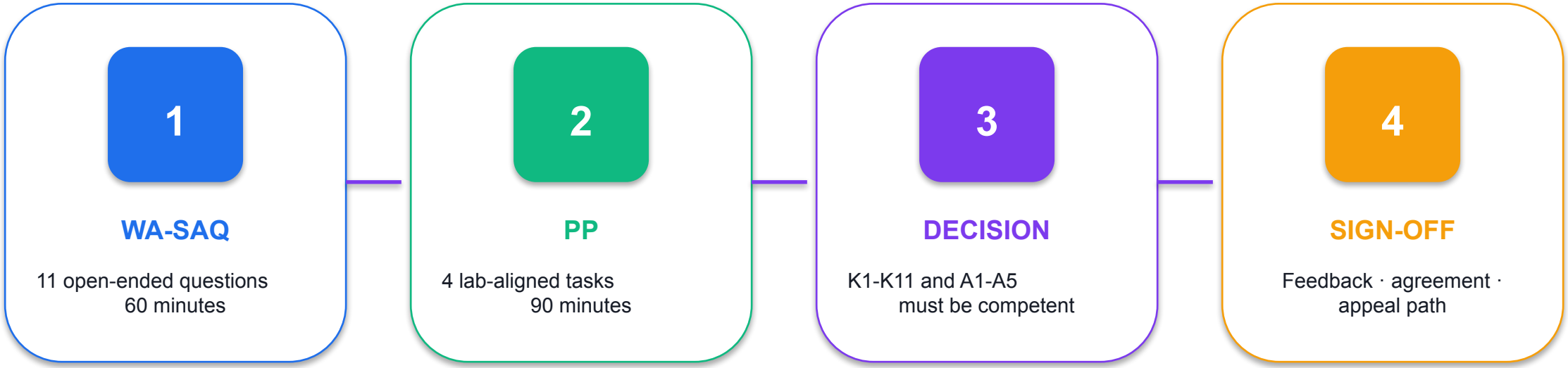
Follow the documented appeal or reassessment process if required.

Control: define the threshold or failure that routes to human review.

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Assessment Flow



If the candidate disagrees, do not sign; notify the assessor and use the documented appeal process within 7 days.

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TRAQOM · MANDATORY

Digital Attendance (Assessment)

1

SCAN

Use the SSG QR code shown by the trainer.

2

SUBMIT

Complete assessment attendance in Singpass.

3

VERIFY

Show confirmation before leaving.

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MICROSOFT AZURE AI FUNDAMENTALS · COMPLETE

Thank You

Support: enquiry@tertiaryinfotech.com · +65 6100 0613

Retain your redacted evidence pack and confirm all temporary Azure resources are deleted or stopped.

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